

# The Lain Programming Language

Language Specification

Version 1.1

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*This document is the authoritative specification of the Lain programming language. It defines the syntax, semantics, and constraints that every conforming implementation shall enforce.*

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# Foreword

This document specifies the Lain programming language, Version 1.1.

Lain is a statically-typed, compiled programming language for systems programming. It guarantees memory safety, type safety, and resource safety entirely at compile time with zero runtime overhead.

This specification is structured analogously to ISO/IEC 9899 (the C standard). Clauses 1 through 4 are administrative; Clauses 5 through 20 are normative. Annexes A and B are normative; Annexes C and D are informative.

Text marked **Constraint** describes a condition whose violation requires an implementation to issue a diagnostic message. Text marked *implementation-defined*<sup>1</sup> denotes behavior that is valid but whose exact result is determined by the implementation and shall be documented by it. Text marked **undefined behavior**<sup>2</sup> denotes a situation for which this specification imposes no requirements.

## Note

Undefined behavior in Lain can arise only inside **unsafe** blocks or through direct C interoperability. All code in the safe fragment of Lain has defined behavior.

---

<sup>1</sup>Implementation-defined behavior:

<sup>2</sup>Undefined behavior:



# Introduction

## Design philosophy

Lain is built on five foundational pillars:

- P1 — Assembly-Speed Performance** Every language feature compiles to C code that is equivalent in performance to hand-written C. There are no hidden allocations, no garbage collector, and no reference-counting overhead.
- P2 — Zero-Cost Memory Safety** Memory safety is enforced entirely at compile time through a linear type system, a borrow checker, and value range analysis. A conforming implementation that rejects all constraint violations guarantees, for safe programs, the absence of use-after-free, double-free, null dereference, and buffer overflow at runtime.
- P3 — Static Verification** The verification properties required by this specification are decidable and polynomial in program size. No SMT solver or unbounded fixpoint iteration is required.
- P4 — Determinism** Pure functions (`func`) are referentially transparent and guaranteed to terminate. The same inputs always produce the same outputs.
- P5 — Syntactic Simplicity** Lain has no implicit copies, no hidden conversions (beyond the defined integer-widening rules), and no magic methods. Ownership is always explicit.

## Relationship to C

Lain compiles to C99. The C code generated by a conforming Lain implementation is the *translation output*; this specification does not prescribe the form of that output beyond the observable behavior of the resulting program.

## How to read this specification

Each normative clause is structured as follows: a prose description of the syntax (with a reference to the formal grammar in Annex A), followed by the semantics, followed by **Constraint** blocks listing every condition that requires a diagnostic message. Non-normative **Note** and **Example** environments appear throughout to aid understanding.



## Chapter 1

# Scope

- 1 This document specifies the Lain programming language. It defines the syntax, the semantic rules, and the constraints that every conforming Lain implementation shall enforce. It serves as the single authoritative reference for language users and implementors alike.
- 2 Lain is a statically-typed, compiled programming language designed for systems programming, embedded systems, and safety-critical software. It achieves memory safety, type safety, and resource safety entirely at compile time with zero runtime overhead.
- 3 This specification covers:
  - lexical structure and grammar (§5);
  - type system and type inference (§7);
  - variable declarations, scoping, and initialization (§6, §10);
  - expressions, operators, and evaluation order (§8);
  - statements and control flow (§9);
  - functions, procedures, and purity (§12);
  - ownership, borrowing, and the linear type system (§11);
  - value range constraints and static verification (§13);
  - compile-time evaluation and generics (§14);
  - pattern matching and exhaustiveness checking (§15);
  - module system (§16);
  - C interoperability (§17);
  - unsafe code boundaries (§18);
  - memory model (§19);
  - diagnostic messages (Annex B);
  - standard library (§20).
- 4 This specification does *not* cover:
  - the internal architecture of any particular compiler implementation;
  - build system integration or tooling;
  - IDE or editor support;
  - operating system interfaces beyond what the standard library provides.

## 1.1 Compilation model

- 1 Lain uses a source-to-C transpilation model. A Lain source file (extension `.ln`) is processed by a conforming implementation to produce C99 source code, which is subsequently compiled by a C99 compiler to produce an executable. The observable behavior of the program is defined by this specification; the form of the intermediate C99 code is *implementation-defined*<sup>1</sup>.
- 2 A conforming implementation shall perform the following translation phases in the following order:
  1. **Lexical analysis** — source text is converted to a token stream as described in §5.
  2. **Parsing** — the token stream is reduced to an abstract syntax tree according to the grammar in Annex A.
  3. **Name resolution** — identifiers are resolved to declarations as described in §6.
  4. **Type checking** — types are inferred and verified as described in §7 through §10.
  5. **Exhaustiveness checking** — pattern matching completeness is verified as described in §15.
  6. **Use analysis (NLL)** — last-use points are computed for all variables.
  7. **Linearity and borrow checking** — ownership and borrowing rules are verified as described in §11.
  8. **Value range analysis** — bounds safety and numeric constraints are verified as described in §13.
  9. **Code emission** — well-formed programs are translated to C99.
- 3 A program that completes all phases without a diagnostic message is **well-formed** and its translation output is defined.

### Note

Phases 3–8 collectively form *semantic analysis*. An implementation may interleave these phases as long as the observable behavior is equivalent to executing them in the order listed.

---

<sup>1</sup>Implementation-defined behavior: the generated C99 code need not be human-readable

## Chapter 2

# Normative references

- <sup>1</sup> The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies.

### **ISO/IEC 9899:2018 (C17)**

*Information technology — Programming languages — C.*

International Organization for Standardization, Geneva, 2018.

Referenced for: C99/C17 calling conventions (§ 17), standard integer types and their representations (§ 7), and undefined behavior categories (§ 18).

### **IEC 60559:2020 (IEEE 754-2019)**

*Floating-point arithmetic.*

International Electrotechnical Commission, Geneva, 2020.

Referenced for: semantics of `f32` and `f64` arithmetic (§ 7).

- <sup>2</sup> The following documents are informative references. They are not required for conformance but provide useful background.

### **ISO/IEC 14882:2020 (C++20)**

*Programming languages — C++.*

Referenced for: vocabulary and structure conventions used in this document.

### **The Rust Reference**

*The Rust Reference*, <https://doc.rust-lang.org/reference/>.

Referenced for: linear type and ownership terminology.



## Chapter 3

# Terms, definitions, and symbols

- <sup>1</sup> For the purposes of this document, the following terms and definitions apply. Terms defined here are used with these precise meanings throughout all normative clauses. Where a term is used in its ordinary English sense rather than as a defined term, it appears without emphasis.

### Note

Cross-references of the form (*see §N*) point to the clause where the concept is treated normatively.

### 3.1 General language terms

#### abstract syntax tree (AST)

The in-memory tree representation of a Lain program produced by the parser and consumed by semantic analysis.

#### behavior

External appearance or action of a program as observed by a user or another program. See also: *defined behavior*, *implementation-defined behavior*, *unspecified behavior*, *undefined behavior*.

#### conforming implementation

An implementation that correctly processes all programs that satisfy the rules of this specification and issues a diagnostic message for every program that violates a **Constraint** (see §4).

#### conforming program

A program that uses only the syntax and semantics defined in this specification, does not rely on undefined behavior, and does not depend on implementation-defined extensions.

#### constraint

A requirement imposed on the source text. If a program violates a constraint, the implementation *shall* emit a diagnostic message. Constraints appear in **Constraint** environments throughout this specification.

#### defined behavior

Behavior that is fully and precisely specified by this document for every conforming program. Contrast with *undefined behavior* and *implementation-defined behavior*.

### **diagnostic message**

A human-readable message issued by the implementation to report a constraint violation. Every diagnostic shall include a source location (file name, line number) and an error code of the form [EXXX] or [VRA] (see Annex B).

### **ill-formed program**

A program that violates one or more of the syntactic or semantic rules of this specification. An ill-formed program is not a conforming program and a conforming implementation shall reject it with a diagnostic.

### **implementation**

The system (compiler, linker, runtime) that translates a Lain source program into an executable. The reference implementation is the *lain* compiler.

### **implementation-defined behavior**

Behavior that is valid but not fully specified by this document; the implementation shall determine the exact behavior and document it.

### **program**

A collection of one or more Lain source files (`.ln`) that together constitute a translation unit or a set of modules.

### **source file**

A UTF-8 encoded text file with the extension `.ln` containing Lain source text.

### **translation output**

The C99 source code emitted by a conforming implementation for a well-formed Lain program.

### **undefined behavior**

Behavior for which this specification imposes no requirements. A program that exhibits undefined behavior is erroneous. In the safe fragment of Lain, no operation produces undefined behavior. Undefined behavior may arise only inside `unsafe` blocks or through direct C interoperability.

### **unspecified behavior**

Behavior for which this specification provides two or more possibilities without prescribing which shall occur. A conforming program shall not depend on unspecified behavior.

### **well-formed program**

A program that satisfies all syntactic and semantic rules of this specification and is therefore accepted by a conforming implementation without diagnostic.

## 3.2 Type system terms

### algebraic data type (ADT)

A composite type that is either a product type (*struct*) or a sum type (*enum*).

### constrained type

A type annotated with a value range constraint of the form *T op bound*, e.g. `int >= 0` (see §7).

### integer rank

A total ordering on integer types that determines the result type of mixed-type arithmetic. The rank system is defined in §7.

### integer widening

An implicit conversion from an integer type of lower rank to one of higher rank, permitted in arithmetic and assignment contexts (see §7).

### linear type

A type whose values must be used exactly once. Struct types and resource types are linear. Primitive types (`int`, `bool`, etc.) are unrestricted (see §11).

### nominal type

A type identified by its declared name rather than its structure.

### ownership mode

An annotation on a variable or parameter that specifies the kind of access the holder has: *shared* (read-only), *var* (read-write), or *owned* (no annotation on a linear type, transferred with *mov*) (see §11).

### primitive type

One of the built-in types: `int`, `u8`, `u16`, `u32`, `u64`, `usize`, `i8`, `i16`, `i32`, `i64`, `isize`, `f32`, `f64`, `bool`. Primitive types are unrestricted (non-linear).

### structural type equivalence

Two types are structurally equivalent if they have the same structure. Lain uses nominal equivalence for user-defined types; the name is part of the type identity.

### unrestricted type

A type whose values may be used any number of times without explicit consumption. All primitive types are unrestricted. Contrast with *linear type*.

### 3.3 Ownership and memory terms

#### active borrow

A borrow that is currently in the ACTIVE phase; its subject cannot be moved or mutated while the borrow is active (see § 11).

#### arena

A memory allocation strategy in which all allocations are made from a contiguous block; the entire block is freed at once. Lain’s compiler allocates all AST nodes from arenas.

#### borrow

A temporary access to a value without taking ownership. A shared borrow grants read-only access; a mutable borrow grants read-write access (see § 11).

#### borrow phase

Either RESERVED (the borrow has been registered but is not yet active, during argument evaluation) or ACTIVE (the borrow is in force) (see § 11).

#### consumed variable

A linear variable whose ownership has been transferred (via `mov`). Reading or transferring a consumed variable is a constraint violation.

#### linear state

The tracking state of a linear variable: one of *Uninit*, *Unconsumed*, or *Consumed* (see § 11).

#### move

The transfer of ownership of a value from one variable or storage location to another, performed by the `mov` operator. After a move, the source is in the Consumed state.

#### non-lexical lifetime (NLL)

A lifetime that ends at the variable’s *last use* rather than at the end of its lexical scope. Lain uses NLL to minimize the duration of borrows (see § 11).

#### read-write lock invariant

The aliasing invariant: at any program point, for any owned value, there exist either  $N$  shared borrows ( $N \geq 0$ ) and no mutable borrow, or exactly one mutable borrow and no shared borrows (see § 11).

#### resource

A value that represents an external entity (file handle, socket, etc.) whose lifetime must be explicitly managed. Resources are modelled as linear types.

#### two-phase borrow

A borrow that begins in the RESERVED phase during argument evaluation and is promoted to ACTIVE after the call expression is complete (see § 11).

### 3.4 Control flow and purity terms

#### decreasing measure

An expression in a **while decreasing** statement that the implementation shall verify is non-negative when the loop condition holds and strictly decreases on every iteration (see § 9).

#### func

A function. Its effects are those its **effect row** names, and the row's default is empty — pure and terminating. A function shall not have an effect its row does not name (see § 12).

#### purity

The property of a function with an empty row: its result depends only on its inputs and it has no observable side effects on any state outside its own stack frame.

#### effect row

The set of effects a function may have, drawn from **io**, **diverge**, **raises** and **alloc**. Inferred from the body, empty by default, widened by an **effects** clause, and a complete upper bound in either case (see § 12.1).

#### proc

*Removed.* An introducer that granted effects rather than stating them; the effect row replaced it. The keyword is still reserved and diagnoses the replacement (see § 10.5).

#### termination

The property that a function always reaches a **return** statement in a finite number of steps. All **funcs** shall be verified to terminate by the implementation (see § 12).

### 3.5 Verification terms

#### bounds check

A runtime test that verifies an array index is within the valid range  $[0, N-1]$ . A conforming implementation shall eliminate bounds checks when VRA can statically prove the index is in range.

#### comptime

Compile-time evaluation. A **comptime** expression or function is evaluated by the compiler at compile time rather than at runtime (see § 14).

#### exhaustiveness

The property of a **case** expression in which every possible value of the scrutinee type is covered by at least one arm (see § 15).

#### in guard

A use of the **in** operator as a loop or conditional guard. An in guard of the form **idx in arr** proves to the VRA that  $\text{idx} \in [0, \text{arr.len}-1]$  in the guarded body (see § 8, § 13).

### **interval**

A pair  $[\ell, u]$  representing the set of integers  $\{x \mid \ell \leq x \leq u\}$ . VRA represents value ranges as intervals with  $-\infty$  and  $+\infty$  sentinels.

### **monomorphization**

The process by which a generic function is instantiated for each concrete set of type arguments, producing a separate, non-generic translation for each instantiation (see § 14).

### **return constraint**

A value range constraint on the return value of a function, declared in the function signature as *T op bound* (see § 13).

### **value range analysis (VRA)**

A static analysis that tracks integer value ranges through the program using interval arithmetic, used to eliminate bounds checks and prove constraint satisfaction (see § 13).

## **3.6 Module and name resolution terms**

### **import**

A directive (`import module.path`) that makes the names defined in another module available in the current module (see § 16).

### **module**

The unit of compilation in Lain. Each source file defines one module.

### **qualified name**

A name of the form *module.identifier* that refers to a declaration in a specific module (see § 6).

### **Universal Function Call Syntax (UFCS)**

A syntactic sugar by which `x.f(args)` is desugared to `f(x, args)` before name resolution. UFCS allows methods to be defined as free functions (see § 12).

## Chapter 4

# Conformance

### 4.1 Conforming implementation

- <sup>1</sup> A **conforming implementation** shall:
  1. accept, without diagnostic, every *conforming program*;
  2. reject, with at least one diagnostic message, every program that violates any constraint stated in this specification;
  3. translate accepted programs such that their observable behavior conforms to the semantic rules stated in this specification;
  4. implement all features described as normative in Clauses 5 through 20;
  5. document every behavior described in this specification as *implementation-defined*<sup>1</sup>.
- <sup>2</sup> A conforming implementation may:
  1. issue additional diagnostic messages (warnings) beyond those required by this specification, provided they do not cause a conforming program to be rejected;
  2. accept programs through implementation-defined extensions, provided those extensions are documented and do not alter the semantics of any conforming program;
  3. omit runtime bounds checks when value range analysis (§ 13) proves the access is within bounds.

#### Note

The reference implementation of Lain (*lain* compiler, version corresponding to this specification) is a conforming implementation. Where this specification marks a behavior as implementation-defined, the reference implementation's documented behavior takes precedence for programs compiled by that implementation.

### 4.2 Conforming program

- <sup>1</sup> A **conforming program** is a program that:
  1. consists only of syntax and semantics defined in this specification;
  2. does not rely on undefined behavior;

---

<sup>1</sup>Implementation-defined behavior: the exact value or behavior is implementation-defined

3. does not rely on unspecified behavior to produce a specific outcome;
  4. does not depend on any implementation-defined extension.
- 2 A **strictly conforming program** additionally:
1. does not use any behavior documented as implementation-defined, even when that behavior is deterministic on a given implementation.

### 4.3 Diagnostic messages

- 1 A diagnostic message issued for a constraint violation shall include:
1. the source file name;
  2. the line number of the offending construct;
  3. the diagnostic code in the form [E`XXX`] (see Annex B);
  4. a human-readable description of the violation.
- 2 A conforming implementation may issue additional diagnostic information (column number, source context, suggested fix) beyond the minimum required by §4.3.

#### Note

The reference implementation additionally prints the source line and a caret (^) pointing to the column of the offending token.

### 4.4 Implementation limits

- 1 A conforming implementation shall support at least the following minimum translation limits. Programs that require more than these limits invoke *implementation-defined*<sup>2</sup>.

| Resource                                    | Minimum      | Clause |
|---------------------------------------------|--------------|--------|
| Identifier length (bytes)                   | 255          | §5     |
| Block nesting depth                         | 64           | §9     |
| Function parameters                         | 127          | §10    |
| Array elements                              | $2^{31} - 1$ | §7     |
| Module import depth                         | 32           | §16    |
| <code>case</code> arms per match expression | 256          | §15    |
| Struct fields                               | 256          | §7     |
| Enum variants                               | 256          | §7     |
| Comptime evaluation depth                   | 256          | §14    |
| Scope nesting depth                         | 64           | §6     |

---

<sup>2</sup>Implementation-defined behavior: the behavior when a translation limit is exceeded

## 4.5 Safe and unsafe fragments

- 1 The **safe fragment** of a Lain program consists of all code that does not appear inside an **unsafe** block and does not use raw pointer operations.
- 2 A conforming implementation shall guarantee, for any well-formed program in the safe fragment, that execution produces no:
  - use of a variable after its ownership has been transferred (use-after-move);
  - double transfer of ownership of the same value;
  - access to an uninitialized variable;
  - violation of the read-write lock invariant (§ 3.3);
  - buffer overrun that has been statically proven safe by value range analysis.
- 3 The **unsafe fragment** comprises all code inside **unsafe** blocks. Within the unsafe fragment, the programmer is responsible for upholding the memory safety invariants listed in § 18; the implementation shall not be required to verify them.

### Constraint

The constraints of the ownership and linearity system (§ 11) are *not* suspended inside **unsafe** blocks. Only raw pointer operations, direct ADT field access, and the bounds-check requirement are relaxed. An implementation shall continue to enforce the ownership and linearity diagnostics ([E001]–[E008] and [E016]) inside **unsafe** blocks.



## Chapter 5

# Lexical conventions

- 1 A Lain source file is translated in two conceptual steps: (1) the source text is decomposed into tokens as described in this clause; (2) the token stream is parsed into an abstract syntax tree according to the grammar. Implementation may combine these steps.

### 5.1 Source file encoding

- 1 A Lain source file shall be a sequence of bytes encoded in UTF-8 (Unicode Standard, Annex D). The byte order mark (U+FEFF) shall not appear in a source file. If it does, behavior is *implementation-defined*<sup>1</sup>.
- 2 Multi-byte UTF-8 sequences are permitted inside string literals and comments but shall not appear in identifiers. Identifiers shall consist solely of ASCII characters (see §5.6).
- 3 Line endings are normalized: a carriage return (U+000D) followed immediately by a line feed (U+000A) is treated as a single line feed. A lone carriage return is treated as a line feed.

### 5.2 Comments

- 1 A *line comment* begins with `//` and extends to the first following line feed character. The line feed is not part of the comment.
- 2 A *block comment* begins with `/*` and ends with the nearest following `*/`. Block comments may be nested: each opening `/*` inside a block comment opens a new nesting level requiring its own closing `*/`.
- 3 Comments are stripped during lexical analysis and have no effect on the semantics of the enclosing program.

#### Constraint

An unterminated block comment (one for which no matching `*/` is found before the end of the source file) is *ill-formed* ([E100]).

### 5.3 Tokens

- 1 The result of lexical analysis is a sequence of tokens. Whitespace and comments serve solely to separate tokens. The grammar production for *token* is given in Annex A (§5.3).
- 2 Where the token stream is ambiguous, the lexer shall apply the *maximal munch* rule: at each position, the longest sequence of characters that forms a valid token is consumed.

#### Example

```
1 // SYNOPSIS: tokens, not a program
```

---

<sup>1</sup>Implementation-defined behavior: whether a leading BOM is silently skipped or causes a diagnostic

```

2  ..=      // one token: range-inclusive, NOT '.' followed by '.*'
3  >=      // one token: greater-than-or-equal, NOT '>' followed by
      '=>'

```

- 3 Keywords (§5.5) take priority over identifiers: the character sequence **func** at a point where an identifier is expected is always the keyword **func**.

## 5.4 Whitespace

- 1 The whitespace characters are: space (U+0020) and horizontal tab (U+0009). They serve only to separate adjacent tokens.
- 2 Line feeds (U+000A), after normalization (§5.1), are significant: they act as statement terminators in contexts where a statement may follow (see §9). Multiple consecutive line feeds are treated as a single statement boundary.
- 3 A semicolon (;) may be used as a statement separator, equivalent to a line feed. Its purpose is to place more than one statement or declaration on a single line; a newline is the normal terminator, and no semicolon is needed at the end of a line. Two statements on one line without a separator are ill-formed ([E100]).

### Rationale

Newlines as the primary terminator eliminate the class of bugs caused by *missing* terminators. A semicolon is available only when the programmer deliberately wants several statements on one line (e.g. `a = 1; b = 2`), making a multi-statement line an explicit, opt-in choice rather than something that can happen by accident.

## 5.5 Keywords

- 1 The following identifiers are reserved as keywords. They shall not be used as user-defined identifiers. The grammar production is *keyword* (Annex A).

| Keyword           | Purpose                          | Clause |
|-------------------|----------------------------------|--------|
| <b>and</b>        | Logical AND operator             | §8     |
| <b>as</b>         | Type cast                        | §8     |
| <b>break</b>      | Loop break                       | §9     |
| <b>c_include</b>  | Foreign header include           | §17    |
| <b>case</b>       | Pattern matching                 | §15    |
| <b>comptime</b>   | Compile-time evaluation          | §14    |
| <b>continue</b>   | Loop continue                    | §9     |
| <b>decreasing</b> | Bounded-loop termination measure | §9     |
| <b>defer</b>      | Deferred execution               | §9     |
| <b>else</b>       | Alternative branch               | §9     |
| <b>extern</b>     | Foreign function declaration     | §17    |
| <b>false</b>      | Boolean literal                  | §5.7   |
| <b>for</b>        | Range-based iteration            | §9     |
| <b>func</b>       | Pure function declaration        | §12    |
| <b>if</b>         | Conditional                      | §9     |
| <b>import</b>     | Module import                    | §16    |

(continued)

| Keyword        | Purpose                               | Clause |
|----------------|---------------------------------------|--------|
| <b>in</b>      | Range iteration / bounds guard        | § 8    |
| <b>mov</b>     | Ownership transfer                    | § 11   |
| <b>or</b>      | Logical OR operator                   | § 8    |
| <b>proc</b>    | <i>reserved</i> ; removed, see [E100] | § 10.5 |
| <b>effects</b> | Effect row                            | § 12.1 |
| <b>return</b>  | Return statement                      | § 9    |
| <b>true</b>    | Boolean literal                       | § 5.7  |
| <b>type</b>    | Type / struct / enum declaration      | § 10   |
| <b>unsafe</b>  | Unsafe block                          | § 18   |
| <b>use</b>     | Local-scope import / alias            | § 16   |
| <b>var</b>     | Mutable binding / mutable borrow      | § 10   |
| <b>while</b>   | Loop statement                        | § 9    |

**Constraint**

An identifier that is identical to a keyword shall not be used in any context that requires an identifier. Such use is *ill-formed* ([E100]).

**5.6 Identifiers**

- 1 An identifier consists of ASCII letters, decimal digits, and underscores, and shall begin with a letter or underscore. The grammar production is *identifier* (Annex A).
- 2 Identifiers are case-sensitive: `foo`, `Foo`, and `F00` are three distinct identifiers.
- 3 A conforming implementation shall support identifiers of at least 255 characters. The behavior when an identifier exceeds this length is *implementation-defined*<sup>2</sup>.

**Constraint**

An identifier shall not be identical to a keyword (§ 5.5).

**Constraint**

An identifier shall begin with a letter or underscore, not a digit. A sequence of characters that begins with a digit is an integer literal (§ 5.7), not an identifier.

**5.7 Literals****5.7.1 Integer literals**

- 1 An integer literal denotes a constant integer value. Four notations are supported (grammar: *integer-literal*):

| Notation | Prefix | Example       |
|----------|--------|---------------|
| Decimal  | (none) | 42, 1_000_000 |

<sup>2</sup>Implementation-defined behavior: whether oversized identifiers are truncated or cause a diagnostic

| Notation    | Prefix | Example       |
|-------------|--------|---------------|
| Hexadecimal | 0x     | 0xFF, 0x1A_3B |
| Binary      | 0b     | 0b1010_0101   |
| Octal       | 0o     | 0o755         |

- 2 An underscore (`_`) may appear between any two digits of an integer or floating-point literal as a visual separator. It has no effect on the value: `1_000.5` is `1000.5`.
- 3 A hexadecimal, octal or binary literal denotes a *bit pattern* of at most 64 bits, read as a two's-complement `i64`: `0x7FFF_FFFF_FFFF_FFFF` is the largest `i64` and `0x8000_0000_0000_0000` the smallest. A pattern with the top bit set is therefore negative, and a `u64` context does not accept it implicitly; `as%` takes the same bits as the unsigned value, e.g. `0xcbf29ce48422325 as% u64`. A decimal literal denotes its value, which shall fit in `i64`.

#### Constraint

A hexadecimal, octal or binary literal of more than 64 bits, or a decimal literal outside the range of `i64`, is *ill-formed* (`[E086]`).

- 4 The type of an integer literal is inferred from context (see §7). In the absence of type context, the default type is `int`.

#### Constraint

An integer literal whose value does not fit in the type determined by context is *ill-formed* (`[E086]`) — it is diagnosed by the same range mechanism that rejects arithmetic overflow (§13.10).

#### Example

```

1 var a = 255           // int, value 255
2 var b u8 = 255        // u8, value 255 -- OK
3 var c u8 = 256        // [E086]: 256 does not fit in u8
4 var d = 0xFF          // int, value 255
5 var e = 0b1111_0000   // int, value 240
6 var f = 1_000_000     // int, value 1000000

```

### 5.7.2 Floating-point literals

- 5 A floating-point literal consists of a decimal integer part, a decimal point, and a fractional part. An optional exponent part `eN` or `EN` may follow (grammar: *float-literal*).
- 6 The default type of a floating-point literal is `f64`.

#### Example

```

1 var x = 3.14          // f64
2 var y f32 = 1.0       // f32
3 var z = 2.5e3          // f64, value 2500.0

```

### 5.7.3 Boolean literals

- 7 The boolean literals are `true` and `false` (grammar: *bool-literal*). Both have type `bool`.

### 5.7.4 Character literals

- 8 A character literal is a single byte value enclosed in single quotes. It has type `u8`. Escape sequences (§5.7.6) are permitted.

#### Example

```
1 var a = 'A'    // u8, value 65
2 var n = '\n'   // u8, value 10
3 var z = '\0'   // u8, value 0
```

#### Constraint

A character literal shall contain exactly one character or one escape sequence. An empty or multi-character character literal is *ill-formed* ([E100]).

### 5.7.5 String literals

- 9 A string literal is a sequence of zero or more characters enclosed in double quotes. Escape sequences (§5.7.6) are permitted (grammar: *string-literal*).
- 10 The type of a string literal of  $N$  visible characters is `Fixed_u8_N+1` (an array of  $N+1$  bytes, null-terminated). This type is implicitly coercible to `u8[]` (a byte slice).
- 11 The `.len` property of a string literal value equals  $N$  (excluding the null terminator).

#### Example

```
1 var s = "hello" // type: Fixed_u8_6 (5 chars + null)
2           // s.len == 5
```

- 12 String literals are stored in the read-only data segment of the translation output. Modifying a string literal through a pointer is **undefined behavior**<sup>3</sup>.

### 5.7.6 Escape sequences

- 13 The following escape sequences are recognized in string and character literals:

| Escape          | Byte value | Description         |
|-----------------|------------|---------------------|
| <code>\n</code> | 0x0A       | Line feed (newline) |
| <code>\t</code> | 0x09       | Horizontal tab      |
| <code>\r</code> | 0x0D       | Carriage return     |
| <code>\0</code> | 0x00       | Null byte           |
| <code>\\</code> | 0x5C       | Backslash           |
| <code>\"</code> | 0x22       | Double quote        |
| <code>\'</code> | 0x27       | Single quote        |

#### Constraint

A backslash not followed by one of the characters in the table above is an unrecognized escape sequence and is *ill-formed* ([E100]).

<sup>3</sup>Undefined behavior: the result of writing through a pointer derived from a string literal

## 5.8 Operators and punctuators

- <sup>1</sup> The complete set of operator and punctuator tokens is listed below. The grammar productions *operator* and *punctuator* are in Annex A.

### 5.8.1 Arithmetic operators

| Token                 | Meaning                                                  |
|-----------------------|----------------------------------------------------------|
| <code>+</code>        | Addition; unary identity                                 |
| <code>-</code>        | Subtraction; unary negation                              |
| <code>*</code>        | Multiplication                                           |
| <code>/</code>        | Division                                                 |
| <code>%</code>        | Modulo (remainder)                                       |
| <code>++ -- **</code> | Checked add/sub/mul ( <a href="#">panic</a> on overflow) |
| <code>+% -% **</code> | Wrapping add/sub/mul (modular)                           |
| <code>+  -  **</code> | Saturating add/sub/mul (clamp)                           |

- <sup>2</sup> The plain `+`, `-`, `*` auto-widen their result so they cannot overflow (§7.2.5); the `++/%|` variants keep the operand type and select an explicit overflow policy (§7.2.6).

### 5.8.2 Comparison operators

| Token              | Meaning               |
|--------------------|-----------------------|
| <code>==</code>    | Equal                 |
| <code>!=</code>    | Not equal             |
| <code>&lt;</code>  | Less than             |
| <code>&gt;</code>  | Greater than          |
| <code>&lt;=</code> | Less than or equal    |
| <code>&gt;=</code> | Greater than or equal |

### 5.8.3 Logical operators

| Token            | Meaning                     |
|------------------|-----------------------------|
| <code>and</code> | Logical AND (short-circuit) |
| <code>or</code>  | Logical OR (short-circuit)  |
| <code>!</code>   | Logical NOT (prefix)        |

### 5.8.4 Bitwise operators

| Token              | Meaning     |
|--------------------|-------------|
| <code>&amp;</code> | Bitwise AND |

| Token | Meaning     |
|-------|-------------|
|       | Bitwise OR  |
| ^     | Bitwise XOR |
| «     | Left shift  |
| »     | Right shift |

### 5.8.5 Assignment operators

| Token | Meaning                |
|-------|------------------------|
| =     | Assignment             |
| +=    | Add and assign         |
| -=    | Subtract and assign    |
| *=    | Multiply and assign    |
| /=    | Divide and assign      |
| %=    | Modulo and assign      |
| &=    | Bitwise AND and assign |
| =     | Bitwise OR and assign  |
| ^=    | Bitwise XOR and assign |
| «=    | Left shift and assign  |
| »=    | Right shift and assign |

### 5.8.6 Other operators and punctuators

| Token | Meaning                              |
|-------|--------------------------------------|
| .     | Member access; module path separator |
| ..    | Exclusive range (half-open interval) |
| ->    | Function return type separator       |
| (     | Open parenthesis                     |
| )     | Close parenthesis                    |
| {     | Open brace                           |
| }     | Close brace                          |
| [     | Open bracket                         |
| ]     | Close bracket                        |
| ,     | List separator                       |
| :     | Type annotation; match arm separator |

- <sup>3</sup> Attributes use bracket syntax `[name]` or `[name(args)]` before a declaration (§10). The brackets `[` and `]` listed above serve both as array/slice subscripts and as attribute delimiters; the context (declaration position vs. expression position) determines the role.

## 5.9 Token disambiguation

- 1 Several token sequences have context-dependent meaning that is resolved by the parser, not the lexer. The lexer produces the same token regardless of context:

| Symbol             | Context 1               | Context 2                                                  |
|--------------------|-------------------------|------------------------------------------------------------|
| <code>*</code>     | Multiplication (binary) | Pointer dereference (unary prefix, only in <b>unsafe</b> ) |
| <code>-</code>     | Subtraction (binary)    | Negation (unary prefix)                                    |
| <code>.</code>     | Member access           | Module path separator                                      |
| <code>&amp;</code> | Borrowed match prefix   | Bitwise AND                                                |

## 5.10 Integer arithmetic and overflow

- 1 Integer arithmetic in Lain uses two's complement representation for all signed integer types.
- 2 A program in which an integer operation can overflow is *ill-formed*. The plain operators `+`, `-` and `*` carry a proof obligation: the result shall be shown to lie within its type. Where the implementation cannot show it, the program is *ill-formed* ([E086]). Overflow is therefore a diagnosable constraint violation, not a runtime behaviour.
- 3 `+`, `-` and `*` *widen*: the result of an operation on two operands of width  $N < 64$  has a type wide enough to represent every result, so the operation itself cannot overflow. The obligation attaches instead to the *narrowing* — the point at which that result is stored into, passed as, or returned as a narrower type. At width 64 no wider type exists and the obligation attaches to the operation.
- 4 Overflow that is intended shall be written as such. Each form is a distinct operator, and each is well-formed by construction:
  - `+%`, `-%`, `*%` — *modular*: the result wraps modulo  $2^N$ .
  - `+=`, `-=`, `*=` — *saturating*: the result is clamped to the bounds of its type.
  - `++?`, `--?`, `*?*`, `as?` — *checked*: the operation shall be handled by an **else** arm supplying a fallback value, propagating with **return**, or **panicing** ([E067]).
- 5 Within an **unsafe** block the obligation is waived for every operation lexically inside it, including the narrowing. Responsibility passes to the program.
- 6 A conforming implementation that emits C99 shall compile the translation output with `-fwrapv` or equivalent. This is a requirement on the *translation output*, not a licence for the source language: it ensures that the wrapping operators have their defined meaning and that no operation the implementation has accepted becomes undefined behaviour downstream.

### Note

Unsigned arithmetic in ISO C is modular and signed overflow is undefined. Lain admits neither as a source-language behaviour for the plain operators: a program that would rely on either is rejected, and a program that wants one says which. See §13.10.

## Chapter 6

# Basic concepts

### 6.1 Declarations and definitions

- 1 A *declaration* introduces a name into a scope and associates it with an entity: a variable, function, procedure, type, or module. Every name shall be declared before it is used, with the exception of names that are resolved by Universal Function Call Syntax (§12).
- 2 A *definition* is a declaration that also allocates storage (for variables) or provides a body (for functions and procedures). In Lain, every declaration at the local level is also a definition.

#### Constraint

Redeclaring a name already visible in the current or an enclosing scope — a duplicate parameter name, or a local that repeats or shadows an outer binding — is *ill-formed* ([E013]). Lain forbids shadowing (§6.2).

#### Constraint

Use of an undeclared identifier is *ill-formed* ([E106]). The reference implementation issues this diagnostic when the translation output is generated — after name resolution, monomorphization, and UFCS desugaring are complete — so that a name still unbound at that point is unambiguously undeclared.

#### Example

```
1 func f() {  
2     var x = 1  
3     var x = 2    // [E013]: redeclaration of 'x' is forbidden  
4 }
```

### 6.2 Scopes

- 1 A *scope* is a region of program text within which a declared name can be used. Lain has the following scope levels:

**Module scope** Names declared at the top level of a source file are in module scope. Module-scope names are accessible throughout the module and, by default, accessible from other modules that import this module.

**Function/procedure scope** Parameter names are in the scope of the function or procedure body.

**Block scope** Names declared inside a block (`{ }`) are in the scope of that block. They are not accessible outside the enclosing block.

- 2 Scopes are *lexically nested*. Lain *forbids shadowing* entirely: a declaration whose name is already visible in the current scope or in any enclosing scope — a function parameter shadowed by a local, or a name redeclared in a nested block — is *ill-formed* ([E013]).

#### Rationale

Disallowing shadowing removes a class of bugs in which an inner binding silently captures a name the reader expects to denote an outer one. Every identifier in a function body therefore resolves to exactly one declaration.

- 3 A conforming implementation shall support at least 64 levels of nested block scope (§ 4.4).

### 6.3 Name resolution

- 1 Name resolution maps an identifier in source text to its declaration. The lookup algorithm is:
  1. Search the innermost block scope.
  2. If not found, search progressively wider enclosing block scopes.
  3. If not found, search the function/procedure scope (parameters).
  4. If not found, search the module scope.
  5. If not found, search imported module scopes (in import order).
  6. If not found, attempt UFCS resolution (§ 12).
  7. If still not found, the identifier is undeclared and the program is *ill-formed* ([E106]).
- 2 Because shadowing is forbidden (§ 6.2), a name resolves to at most one declaration along the scope chain: a lookup that would find a second binding of the same name is rejected at the point of the redeclaration ([E013]), not silently resolved by an innermost-wins rule.

### 6.4 Storage duration

- 1 Every object has a *storage duration* that determines the lifetime of its storage.

**Automatic storage duration** Variables declared inside a block or function body. Storage is allocated on entry to the block and deallocated on exit.

**Static storage duration** Variables declared at module scope. Storage persists for the lifetime of the program. *implementation-defined*<sup>1</sup>.

**Arena storage duration** Objects allocated from an explicit **Arena**. Storage persists until the arena is freed as a whole.

---

<sup>1</sup>Implementation-defined behavior: whether module-scope variables are zero- initialized before `main` is called

## 6.5 Objects and values

- 1 An *object* is a region of storage interpreted according to a type. Every declared variable corresponds to an object.
- 2 Every variable has:
  - a *declared type* (§ 7);
  - an *ownership mode*: `owned` (default for linear types), `shared` (read-only reference), or `var` (mutable reference) (§ 11);
  - a *linear state*: `Uninit`, `Unconsumed`, or `Consumed` (for linear types only) (§ 11).

## 6.6 Lvalues and rvalues

- 1 An *lvalue* is an expression that designates an object with a persistent storage location: a variable identifier, a field access on an lvalue, or an array index expression on an lvalue.
- 2 An *rvalue* is an expression whose value has no persistent storage location: literals, the result of arithmetic operations, or the result of a function call.
- 3 Assignment requires the left-hand side to be an lvalue.

### Constraint

An assignment whose left-hand side is an rvalue is *ill-formed* ([E100]).

## 6.7 Alignment

- 1 Every type has a required alignment. The alignment of a type is *implementation-defined*<sup>2</sup>. Struct layout follows C99 struct layout rules, including padding (§ 7).
- 2 Stack-allocated variables are aligned to their type's required alignment. The implementation shall not insert additional padding between local variables unless required by the hardware.

<sup>2</sup>Implementation-defined behavior: the same as the alignment of the corresponding C99 type



## Chapter 7

# Types

- 1 Lain is statically typed. Every variable, expression, and parameter has a type determined at compile time. There is no runtime type information (RTTI). The grammar productions for types are in Annex A (*type*).

### 7.1 Type taxonomy

- 1 The types of Lain are organized as follows:
  - *Primitive types*: integer, floating-point, boolean, void (§ 7.2);
  - *Array types*: fixed-size, compile-time-sized (§ 7.3);
  - *Slice types*: dynamically-sized views into contiguous memory, including sized slice types with static length constraints (§ 7.4);
  - *Vector types*: a fixed number of lanes, operated on lane by lane (§ 7.5);
  - *Struct types*: named product types (§ 7.6);
  - *Enum types*: simple enumerations (§ 7.7);
  - *ADT types*: sum types with associated data (§ 7.8);
  - *Pointer types*: raw memory addresses, unsafe only (§ 7.9);
  - *Constrained types*: types with value range annotations (§ 7.10).
- 2 Types are classified as *linear* or *unrestricted*. Linear types must be consumed exactly once (see § 11). Unrestricted types may be used any number of times.
- 3 All primitive types are unrestricted. A struct type is linear if and only if at least one of its fields is of a linear type. This property is transitive.

### 7.2 Primitive types

#### 7.2.1 Integer types

- 1 The integer types are listed below. Sizes marked (*platform*) are *implementation-defined*<sup>1</sup>.

| Type            | Signedness | Bit width | Range          | C99 type             |
|-----------------|------------|-----------|----------------|----------------------|
| <code>u8</code> | unsigned   | 8         | $[0, 2^8 - 1]$ | <code>uint8_t</code> |

---

<sup>1</sup>Implementation-defined behavior: determined by the target platform as in the C99 standard

(continued)

| Type               | Signedness | Bit width  | Range                     | C99 type               |
|--------------------|------------|------------|---------------------------|------------------------|
| <code>u16</code>   | unsigned   | 16         | $[0, 2^{16} - 1]$         | <code>uint16_t</code>  |
| <code>u32</code>   | unsigned   | 32         | $[0, 2^{32} - 1]$         | <code>uint32_t</code>  |
| <code>u64</code>   | unsigned   | 64         | $[0, 2^{64} - 1]$         | <code>uint64_t</code>  |
| <code>usize</code> | unsigned   | (platform) | $[0, 2^N - 1]$            | <code>size_t</code>    |
| <code>i8</code>    | signed     | 8          | $[-2^7, 2^7 - 1]$         | <code>int8_t</code>    |
| <code>i16</code>   | signed     | 16         | $[-2^{15}, 2^{15} - 1]$   | <code>int16_t</code>   |
| <code>i32</code>   | signed     | 32         | $[-2^{31}, 2^{31} - 1]$   | <code>int32_t</code>   |
| <code>i64</code>   | signed     | 64         | $[-2^{63}, 2^{63} - 1]$   | <code>int64_t</code>   |
| <code>isize</code> | signed     | (platform) | $[-2^{N-1}, 2^{N-1} - 1]$ | <code>ptrdiff_t</code> |
| <code>int</code>   | signed     | 32         | $[-2^{31}, 2^{31} - 1]$   | <code>int32_t</code>   |

- 2 `int` is a documented alias of `i32`, identical in every semantic respect (boundary checks, niche optimization, codegen). Use either spelling — `int` when bit-width is incidental, `i32` when explicit width matters.
- 3 Generalized integer types `iN/uN` are admitted for any  $N \in [1, 64]$ . Hardware-aligned widths ( $N \in \{1, 8, 16, 32, 64\}$ ) map to standard C99 `stdint` types. Logical widths (other  $N$ ) occupy the smallest hardware-aligned container that fits  $N$  bits when standalone; in a `[packed]` struct they occupy exactly  $N$  bits.
- 4 `usize` and `isize` are the platform-native unsigned and signed integer types whose width equals the pointer width. Their exact width is *implementation-defined*<sup>2</sup>.

## 7.2.2 Hardware-aligned vs logical integer types

- 5 The `iN/uN` types fall into two semantic categories:
- Hardware-aligned** ( $N \in \{1, 2, 4, 8, 16, 32, 64\}$ , with the common subset being  $\{8, 16, 32, 64\}$ ): these correspond to fundamental C99 `stdint` widths (`uint8_t`, `int32_t`, etc.). They are the canonical choice for C interop and for any context where exact storage size matters at the byte boundary.
- Logical** (other  $N$ ): these are *bit-width quantities*. Their semantics are context-dependent:
- **Standalone**: storage uses the smallest hardware-aligned container fitting  $N$  bits (e.g., `u12` maps to `uint16_t`). The natural range is  $[-2^{N-1}, 2^{N-1} - 1]$  signed or  $[0, 2^N - 1]$  unsigned.
  - **Inside a packed struct** (future feature, §7.2.4): the type occupies exactly  $N$  bits of storage with bit-exact layout.
- 6 **Distinction from refinement.** Logical `iN/uN` types differ from a refinement constraint on `int`:
- `int >= 0 and <= 7` declares an `int` with a range constraint. Storage is the `int` default (`int32_t`); bit-width is not specified.
  - `u3` declares a 3-bit quantity. Standalone its range is  $[0, 7]$  and its storage is `uint8_t`; in a `[packed]` struct (§7.2.4) it occupies exactly 3 bits.

The two forms are interchangeable when only the range matters; they differ when bit-exact storage layout matters.

<sup>2</sup>Implementation-defined behavior: the pointer width on the target platform

**Example**

```

1 // A refinement constraint on int, named: range [0, 4095],
  container = int32_t.
2 // (A constraint is written on a parameter, a field or an alias
  like this one,
3 // not on a local's declaration.)
4 type Twelve = int >= 0 and <= 4095
5 type Pressure = int >= 0 and <= 1000
6
7 func main() int {
8     var x u12 = 4095           // logical bit-width: a 12-bit
                                quantity, container = uint16_t
9     var y Twelve = 4095       // the same range as an int
                                refinement
10    var p Pressure = 500
11    return 0
12 }
```

**7.2.3 Refinement type aliases**

- 7 A type alias may carry a refinement constraint that narrows the underlying type's value range:

```

1 type Percent = int >= 0 and <= 100
2 type Pressure = int >= 0 and <= 1000
3 type NonZero = int != 0
```

- 8 Variables declared with a refinement type alias inherit its constraint. Assignments whose source value range violates the alias constraint are *ill-formed* ([E086]).

**Example**

```

1 type Percent = int >= 0 and <= 100
2
3 func main() int {
4     var ok Percent = 75           // accepted: 75 fits [0,
                                    100]
5     // var bad Percent = 150      // refused, E086: 150 > 100
6     return 0
7 }
```

**7.2.4 Packed structs**

- 9 The **[packed]** attribute (§10.1) on a struct declaration enables bit-exact memory layout. Each **iN/uN** field occupies exactly *N* bits within a single scalar container.

```

1 [packed]
2 type GpioModer {
3     pin0 u2      // bit 0-1
4     pin1 u2      // bit 2-3
5     pin2 u2      // bit 4-5
6     pin3 u2      // bit 6-7
7 }
8 // total: 8 bits -> uint8_t container, bit-exact layout
```

- 10 **Layout rules:**

- Fields are laid out in declaration order, starting at bit 0.

- Each field occupies exactly its declared bit width.
- Total bit width must be  $\leq 64$ ; container is the smallest of `uint8_t`, `uint16_t`, `uint32_t`, or `uint64_t` that fits the total.

#### 11 Access semantics:

- Construction: `GpioMder(1, 2, 3, 0)` packs the four 2-bit fields into the container.
- Read: `r.pin1` extracts bits 2-3 via shift-and-mask; an `iN` field is sign-extended.
- Write: `r.pin1 = v` updates bits 2-3, leaving other fields unchanged.
- The whole struct is the container integer: it is copied, passed and returned as one.

#### Constraint

Inside a **[packed]** struct, fields shall be `iN/uN` types only. Nested structs, pointers, and slices are not permitted. Violation is *ill-formed* ([E121]).

#### Constraint

The total bit width of fields in a **[packed]** struct shall not exceed 64. Otherwise *ill-formed* ([E121]).

#### Constraint

A field of a **[packed]** struct has no address. A `var` reference to one (`f(var r.pin1)`, `var x = var r.pin1`) or its address (`&r.pin1`) is *ill-formed* ([E121]), inside `unsafe` as well: this is a property of the representation, not a safety policy. A `var` reference to the whole struct is well-formed.

#### Note

This feature targets embedded use cases: hardware registers, packed network protocols, codec implementations. The reference implementation emits the container as a scalar typedef and every access inline: shift-and-mask reads, read-modify-write stores, and construction by OR-ing the shifted fields. The layout can be asserted where the type is declared: `assert @sizeof(GpioMder)== 1` (§10.7).

### 7.2.5 Integer rank and widening

12 The integer types are partially ordered by *rank*:

| Rank | Types                      | Bit width |
|------|----------------------------|-----------|
| 1    | <code>u8, i8</code>        | 8         |
| 2    | <code>u16, i16</code>      | 16        |
| 3    | <code>u32, i32, int</code> | 32        |
| 4    | <code>u64, i64</code>      | 64        |
| 5    | <code>usize, isize</code>  | platform  |

13 **Implicit widening:** a value of a lower-rank integer type may be used where a higher-rank integer type is expected, without an explicit cast.

- 14 **Result type of arithmetic (auto-widening).** The result type of a plain `+`, `-`, or `*` is *wide enough to hold every representable result*, so the operation itself cannot overflow: `+/-` widens by one bit (with a sign bit when an unsigned operand feeds a signed result) and `*` sums the operand widths — e.g. `u8 + u8` yields `u9`, `u8 - u8` yields `i10`, `i32 + i32` yields `i33`, and `i32 * i32` yields `i64`. When the required width would exceed 64 bits (the `i128` ceiling), the result type is the higher-rank operand and the operation is subject to the overflow diagnostic below. A *signed* `/` widens by one bit — `i32 / i32` yields `i33` — because exactly one quotient does not fit its operands' type, `TYPE_MIN/-1`; an unsigned `/` and every `%` keep the higher-rank operand type (a remainder is smaller in magnitude than its divisor). The other binary operators (`&`, `|`, `^`, `<<`, `>>`) yield the higher-rank operand type. Overflow therefore becomes possible only at a *narrowing* (§7.2.6).

### Example

```

1 var a u8 = 42
2 var b int = a           // u8 -> int: implicit widening, OK
3 var c = a + a           // u8 + u8 -> u9 (holds 0..510; cannot
    overflow)
4 var d = c as u8         // narrowing back to u8 is checked (see
    below)

```

### Constraint

An implicit conversion from an integer type to one that cannot represent every value of the source — a narrowing, or a signedness change — is *ill-formed* unless value-range analysis proves it fits (§13). Such a conversion otherwise requires an explicit cast tier (`as`, `as?`, `as%`, or `as|`; §8.8).

## 7.2.6 Integer overflow

- 15 Because `+`, `-`, and `*` auto-widen (§7.2.5), overflow cannot occur at the operation. It can occur only when a value flows into a type that cannot hold it — a *narrowing* at an assignment, return, argument, or explicit cast. A narrowing whose value range is not proven to fit the target is *ill-formed* ([E086]); the programmer resolves it by proving the range, widening the target, or choosing an explicit overflow policy:

| Policy                        | Form                                                                                                         |
|-------------------------------|--------------------------------------------------------------------------------------------------------------|
| proven (default)              | <code>x as T</code> (narrows only if proven, else [E086])                                                    |
| checked ( <code>else</code> ) | <code>x as? T</code> , <code>a +? b</code> , <code>a -? b</code> , <code>a *? b</code>                       |
| wrapping (modular)            | <code>x as% T</code> , <code>a +% b</code> , <code>a -% b</code> , <code>a *% b</code> , <code>a /% b</code> |
| saturating (clamp)            | <code>x as  T</code> , <code>a +  b</code> , <code>a -  b</code> , <code>a *  b</code> , <code>a /  b</code> |

- 16 A checked form performs a run-time test and, on overflow, evaluates its `else` arm — a fallback value, a `return`, or `panic(...)`, which terminates the program; a checked form without `else` is *ill-formed* ([E067]). A wrapping form uses two's-complement modular arithmetic. A saturating form clamps to the target type's minimum or maximum. Every form is defined at every integer width, odd widths included: a wrapping result is reduced modulo  $2^N$  for the type's own  $N$ , not for the width of the machine type that holds it. The checked and wrapping/saturating *operators* keep the operand type (they do not auto-widen). For division the only overflowing case is `TYPE_MIN/-1`: `/%` yields `TYPE_MIN` and `/|` yields `TYPE_MAX`; every other quotient is the ordinary one.

**Constraint**

Where no wider type exists — a signed / or % on 64-bit operands — the operation itself must be proven not to divide `TYPE_MIN` by `-1`: the dividend's range excludes `TYPE_MIN` or the divisor's excludes `-1`. Otherwise the program is *ill-formed* ([E086]). There is no implicit run-time check; /% and /| state what the programmer means.

**Constraint**

An implicit conversion between integer types and floating-point types is *ill-formed*. Such conversions require an explicit `as` cast.

**Constraint**

An implicit conversion between integer types and `bool` is *ill-formed*. Such conversions require an explicit `as` cast.

**7.2.7 Floating-point types**

- 17 `f32` and `f64` are 32-bit and 64-bit floating-point types conforming to IEC 60559 (IEEE 754). They correspond to C's `float` and `double` respectively. `float` is a documented alias of `f32`.  
 18 The default type of a floating-point literal is `f64` (§5.7.2).  
 19 Arithmetic on `f32` and `f64` follows IEEE 754 semantics: overflow produces  $\pm\infty$ ; the result of `0.0/0.0` and similar operations is NaN. These are defined behaviors in Lain.

**7.2.8 Boolean type**

- 20 `bool` has exactly two values: `true` and `false`. Every boolean the language makes has this type: the literals, the comparisons, `in`, and `and`, `or` and `!` (§8.5); none of them is an integer. `bool` is stored as a single byte; its exact representation is *implementation-defined*<sup>3</sup>.

**Constraint**

There are no implicit **value** conversions between `bool` and any integer type for general arithmetic or assignment.

- 21 The condition expression of `if` and `while` accepts either a value of type `bool` or any integer type. When an integer value is used in condition position, zero is treated as `false` and any non-zero value is treated as `true`. This relaxation is ergonomic only and does not extend to other contexts.

**Example**

```
1 var n int = read_count()
2 if n { ... }           // OK: integer in condition, non-zero is
   true
3 var b bool = n         // [E012]: not a value conversion; write n
   != 0
```

**7.2.9 The void type**

- 22 `void` represents the absence of a value. It is used exclusively in pointer types (`*void`) for C interoperability (see §17).

<sup>3</sup>Implementation-defined behavior: whether `true` is 1 or any non-zero value

**Constraint**

A variable declaration with type `void` is *ill-formed*.

**7.3 Array types**

- 1 An array type `T[N]` denotes a contiguous sequence of  $N$  values of type `T`.  $N$  shall be a compile-time constant integer greater than zero (grammar: *array-type*).
- 2 The `.len` property of an array type yields  $N$  as a compile-time constant of type `int`.

**Note**

**Runtime-length local arrays (VLAs).** A *local* declaration `var a T[n]` whose length `n` is a runtime value and which has no initializer is an owned, runtime-sized stack array (emitted as a C variable-length array `T a[n]`), with `a.len == n` and native bounds-checked indexing. It is distinct from a sized-slice *view* of the same spelling: a parameter `f(a T[n])`, or an *initialized* local `var s = arr[i..j]`, is a borrowed `{len, data}` slice, not owned storage. Indexing a VLA under a proven guard (`i < n`) needs no `unsafe`; an index VRA cannot bound is `[E085]` as for any array.

- 3 Array elements are laid out contiguously in memory in increasing index order, with element 0 at the base address. The stride between elements is *implementation-defined*<sup>4</sup>.

**Constraint**

An array size  $N \leq 0$  is *ill-formed* (`[E100]`).

**Constraint**

Every array index expression shall satisfy  $0 \leq idx < arr.len$ . An access for which value range analysis cannot prove this is an out-of-bounds violation (`[E085]`).

**Example**

```
1 var arr int[5] = [1, 2, 3, 4, 5]
2 var x = arr[2]           // OK: 2 is statically in [0, 4]
3 // var y = arr[5]       // [E085]: 5 >= arr.len
```

**7.4 Slice types**

- 1 A slice type `T[]` is a reference to a contiguous sequence of values of type `T` whose length is determined at runtime. Its runtime representation is a *fat* pair (`len: usize, data: *T`) — a length followed by a data pointer.

**Constraint**

**The `*` rule.** A `*` prefix denotes a *raw pointer*, and only that. Arrays and slices are always passed by reference, so they never carry a `*`: a slice is `T[]`, a sized slice `T[n]`, a fixed array `T[N]`, a sentinel string `T[:S]`. A `*` is permitted only before a non-array, non-slice type — `*T`, `**T`, `*void`, and function-pointer types (§17). A `*` before any bracket form is *ill-formed* (`[E100]`).

- 2 The reference/pointer type family and its C mapping (parameter position):

<sup>4</sup>Implementation-defined behavior: equal to `sizeof(T)` in C99, including any intra-element padding

| Lain               | What it is         | C parameter                         | Cost                        |
|--------------------|--------------------|-------------------------------------|-----------------------------|
| <code>u8</code>    | scalar             | <code>uint8_t</code> (by value)     | 1 word                      |
| <code>T[N]</code>  | inline fixed array | <code>const Fixed_T_N*</code>       | $N \times \text{sizeof}(T)$ |
| <code>T[]</code>   | fat slice          | <code>size_t __len, const T*</code> | 2 words                     |
| <code>T[n]</code>  | sized slice        | <code>size_t, const T*</code>       | 2 words                     |
| <code>T[:S]</code> | sentinel string    | <code>struct { T* data; }</code>    | 1 word                      |
| <code>*T</code>    | raw pointer        | <code>const T*</code>               | 1 word                      |
| <code>*void</code> | opaque C pointer   | <code>const void*</code>            | 1 word                      |

**Note**

A `*` in the emitted C has *two* independent sources: (1) a `*` written in the Lain type, which is a raw pointer; and (2) the implicit by-reference passing of an *aggregate* (array, slice, struct) parameter — a hidden `const T*` that is part of the calling convention, not the type. The same aggregate is a plain value as a local (`var p Point` emits `Point p`) but a pointer as a parameter (`const Point*`). This is why an aggregate type carries no `*` even though it is passed as a pointer: the `*` names raw pointers, not the borrow.

- 3 A slice has two properties accessible via dot notation:
  - `.len`: the number of elements, type `usize`;
  - `.data`: a pointer to the first element, type `*T`.
- 4 A string literal of  $N$  characters has static type `+1Fixed_u8_N` (a null-terminated fixed-size array of  $N+1$  bytes) and is implicitly coercible to `u8[]`.

**7.4.1 Sized slice types**

- 5 A *sized slice type* is a slice type annotated with a size constraint on its length. The constraint is part of the parameter or variable declaration, not of the nominal type itself. The syntax is:

| Syntax                  | Meaning                                       |
|-------------------------|-----------------------------------------------|
| <code>T[]</code>        | unsized dynamic slice (no length constraint)  |
| <code>T[n]</code>       | length exactly $n$ ; $n$ is a size expression |
| <code>T[&gt;= n]</code> | length $\geq n$ ; $n$ is a size expression    |
| <code>T[&gt; n]</code>  | length $> n$ ; $n$ is a size expression       |

- 6 A *size expression* (`size_expr`) may be any of:
  - an integer literal (e.g., `3`, `0`);
  - an identifier that names a numeric variable in scope (e.g., `n`);
  - a member expression of the form `p.len`, where `p` is a slice or array parameter in scope (e.g., `a.len`, `src.len`);
  - a binary arithmetic expression over the above (e.g., `a.len + b.len`, `src.len - 1`).

**Example**

```

1 // Exact-length constraint: out must have length == a.len +
  b.len
2 // (the bounds keep a.len + b.len from overflowing a usize)
3 func concat(a i32[<= 1000000], b i32[<= 1000000], out i32[a.len
  + b.len]) {
4     for j in 0..a.len { out[j] = a[j] }
5     for j in 0..b.len { out[j + a.len] = b[j] }
6 }
7
8 // Lower-bound constraint: arr must have at least one element
9 func first(arr i32[> 0]) i32 { return arr[0] }
10
11 // Derived constraint using member .len (src is not empty, so
   src.len - 1 cannot underflow)
12 func slide(src i32[>= 1], out i32[src.len - 1]) {
13     for i in 0..out.len { out[i] = src[i + 1] -% src[i] }
14 }

```

**7 Constraint semantics.**

- $T[\text{expr}]$ : the value range analysis (VRA, §13) records the constraint  $\text{len} == \text{expr}$  and uses it to prove subsequent bounds checks inside the function body.
- $T[\geq \text{expr}]$ : VRA records  $\text{len} \geq \text{expr}$ .
- $T[> \text{expr}]$ : VRA records  $\text{len} > \text{expr}$ , i.e.  $\text{len} \geq \text{expr} + 1$ .

8 **Call-site verification.** At every call site, the compiler verifies that the argument satisfies the formal parameter's size constraint ([E087]); see §13.12.

9 **C ABI decomposition (Fase 7).** When a function parameter has a sized or unsized dynamic slice type, the compiler decomposes it into a pair of C parameters:

- a length parameter `size_t`;
- a data pointer `T* restrict`.

Whether the callee may write through the pointer is stated by the GCC attribute `access(read_only, p, n)` or `access(read_write, p, n)`, where  $p$  and  $n$  are the positions of the pointer and its length. A slice whose length is an expression over other parameters' lengths (`out i32[a.len + b.len]`) is passed the same way: its length is an ordinary parameter, checked against the expression at every call (§13.12).

**Example**

```

1 // Source:
2 func vadd(var out i32[], a i32[out.len], b i32[out.len]) {
3     for i in 0..out.len { out[i] = a[i] +% b[i] }
4 }
5
6 // C emission (the name is prefixed with its module's):
7 // __attribute__((access(read_write, 2, 1)))
   __attribute__((access(read_only, 4, 3)))
8 // __attribute__((access(read_only, 6, 5)))
9 // void vadd(size_t, int32_t* restrict, size_t, int32_t*
   restrict,
10 //             size_t, int32_t* restrict);

```

## 7.5 Vector types

- 1 `Vec(N, T)` is a value of  $N$  lanes of type  $T$ , held and passed whole.  $T \times N$  is the same type: `u8x16` is `Vec(16, u8)`, `i32x4` is `Vec(4, i32)`. A vector may be a local, a parameter, a return value or a struct field.

### Constraint

$N$  shall be a power of two from 1 to 64, and  $T$  an 8, 16, 32 or 64-bit integer type, or `f32` or `f64` ([E100]).

- 2 A vector is built from an array literal of exactly  $N$  elements where a vector is expected: a declaration, an argument, a field, a return, or an operand beside a vector (`a - [1, 1, 1, 1]`). `@splat(V, x)` is the vector of type  $V$  with  $x$  in every lane.
- 3 The arithmetic, bitwise and shift operators and the comparisons apply lane by lane, and a scalar operand applies to every lane. Integer lanes *wrap*: a lane's result is taken modulo  $2^w$  for a  $w$ -bit lane, including unary `-`, so vector arithmetic carries no overflow obligation. Two obligations remain, as on scalars:
  - a division or remainder's divisor shall be provably non-zero ([E015]); a scalar divisor is judged like a scalar's, while a vector divisor cannot be proven, because the analysis holds no per-lane facts;
  - a shift's amount shall be a scalar provably in  $[0, w - 1]$  ([E086]); an amount given per lane is refused for the same reason.
- 4 A comparison yields a *mask* of the operands' vector type: every bit of a lane is set where the comparison holds and clear where it does not (255 and 0 for `u8` lanes).
- 5 `v[k]` is lane  $k$  of  $v$ , read or written in place.  $k$  shall be provably less than  $N$  ([E085]).
- 6 `@load(V, b, i)` reads the  $N$  elements of  $b$  starting at element  $i$  into a vector of type  $V$ , and `@store(b, i, v)` writes  $v$ 's lanes there. For an array or slice  $b$  the obligation is  $i + N \leq b.len$  ([E085]). A raw pointer has no length, so a load or store through one is allowed only inside `unsafe`.
- 7 `@movemask(v)` is the `u32` whose bit  $k$  is the top bit of lane  $k$ : of a mask, the lanes where the comparison held.

### Constraint

The operand of `@movemask` shall be a vector of at most 32 lanes of 8 bits ([E100]).

- 8 `@shuffle(t, idx)` is a table lookup: lane  $k$  of the result is `t[idx[k]]` when `idx[k] < N`, and 0 otherwise. It is total: an index out of range reads 0, never another lane and never an undefined value, so it states no obligation. This is the behaviour of ARM NEON's `tbl` and of Rust's `swizzle_dyn`; on x86 with SSSE3 it is `pshufb`. The classifier idiom `@shuffle(tbl, (x >> 4) & 15)` indexes a 16-entry table by each byte's high nibble.

### Constraint

$t$  shall be a vector of integer lanes and  $idx$  a vector of as many lanes of the unsigned integer type of the same width: `u8x16` for a `u8x16` or `i8x16` table, `u32x4` for a `u32x4` table ([E100]).

- 9 The language names no instruction set. A vector is translated to a C `vector_size` type; the C compiler's target flags choose the instructions, and the same program builds for x86-64 and AArch64. `@movemask` uses the SSE2 or AVX2 intrinsic where the target defines it and a portable loop elsewhere; `@shuffle` uses `__builtin_shuffle` under GCC and a lane loop elsewhere.

## 7.6 Struct types

- 1 A struct type groups named fields into a single named product type. The grammar production is *struct-declaration* (Annex A).
- 2 Field order is the declaration order. The memory layout of a struct is *implementation-defined*<sup>5</sup>.
- 3 A struct type  $S$  is linear if at least one field of  $S$  has a linear type (transitively).

### Constraint

**Q-008.** Every field of a struct or enum variant whose declared type is linear shall be annotated `mov`. Omitting `mov` on a linear field is *ill-formed* ([E083]). This requirement makes linearity locally visible in the declaration: a reader of a struct definition does not need to inspect the field's type definition to know whether the struct is linear.

### Example

```

1  type Handle {
2      mov ptr *int           // explicit 'mov' on linear field
3  }
4
5  type Session {
6      mov h Handle           // OK: Handle is linear, mov
7      name u8[]              // unrestricted field, no mov needed
8  }
9
10 // Ill-formed: missing mov on a linear field
11 type Bad {
12     h Handle                // [E083]
13 }
```

- 4 **Struct construction:** all fields shall be provided in declaration order. The type of each argument shall be compatible with the declared field type (§7.11).

### Constraint

A struct constructor expression with a number of arguments different from the number of fields is *ill-formed* ([E100]).

### Constraint

Writing to a field of a struct variable requires the variable to be declared as `var` (mutable). Writing to an immutable binding's field is *ill-formed*.

### Constraint

Using `==` or `!=` on a struct type is *ill-formed*. Field-by-field comparison must be performed explicitly.

### 7.6.1 Field invariants

- 5 A field declaration may state a fact about the field's value for the whole life of every value of the struct type. Three forms are provided:

<sup>5</sup>Implementation-defined behavior: equal to the C99 struct layout, including compiler-inserted padding for alignment

- `pos usize in text` — an *index* invariant: `pos` is a valid index into the slice or array field `text` (`pos < text.len`);
- `pos usize <= src.len` — a *position* invariant: `pos` compared with the length of the slice or array field `src`, by `<`, `<=`, `>` or `>=`;
- `len usize <= cap` — a relation between two integer fields, by the same four comparisons;
- `src u8[<= 4294967295]` — a *length bound* on a slice field: its length compared with an integer literal, written inside the brackets as for a parameter (`a i32[>= 2]`), by `<`, `<=`, `>`, `>=` or `==`.

A refinement against a literal (`pct i32 <= 100`) is a plain field refinement (§7.10) and is not a field invariant.

- 6 A length bound is what a *narrow* position needs: `pos u32 <= src.len` does not make `pos + 1` fit a `u32` when `src.len` is a `usize`, since a longer source admits `pos = 232 - 1`; bounding `src` to at most `232 - 1` bytes does. A slice field's length changes only when the field (or the whole struct) is written — a `var` reference cannot rebind a slice ([E009]) and an element write keeps the length — so those writes are where it is checked.
- 7 An index invariant cannot hold for an empty container. A position invariant can: `pos usize <= src.len` admits `pos == 0` over an empty source and `pos == src.len` at its end, which is the state a scanner is in at end of input.

### Constraint

A field invariant shall hold wherever a value of the struct type is built, and after every write to either field it relates. A construction or a field write at which it is not proven is *ill-formed* ([E121]).

### Constraint

A `var` reference to a field that a field invariant relates — a `var` argument (`f(var s.pos)`) or a reference binding (`var r = var s.pos`) — is *ill-formed* ([E121]): a write through it would not be checked against the invariant. Pass the whole struct as `var` instead; writes to its fields are checked.

### Constraint

A field refinement naming another field in any other shape — another comparison, a field of another kind, an expression — is *ill-formed* ([E132]). So is a slice field whose length is bounded by anything but a literal (other than the struct's own generic parameter, `type Buf(N usize){ data u8[N] }`), and a refinement written after a slice type (`src u8[ <= 4096]`), which bounds nothing a slice has.

- 8 Because every write that could break a field invariant is checked, the invariant is a fact at every read of the field: `c.text[c.pos]` needs no bounds check under `pos usize in text`, and `b.cap - b.len` needs no underflow check under `len usize <= cap`.

### Example

```

1 type Lexer {
2     src u8[]
3     pos usize <= src.len          // position: 0 <= pos <=
        src.len
4 }
5 func lexer_new(text u8[:0]) Lexer { return Lexer(text, 0) }
        // OK even for ""
6 func advance(l var Lexer) {
```

```

7   if l.pos < l.src.len { l.pos = l.pos + 1 }
   // keeps the invariant
8 }

1  type Lexer {
2     src u8[]
3     pos usize <= src.len
4 }
5  func bad(l var Lexer) { l.pos = l.pos + 1 } // ERROR: E121
   (and E086), unguarded

```

## 7.7 Enum types

- 1 An enum type is a type with a finite, named set of values. Each variant carries no associated data. The grammar production is *enum-declaration* (Annex A).
- 2 The underlying representation of an enum type is *implementation-defined*<sup>6</sup>.
- 3 Enum values are accessed as `TypeName.VariantName`.

### Constraint

Using `==` or `!=` to compare enum values is *ill-formed*. Enum values shall be compared using `case` pattern matching (§15).

### Note

The constraint on `==` for enums reflects that the underlying tag representation is implementation-defined. Pattern matching is always sound.

## 7.8 Algebraic data types (ADTs)

- 1 An ADT is a sum type in which each variant may carry associated fields. A type declaration is classified as an ADT if at least one variant has fields. The grammar production is *enum-declaration* (Annex A, using the record-variant form).
- 2 ADTs use a niche-based representation whenever a zero-cost encoding is available; otherwise they fall back to a tagged union (a tag field of implementation-defined integer type plus a union of variant payload structs).

### Note

**D-Niche aggressive multi-slot.** When the variant payload admits a *niche*—a bit pattern that cannot occur in any valid value of the payload type—that niche is used to encode tag-less variants. Niche sources, in priority order: zero-page pointer slots for pointers/slices; VRA-proven unused values for constrained integers; bit patterns outside the declared variants for `bool` or small enums; niche of any recursive field. See §19.9 for the full algorithm.

If no niche or only a partial niche can be derived, the compiler emits **[W120]** (best-effort warning, never an error) and falls back to a tag+union representation. The programmer may resolve the warning by constraining the payload, by changing to a payload type with larger sentinel space, or by accepting the tag byte. See §19.9 for the full layout algorithm and target dependencies.

<sup>6</sup>Implementation-defined behavior: an integer type large enough to hold all variant tags; the reference implementation uses `int`

- 3 ADT values are constructed as `TypeName.VariantName(args)` for variants with fields, or `TypeName.VariantName` for unit variants.
- 4 ADT values are destructured via `case` pattern matching (§15).

## 7.9 Pointer types

- 1 A pointer type `*T` or `const *T` is a raw memory address. Pointer operations require an `unsafe` block (§18).
- 2 The ownership modes for pointer types are:

| Lain syntax         | Mode               | C99 type              | Linear? |
|---------------------|--------------------|-----------------------|---------|
| <code>*T</code>     | shared (read-only) | <code>const T*</code> | no      |
| <code>var *T</code> | mutable            | <code>T*</code>       | no      |
| <code>mov *T</code> | owned              | <code>T*</code>       | yes     |

### Constraint

Dereferencing a pointer or taking the address of a variable shall occur only inside an `unsafe` block. Pointer dereference in safe code is *ill-formed* ([E060]).

## 7.10 Constrained types

- 1 A constrained type is a type annotated with a value range constraint:

$$T \text{ op } bound$$

where  $op \in \{>=, <=, >, <, ==\}$  and  $bound$  is an integer literal.

- 2 Constrained types are used in function return types and parameter types to declare value range invariants that the value range analysis (§13) shall enforce.

### Example

```

1 func nonzero_divisor() int >= 1 {
2     return 42    // VRA verifies: 42 >= 1 holds
3 }

```

- 3 The constraint is part of the function signature, not the type system's nominal identity. Two constrained types with the same base type are assignment-compatible.

## 7.11 Type compatibility and equivalence

- 1 Lain uses **nominal type equivalence**: two types are the same if and only if they have the same declared name.
- 2 Two types  $T_1$  and  $T_2$  are *compatible* for an expression context if:
  - $T_1 = T_2$  (same nominal type), or
  - $T_1$  is an integer type of lower rank than  $T_2$  (implicit widening; §7.2.5).

**Constraint**

An expression of type  $T_1$  used in a context requiring type  $T_2$ , where  $T_1$  and  $T_2$  are not compatible, is *ill-formed*.

3 The operators `==` and `!=` are defined for:

- primitive numeric types;
- `bool`;
- pointer types.

They are not defined for struct types, array types, or enum types.



## Chapter 8

# Expressions

- 1 An expression is a syntactic construct that evaluates to a value. Every expression has a type, determined statically at compile time. The grammar production for *expression* is in Annex A.
- 2 Expressions are evaluated strictly left-to-right: for any binary operator *a op b* or function call *f(a, b)*, the left operand (or earlier argument) is fully evaluated before the right operand (or later argument).

### 8.1 Operator precedence and associativity

- 1 The following table lists all binary operators in order of decreasing precedence. All operators listed on the same row have equal precedence.

| Prec.    | Operators                   | Assoc. | Result type         |
|----------|-----------------------------|--------|---------------------|
| 1 (high) | () [] .                     | left   | (context-dependent) |
| 2        | unary - ! <b>mov</b>        | right  | (operand type)      |
| 3        | <b>as</b>                   | left   | target type         |
| 4        | * / % and their %/ /? forms | left   | § 7.2.5             |
| 5        | + - and their %/ /? forms   | left   | § 7.2.5             |
| 6        | << >>                       | left   | left operand type   |
| 7        | &                           | left   | wider operand type  |
| 8        | ^                           | left   | wider operand type  |
| 9        |                             | left   | wider operand type  |
| 10       | < > <= >= <b>in</b>         | left   | <b>bool</b>         |
| 11       | == !=                       | left   | <b>bool</b>         |
| 12       | <b>and</b>                  | left   | <b>bool</b>         |
| 13 (low) | <b>or</b>                   | left   | <b>bool</b>         |

- 2 Parentheses override precedence.

### 8.2 Primary expressions

- 1 A primary expression is one of: an identifier, a literal, a parenthesized expression, a block expression, an **if** expression, a **case** expression, a constructor expression, or a **comptime** expression.
- 2 **Identifier expression:** evaluates to the current value of the named variable.

### Constraint

The identifier shall be declared in scope; use of an undeclared identifier is *ill-formed* ([E106], § 6.3).

### Constraint

The identifier shall have been initialized before use ([E005]).

## 8.3 Arithmetic operators

- 1 The binary arithmetic operators are + (addition), - (subtraction), \* (multiplication), / (division), and % (remainder). Their result types are given in § 7.2.5: +, -, \* and a signed / widen so the operation cannot overflow; the forms with an explicit policy (+%, /!, ..., § 7.2.6) keep the operand type.
- 2 Integer division truncates toward zero. f32/f64 division follows IEEE 754.

### Constraint

Integer division or remainder where the divisor is not provably non-zero (by value range analysis, a != 0 refinement, or a dominating guard; § 13) is *ill-formed* ([E015]).

### Constraint

% shall not be applied to floating-point operands.

- 3 The unary - operator negates its operand. For unsigned integer types the result wraps modulo  $2^N$ .

## 8.4 Comparison operators

- 1 The comparison operators ==, !=, <, >, <=, >= compare two values and yield bool.

### Constraint

Operands of a comparison operator shall have compatible types (§ 7.11).

### Constraint

== and != shall not be applied to struct types, array types, or enum types (§ 7.7).

## 8.5 Logical operators

- 1 and and or are short-circuit operators: the right operand is not evaluated if the result is determined by the left operand alone. ! is prefix unary negation. All yield bool.

### Constraint

Operands of and, or, and ! shall have type bool.

## 8.6 Bitwise operators

- 1 The bitwise operators &, |, ^, <<, >> operate on integer operands. The result type is the wider operand type for &, |, ^; for shift operators, the result type is the left operand type.

**Constraint**

Operands of bitwise operators shall be integer types.

**Constraint**

The shift amount (right operand of `<<` and `>>`) shall satisfy  $0 \leq \text{shift} < N$  where  $N$  is the bit width of the left operand. A shift amount that value range analysis cannot prove to be in range is *ill-formed* ([E086]) — the boundary mechanism (§13.10) that also rejects arithmetic overflow.

## 8.7 The `in` operator

- 1 The expression `idx in arr` evaluates to `true` if and only if  $0 \leq \text{idx} < \text{arr.len}$ . The operand `arr` shall be an array or slice type; `idx` shall be an integer type.
- 2 When `idx in arr` appears as the condition of a `while` or `if` statement (or as the leftmost operand of an `and`-chain condition), it establishes an **in guard**: inside the body of that statement, the implementation shall accept `arr[idx]` without requiring further bounds verification.
- 3 In-guards propagate through `and`-chains: if the left operand of `and` establishes an in-guard, that guard is active when evaluating the right operand.

**Example**

```

1 // In-guard: arr[i] is safe without further proof
2 func sum(arr i32[]) i32 {
3     var s i32 = 0
4     var i usize = 0
5     while i in arr {
6         s = s +% arr[i]    // OK: in-guarded
7         i = i + 1
8     }
9     return s
10 }
11
12 // And-chain propagation:
13 type Lexer {
14     src u8[]
15     pos usize
16 }
17 func skip_to_quote(l var Lexer) {
18     while l.pos in l.src and (l.src[l.pos] as int) != '"' {
19         l.pos = l.pos + 1
20     }
21 }

```

**Constraint**

An in-guard of the form `idx in arr` covers only `arr[idx]` exactly. Accesses `arr[idx + k]` for any  $k \neq 0$  are not covered by this guard and require separate verification or an `unsafe` block.

## 8.8 Cast expression (`as`)

- 1 The expression `expr as T` explicitly converts the value of `expr` to type `T`. An integer cast whose target cannot hold every value of the source — a narrowing, or a signedness change such as `u32` to `i32` or `usize` to `i64` — carries an *overflow policy* chosen by the cast tier (§7.2.6):

| Tier       | Form       | On a value outside T                                   |
|------------|------------|--------------------------------------------------------|
| proven     | <b>as</b>  | narrows only if VRA proves it fits, else [E086]        |
| checked    | <b>as?</b> | run-time bounds check; the <b>else</b> arm on overflow |
| wrapping   | <b>as%</b> | two's-complement modular truncation                    |
| saturating | <b>as </b> | clamp to the minimum or maximum of T                   |

- 2 The following casts are permitted:

| From    | To      | Semantics                                                    |
|---------|---------|--------------------------------------------------------------|
| integer | integer | The same value where T holds it; otherwise the tier's policy |
| integer | float   | Nearest representable value (IEEE 754)                       |
| float   | integer | Truncation toward zero                                       |
| float   | float   | Precision change                                             |
| pointer | pointer | Reinterpretation ( <b>unsafe</b> required)                   |
| integer | pointer | Reinterpretation ( <b>unsafe</b> required)                   |
| pointer | integer | Reinterpretation ( <b>unsafe</b> required)                   |

### Constraint

Casts between pointer types, between a pointer and an integer, or between any type and a pointer shall occur only inside an **unsafe** block. Such a cast outside **unsafe** is *ill-formed* ([E012]).

### Constraint

A cast between non-numeric, non-pointer types (e.g., struct to integer) is *ill-formed*.

## 8.9 Move expression (**mov**)

- 1 **mov** *expr* transfers ownership of the value produced by *expr*. The source is placed in the Consumed state; any subsequent use of the source variable is *ill-formed* ([E001]). See §11.

## 8.10 Borrows at call sites

- 1 Lain has *no* prefix borrow operator (no **&**, no **ref**, no **mut** expression form). A value is borrowed simply by passing it to a function or procedure; the borrow *mode* is written at the call site, matching the parameter's declared mode: nothing for a shared (read-only) borrow, **var** for a mutable borrow, and **mov** to transfer ownership (§8.9). See the call-site mode table in §8.14 and the ownership rules in §11.

## 8.11 Field access

- 1 The expression *expr*.*field* accesses the named field of a struct or ADT value, or a property of an array or slice (*.len*, *.data*).

### Constraint

The field name shall exist on the type of *expr*.

**Constraint**

Writing to a field (`expr.field = v`) requires `expr` to be a mutable lvalue.

**8.12 Index expression**

- 1 The expression `arr[idx]` accesses the element of array or slice `arr` at integer index `idx`.

**Constraint**

The value range analysis shall prove  $0 \leq idx < arr.len$  at every index expression. An index expression for which this cannot be proved is *ill-formed* ([E085]). An `in` guard (§ 8.7) or an `unsafe` block suppresses this requirement.

**8.13 Range index expression (sub-slicing)**

- 1 The expression `arr[a..b]` produces a *sub-slice*: a slice covering elements at indices  $a, a + 1, \dots, b - 1$  of `arr`. The range is **half-open**: the element at index  $b$  is excluded. The length of the result is  $b - a$ .
- 2 The inferred type of `arr[a..b]` is  $T[b-a]$ , a sized slice type (§ 7.4.1) whose length constraint records the exact length of the result.

**Constraint**

The start index `a` shall satisfy  $a \geq 0$ . A start index for which VRA cannot prove non-negativity is *ill-formed* ([E085]).

**Constraint**

The end index `b` shall satisfy  $b \leq arr.len$ . An end index for which VRA cannot prove this is *ill-formed* ([E085]).

**Constraint**

When both `a` and `b` are integer literals, the compiler additionally verifies  $a \leq b$ . If  $a > b$ , the program is *ill-formed* ([E087]).

**Example**

```

1 var xs i32[5] = [10, 20, 30, 40, 50]
2 var s = xs[1..4]           // type i32[3]; covers indices 1, 2, 3
3 var e = xs[3..3]           // OK: empty sub-slice, length 0
4 // var bad = xs[5..2] // [E087]: 5 > 2 (literal ordering
   //                   // violated)

```

**8.14 Function call**

- 1 A function call expression `f(args)` evaluates the arguments and transfers control to the callee. Arguments are evaluated left-to-right.
- 2 Argument ownership modes shall be given explicitly at the call site:

| Syntax                | Meaning                                                          |
|-----------------------|------------------------------------------------------------------|
| <code>f(x)</code>     | shared borrow (default; parameter declared with no mode keyword) |
| <code>f(var x)</code> | mutable borrow (parameter declared <code>var</code> )            |
| <code>f(mov x)</code> | ownership transfer (parameter declared <code>mov</code> )        |

**Constraint**

The mode annotation at the call site shall match the declared parameter mode. A mismatch is *ill-formed*.

**Constraint**

The number of arguments shall equal the number of parameters (unless the callee is a variadic C function declared with `...`).

- 3 **Two-phase borrows:** during argument evaluation, borrows of the receiver of a method call are placed in the RESERVED phase and promoted to ACTIVE after all arguments are evaluated (see §11).

## 8.15 Block expression

- 1 A block expression `{ stmts... expr }` evaluates each statement in order, then evaluates the final expression, whose value is the value of the block. If the final expression is absent, the block has no value.

## 8.16 If expression

- 1 An **if** expression of the form `if cond { e1 } else { e2 }` evaluates *cond*; if true, evaluates *e1*; otherwise, evaluates *e2*.

**Constraint**

The types of *e1* and *e2* shall be compatible. If they are not, the expression is *ill-formed*.

**Constraint**

*cond* shall have type `bool`.

## 8.17 Match expression (case)

- 1 A **case** expression (*match-expression*) selects one of several alternatives based on pattern matching. All arms shall produce values of compatible types; the type of the **case** expression is that common type. See §15 for full exhaustiveness rules.

## 8.18 Constructor expression

- 1 A constructor expression builds a value of a struct or ADT type by supplying all field values. See §7.6 and §7.8 for constructor rules.

## 8.19 Compound assignment

- 1 Compound assignment `x op= e` is syntactic sugar for `x = x op e`. The left-hand side shall be a mutable lvalue. All compound forms of the arithmetic and bitwise operators are supported (see §5.8).

## Chapter 9

# Statements

- 1 A statement is a unit of execution. Statements are terminated by a line feed or a semicolon (§5.4); the semicolon is used to place several statements on one line. The grammar production for *statement* is in Annex A.

### 9.1 Block statement

- 1 A block statement `{ stmts... }` creates a new scope (§6.2) and executes each contained statement in order. Names declared inside a block are not accessible outside it.

### 9.2 Immutable declaration

- 1 `name [Type] = expr` declares and initializes an immutable binding. The type may be omitted (inferred from the initializer). The binding cannot be reassigned after initialization.

```
1 type Point {  
2     x i32  
3     y i32  
4 }  
5  
6 func main() i32 {  
7     x i32 = 42           // explicit type  
8     y = 10               // type inferred (i32)  
9     p Point = Point(1, 2)  
10    return 0  
11 }
```

#### Constraint

An immutable declaration requires an initializer expression: an immutable binding must have a definite value at its point of declaration.

#### Constraint

When a type annotation is present, the type of the initializer shall be compatible with the declared type (§7.11).

#### Note

When the type is omitted, the distinction between declaration and reassignment is resolved by scope: `x = 5` where `x` is not yet declared creates a new immutable binding; where `x` is already declared as `var`, it is a reassignment.

### 9.3 Mutable variable declaration

- 1 **var** name Type = expr declares a mutable variable. Mutable variables can be reassigned. The type may be omitted if it can be inferred from the initializer.

```

1 var counter i32 = 0
2 counter = counter + 1           // OK: var is mutable
3
4 var buf u8[256]                // OK: type present; unassigned
   until written
5                               // (reading before a write is [E005])

```

#### Constraint

A mutable declaration shall include either an explicit type annotation or an initializer expression. A declaration with neither is *ill-formed* ([E100]).

#### Constraint

When both a type annotation and an initializer are present, the type of the initializer shall be compatible with the declared type (§7.11).

### 9.4 Assignment statement

- 1 An assignment statement lvalue = expr stores the value of *expr* into the storage designated by *lvalue*. An assignment target without a type annotation is always a reassignment, never a declaration.

#### Constraint

The left-hand side shall be a mutable lvalue (§6.6). Assigning to an immutable binding is *ill-formed* ([E009]).

#### Constraint

Overwriting a linear variable whose current value has not been consumed would leak that value. Such an assignment is *ill-formed* ([E003]).

#### Note

Assigning to a linear variable after its previous value has been consumed (e.g., loop reassign-and-consume pattern) is permitted. The implementation resets the variable's linear state to Unconsumed after a legal overwrite.

#### Example

```

1 import std.fs.{open_file, close_file}
2
3 func main() int effects io, raises, alloc {
4     var f = open_file("data.txt", "r")
5     close_file(mov f)           // f is now Consumed
6     f = open_file("data.txt", "r") // OK: overwriting Consumed
       state
7     close_file(mov f)
8     return 0
9 }

```

## 9.5 Expression statement

- 1 An expression followed by a statement terminator is an expression statement. The value of the expression is discarded.

## 9.6 If statement

- 1 **if** *cond* { *body* } [**else if** *cond* { *body* }]... [**else** { *body* }] evaluates *cond*; if true, executes the then-body; otherwise, tests the **else if** clauses in order; if none matches, executes the **else** body (if present).

### Constraint

*cond* shall have type `bool`.

- 2 **Range refinement**: inside the then-branch of **if**  $x < N$  { }, the value range analysis (§13) knows that  $x \in (-\infty, N - 1]$ ; inside the else-branch,  $x \in [N, +\infty)$ .
- 3 **In-guard**: when *cond* contains or is an **in** expression (§8.7), the in-guard is active inside the then-branch only.
- 4 **Branch consistency**: if one branch consumes a linear variable, all branches shall consume it. If branches disagree on the linear state of a variable from the enclosing scope, the program is *ill-formed* ([E003]).

## 9.7 While statement

- 1 **while** *cond* { *body* } repeatedly evaluates *cond* and, if true, executes *body*.

### Constraint

A **while** loop inside a **func** shall be bounded: it shall carry a **decreasing** measure, or the implementation shall be able to infer one from its condition (§9.7.2). A loop for which no measure is present and none can be inferred is *ill-formed* ([E011]).

### Constraint

*cond* shall have type `bool`.

- 2 **In-guard**: when *cond* contains or is an **in** expression, the in-guard is active inside the loop body for each iteration.

### 9.7.1 Bounded while statement

- 3 **while** *cond* **decreasing**  $M$  { *body* } is a bounded loop. In addition to the while semantics above, the implementation shall statically verify:

1. **Bounded below** ([E082]): the implementation shall prove that  $M$  cannot decrease without limit.
2. **Analyzable measure** ([E082]):  $M$  shall be in a form the implementation can analyse. A measure it cannot analyse at all — including a constant, or one over variables the body never assigns — is rejected, and so is one that is merely not provably well-founded: both are [E082], because one termination obligation with one source reports them. (Two earlier codes, E080 and E081, separated these sub-cases when a second termination checker existed in the front end; they are withdrawn, see Annex B.)

3. **Strict decrease** ([E082]): the implementation shall prove that every execution of *body* strictly decreases the value of  $M$ , *except* along paths that leave the loop via **break** or **return**. A branch that unconditionally exits the loop never iterates again, so it is exempt from the decrease requirement; only paths that loop back need decrease  $M$ . (A **continue** re-evaluates *cond* and thus loops back, so it is *not* an exit.)
- 4 A bounded **while** loop is permitted inside a **func** because the implementation has statically verified termination.
- 5 The following patterns are accepted by the standard verification algorithm:

| Condition                      | Measure                    | Why                                           |
|--------------------------------|----------------------------|-----------------------------------------------|
| $i < n$                        | $n - i$                    | $i$ increases $\Rightarrow$ measure decreases |
| $i \leq n$                     | $n - i$                    | $i$ increases toward the inclusive bound      |
| $n > 0$                        | $n$                        | $n$ decreases directly                        |
| $i > k$                        | $i - k$                    | $i$ decreases toward $k$                      |
| $a < b$                        | $b - a$                    | either $a$ increases or $b$ decreases         |
| $i$ <b>in</b> <code>arr</code> | <code>arr.len</code> - $i$ | <b>in</b> implies $i < \text{arr.len}$        |

- 6 Beyond the syntactic patterns above, the standard verification algorithm also accepts a measure variable that is set to a constant provably below the loop's guaranteed lower bound (e.g.  $j = 0$  inside **while**  $j > 0$ ), and unsigned measures driven by  $v = X \% v$ ,  $v = v / D$ , or  $v = v \gg C$ .

### 9.7.2 Measure inference

- 7 The **decreasing** clause *may be omitted*. When it is, and the loop appears in a context that requires termination (a **func** body), the implementation shall attempt to *infer* a measure from *cond* using the correspondences above: a relational condition  $i < n$ ,  $i \leq n$ ,  $i > k$ , or  $i \geq k$  yields the candidate  $n - i$  (respectively  $i - k$ ); a two-variable bound  $lo < hi$  yields  $hi - lo$ . A pointer condition  $p$  **in** `arr` yields the pointer-offset measure.
- 8 An inferred measure is subject to the *identical* verification as an explicit one (non-negativity, analyzability, strict decrease). If no candidate can be inferred, or the candidate fails verification, the loop is *ill-formed* ([E011]): the programmer shall then supply an explicit **decreasing** clause or restructure the loop. Inference is therefore a convenience only; it never accepts a loop that an explicit measure could not.

#### Example

```

1 func count(n int) int {
2     var i = 0
3     while i < n {           // measure 'n - i' is inferred
4         i = i + 1
5     }
6     return i
7 }
```

#### Rationale

Bounded loops bridge the gap between pure functions (which must terminate) and the common pattern of finite-input state machines (lexers, parsers) that are provably terminating. For the overwhelmingly common counting and scanning loops the measure is mechanical,

so it is inferred; the **decreasing** keyword remains available to state a termination argument the compiler cannot infer, which it then verifies.

## 9.8 Return statement

- 1 **return** `[expr]` terminates execution of the current function or procedure and transfers the value of `expr` (if present) to the caller.

### Constraint

At the point of a **return**, all linear variables in scope shall have been consumed. A live (Unconsumed) linear variable at **return** is *ill-formed* ([E003]).

- 2 A **return** expression may be prefixed with **mov** or **var**: **return mov** `x` transfers ownership of `x` explicitly, and **return var** `x` returns a mutable borrow of `x` (the pattern used by a **func** whose return type is **var** `T`). Ownership transfer is also implicit from the function's declared return type, so a bare **return** `x` suffices when that type is owned.

### Constraint

A **return var** `x` (or **return mov** `x`) that would escape a borrow of a *local* variable — one whose storage does not outlive the call — is a dangling reference and is *ill-formed* ([E010]). Returning a **var** borrow tied to a **var** parameter is permitted.

## 9.9 Defer statement

- 1 **defer** `{ body }` schedules `body` for execution immediately before the enclosing scope exits, on all exit paths (**return**, **break**, **continue**, or fall-through). Multiple **defer** statements execute in LIFO order.
- 2 **Interaction with linear types**: if a linear variable is consumed inside a **defer** body, the variable is marked as defer-consumed. It is treated as unconsumed for the purposes of the main body (it may be used before the scope exit) and as consumed at scope exit.

### Constraint

A **defer** body shall not contain **return**, **break**, or **continue** statements that would transfer control out of the deferred body.

### Example

```
1 import std.fs.{open_file, close_file}
2
3 func process() int effects io, raises, alloc {
4     var f = open_file("data.txt", "r")
5     defer close_file(mov f)           // f consumed at scope exit
6     // ... use f ...
7     return 0
8 }
```

## 9.10 Unsafe block

- 1 **unsafe** `{ body }` is an unsafe block (see §18). It creates a scope in which certain static checks are relaxed.

## 9.11 Comptime if statement

- 1 `comptime if` `cond` { `then` } [`else` { `else` }] evaluates *cond* at compile time. Only the taken branch is type-checked and emitted; the other branch is discarded. *cond* shall be a compile-time constant expression (§ 14).

### Note

`comptime if` is the Lain equivalent of C preprocessor `#if`: it enables platform-specific or type-specific code selection without dead-code warnings.

## 9.12 For statement

- 1 `for` `i` `in` `start..end` { `body` } iterates with *i* ranging from *start* (inclusive) to *end* (exclusive). The loop variable *i* is immutable; its value range is statically known to be  $[start, end - 1]$  throughout the body, enabling safe array indexing without runtime bounds checks.
- 2 `for` loops are permitted whatever a function's row says: the range is the bound.

### Constraint

*start* and *end* shall have integer types.

## Chapter 10

# Declarations

- 1 This clause specifies top-level declarations. Local variable declarations are specified in §9.3. The grammar productions are in Annex A (*function-declaration*, *struct-declaration*, *enum-declaration*).

### 10.1 Declaration attributes

- 1 A declaration may be preceded by zero or more attributes. An attribute has the form **[name]** or **[name(args)]**, where *name* is a recognized attribute identifier from the closed set in Table 10.1, and *args* (if present) is a comma-separated list of expressions. Multiple attributes on a single declaration are written one per line.

```
1 [fast_math]
2 func dot(a f64[], b f64[a.len]) f64 {
3     var s f64 = 0.0
4     for i in 0..a.len { s = s + a[i] * b[i] }
5     return s
6 }
7
8 [private]
9 func internal_helper() int { return 0 }
10
11 [fast_math]
12 [private]
13 func hot_path() f64 { return 1.0 }
```

- 2 Recognized attribute names (closed set, version 1.1):

Table 10.1: Recognized attribute names

| Attribute          | Applies to      | Purpose                                |
|--------------------|-----------------|----------------------------------------|
| <b>[fast_math]</b> | any function    | Enable FMA fusion (P4 opt-out, local). |
| <b>[private]</b>   | any declaration | Module-private visibility.             |

#### Constraint

**Q-017.** An attribute name not listed in Table 10.1 is *ill-formed* ([E103]). An attribute that omits the identifier (e.g. **[ ]** or **[(arg)]**) is *ill-formed* ([E102]).

**Note**

Attributes are recognized at declaration position only. At expression position, `[` and `]` continue to delimit array and slice subscripts and array literals; the parser disambiguates by context.

## 10.2 Struct declarations

- 1 A struct declaration introduces a named product type with one or more named fields. Fields are listed in declaration order; order is significant for construction and layout.

```

1 type Point {
2     x i32
3     y i32
4 }
```

- 2 A field may be annotated with `mov`, making the field (and therefore the containing struct) a linear type:

```

1 type File {
2     mov handle *FILE
3 }
```

**Note**

A struct with zero fields (a *unit* struct, `type Empty {}`) is permitted: it is a zero-sized marker type, constructed as `Empty()`.

## 10.3 Enum and ADT declarations

- 1 An enum declaration introduces a type whose values are drawn from a fixed, named set of variants. If every variant has no associated fields, the type is a simple enum (§7.7). If at least one variant has associated fields, the type is an algebraic data type (ADT) (§7.8).

```

1 type Direction {                                // simple enum (one variant per
2     line)
3     North
4     South
5     East
6     West
7 }
8
9 type Shape {                                    // ADT
10    Circle    { radius i32 }
11    Rectangle { width i32, height i32 }
12    Point
13 }
```

## 10.4 Function declarations

- 1 A function declaration introduces a `func`: a pure, terminating, side-effect-free callable. Its grammar production is *function-declaration*.
- 2 The signature consists of: the keyword `func`, a name, an optional list of type parameters (§14), a parameter list, and a return type annotation.

### 10.4.1 Parameters

- 3 Each parameter has a name, a type, and an ownership mode. The ownership mode determines how the argument is passed:

| Syntax            | Mode               | ABI (struct T) |
|-------------------|--------------------|----------------|
| name T            | shared (read-only) | const T*       |
| <b>var</b> name T | mutable            | T*             |
| <b>mov</b> name T | owned (transfer)   | T (by value)   |

#### Note

For primitive types (integer, float, bool), the ABI passes the value directly regardless of ownership mode, as for primitive C arguments.

- 4 A parameter whose declared type is the built-in **type** — written name **type**, with no **comptime** prefix — is a compile-time type parameter; it is bound to a concrete type at every call site, by inference from a value argument or explicitly (§14).

### 10.4.2 Return type annotation

- 5 The return type is written after the parameter list. It may carry a value range constraint:

```
1 func abs(x int >= -2147483647) int >= 0 {
2     if x < 0 { return -x }
3     return x
4 }
```

- 6 The constraint's right-hand side is an integer literal, a parameter, a parameter's `.len`, or a sum or difference of those. The function proves it at every return; every caller then knows it of the result. A search can thereby promise an index into its argument:

```
1 func position(a u8[>= 1], c u8) usize < a.len {
2     var i usize = 0
3     while i < a.len {
4         if a[i] == c { return i }
5         i = i + 1
6     }
7     return 0
8 }
9
10 func at_first_space(a u8[>= 1]) u8 {
11     return a[position(a, 32)] // proven: the result is below
12                             a.len
13 }
```

#### Constraint

All return paths of a **func** shall produce a value whose type and value range satisfy the declared return type and constraint ([E086] where a return cannot be shown to). A right-hand side that names anything but a parameter, or the `.len` of anything but a slice parameter, is unsatisfiable.

### 10.4.3 Purity constraints

**Constraint**

A function's `effects` row is an upper bound on what its body does, and its default is  $\emptyset$ . An effect the body has and the row does not name is *ill-formed* ([E011] where the bound is  $\emptyset$ , [E130] where a non-empty row understates it). The three constraints below are consequences of that one rule, not separate rules.

**Constraint**

A function shall not contain an unbounded `while` loop — one whose measure is neither given by `decreasing` nor inferable — unless its row names `diverge` ([E011]/[E082]).

**Constraint**

A function shall not call one whose row names an effect its own row does not ([E011]). The callee's *introducer* is irrelevant: what is refused is the effect, not the keyword that declared it.

**Constraint**

A function shall not participate in mutual recursion unless its row names `diverge` ([E011]); termination cannot be established for a cycle no measure covers. Direct self-recursion *with* an inferred or given measure is permitted.

## 10.5 The entry point

- 1 **`proc` has been removed.** It was an introducer that granted effects instead of stating them, and every effect it was needed for is now named by an `effects` row on an ordinary `func` — including `diverge`, which admits an unbounded loop. The keyword remains *reserved*, and using it is [E100] with the replacement spelled out, so old source fails with an explanation rather than with "expected declaration". The same applies to the function-pointer type `*proc(...)`, replaced by `*func(...)` with a row (§12.1).
- 2 The entry point of every Lain program shall be declared as:

```
1 func main() int { return 0 }      // with 'effects io' if it
   performs any
```

**Constraint**

A complete *program* shall define exactly one `main` returning `int` as its entry point. An individual module (translation unit) compiled on its own — e.g. a library — need not define `main`: the reference compiler accepts a `main`-less source file, and a program missing an entry point is diagnosed when the translation outputs are linked.

## 10.6 Extern declarations

- 1 An extern declaration introduces a C function into the Lain name space without providing a body. See §17 for full semantics.

```
1 extern func printf(fmt *u8, ...) int effects io
2 extern func strlen(s *u8) usize effects
```

**Constraint**

Variadic parameters (...) shall only appear in **extern** declarations. User-defined variadic functions are *ill-formed*.

**10.7 Module-scope assertions**

- 1 A module-scope **assert** states a fact about the program's constants and the layout of its types, where they are declared:

```

1 type Token {
2     kind u8
3     len   u16
4     pos   u32
5 }
6 BUF usize = 64
7 assert @sizeof(Token) == 8
8 assert BUF % 16 == 0

```

- 2 Its operand is a *constant expression*: integer and character literals; named immutable module-level constants whose initialisers are constant expressions; **@sizeof**(T) and **@alignof**(T) (§ 19.8); the arithmetic, bitwise, comparison and logical operators; and **as**. Its value is computed over the integers, not in the width of any one type.

**Constraint**

The operand of a module-scope **assert** shall be a constant expression of type **bool**; otherwise the program is *ill-formed* ([E133]). A value that exists only at run time is asserted inside a function, where it is an obligation the value-range analysis must discharge (§ 13).

**Constraint**

A module-scope assertion whose value is **false** makes the program *ill-formed* ([E134]). When the operand involves **@sizeof** or **@alignof** its value is the target C compiler's, and the implementation shall have that compiler decide it: the reference implementation emits a C **\_Static\_assert** carrying [E134]. Otherwise the implementation decides it itself.

**10.8 Comptime declarations****Note**

Top-level **comptime** { ... } blocks — evaluated by the compiler before any runtime code is emitted, to define compile-time constants or trigger **compileError** assertions — are a *planned* feature not yet accepted by the reference implementation (a top-level **comptime** block is presently [E100]). The **comptime** keyword is nonetheless reserved, and **comptime if** (§ 9.11) is available.



## Chapter 11

# Ownership and linearity

- 1 This clause specifies Lain’s linear type system, ownership modes, borrow checker, and non-lexical lifetime rules. All rules in this clause are enforced at compile time with zero runtime overhead.

### 11.1 Ownership modes

- 1 Every parameter, return value, and struct field has one of three ownership modes:

| Mode    | Param syntax            | Call-site             | Semantics                                        |
|---------|-------------------------|-----------------------|--------------------------------------------------|
| Shared  | <code>name T</code>     | <code>f(x)</code>     | read-only; multiple concurrent borrows allowed   |
| Mutable | <code>var name T</code> | <code>f(var x)</code> | exclusive read-write; no other borrow concurrent |
| Owned   | <code>mov name T</code> | <code>f(mov x)</code> | transfers ownership; source becomes Consumed     |

- 2 Local variables declared with `var name = expr` have mutable storage accessible within their scope; their ownership mode in the type system is Owned. When the value is of a *linear* type, an Owned local must be consumed before its scope exits ([E003]); a local of an unrestricted type carries no such obligation.

### 11.2 Linear states

- 1 Every variable of a linear type has a *linear state* at each program point. The three states are:

**Uninit** The variable has been declared but no value has been assigned.

**Unconsumed** The variable holds a value that has not yet been transferred.

**Consumed** The variable’s value has been transferred via `mov` or a consuming function call.

- 2 The state machine governing transitions is:

| State before | Operation                            | State after                |
|--------------|--------------------------------------|----------------------------|
| Uninit       | first assignment                     | Unconsumed                 |
| Unconsumed   | <code>mov x</code> (transfer)        | Consumed                   |
| Unconsumed   | borrow ( <code>var</code> or shared) | Unconsumed (borrow active) |
| Consumed     | assignment (overwrite)               | Unconsumed                 |
| Consumed     | read or <code>mov</code>             | <i>ill-formed</i>          |
| Uninit       | read or <code>mov</code>             | <i>ill-formed</i>          |

**Note**

The Consumed  $\rightarrow$  Unconsumed transition via overwrite enables the loop-reassign-then-consume pattern: a linear variable declared outside a loop may be consumed and reassigned on each iteration without leaking.

### 11.3 Diagnostics

**Constraint**

Using a variable, or a reference whose owner has been moved, after that value has been transferred (use-after-move; the Consumed state) is *ill-formed* ([E001]). Using a variable *before* it is initialized — the Uninit state — is a different diagnostic, [E005] (below).

**Constraint**

Consuming a struct field that has already been consumed (a double-consume at field granularity) is *ill-formed* ([E002]).

**Constraint**

A linear variable that is in the Unconsumed state when its scope exits (resource leak) is *ill-formed* ([E003]).

**Constraint**

An aliasing violation — attempting a mutable borrow when a shared borrow is active, or a shared borrow when a mutable borrow is active, or a second mutable borrow of the same variable — is *ill-formed* ([E004]).

**Constraint**

Reading a variable, struct field, or array element before it has been initialized (use of an uninitialized value; the Uninit state) is *ill-formed* ([E005]). Definite-assignment analysis tracks initialization per binding and per field.

**Constraint**

Consuming a linear variable or field inside a loop body that may execute more than once is *ill-formed* ([E006]).

**Constraint**

Passing a linear value to an owned (`mov`) parameter without the explicit `mov` keyword at the call site — an implicit move — is *ill-formed* ([E007]). (Consuming a linear value on some branches but not others is a separate diagnostic, [E016], below.)

**Constraint**

Moving a value that is currently borrowed (`var` or shared), or moving a struct after some of its fields have already been consumed, is *ill-formed* ([E008]).

**Constraint**

If a linear value is in state `CONSUMED` on some control-flow paths through an `if/case` statement and in state `UNCONSUMED` on others, the merged state at the join point is undefined and the program is *ill-formed* ([E016]). The same diagnostic applies at field level: if a linear field of a struct is consumed on some paths but not on others, [E016] is issued for that field.

**11.4 Mutable parameters for primitive types (`var T`)**

- 1 A parameter declared `var name T`, where `T` is a primitive type (`iN`, `uN`, `f32`, `f64`, `bool`), is passed **by pointer** in the C ABI: the parameter's C type is `T*`.
- 2 **Write-through semantics:** an assignment to such a parameter inside the function body writes through the pointer and is visible to the caller. Reading the parameter's value automatically dereferences the pointer.

**Example**

```

1 func increment(var n i32) {
2     n = n +% 1    // writes through: caller's variable is updated
3 }
4
5 func main() i32 {
6     var x = 0
7     increment(var x)
8     return x      // x == 1
9 }

```

- 3 **Forwarding:** when a `var T` primitive parameter is passed as argument to another `var T` parameter, the pointer is forwarded directly without dereferencing and re-taking the address.
- 4 This behavior is identical to mutable parameters of struct type, where the by-pointer ABI was already in effect. The distinction is that for primitive types the codegen previously did not dereference reads; this was a bug that has been corrected.

**Note**

The call site shall annotate the argument with `var`: `f(var x)`. Omitting `var` passes a shared (read-only) borrow, which does not permit writes through the callee.

**11.5 The read-write lock invariant**

- 1 **Invariant:** At every program point, for every variable `V`, exactly one of the following holds:
  1. `V` has  $N \geq 0$  active shared borrows and zero active mutable borrows.
  2. `V` has exactly one active mutable borrow and zero shared borrows.
- 2 This invariant is the compile-time analogue of a reader-writer lock. A conforming implementation shall enforce it at every program point.

**11.6 Non-lexical lifetimes (NLL)**

- 1 A borrow's lifetime ends at the borrow's *last use*, not at the end of the variable's lexical scope. The implementation computes last-use points via a use-analysis pre-pass over the AST.

## Example

```

1 type Data {
2     v i32
3 }
4
5 func main() i32 {
6     var data = Data(42)
7     var r = var data.v           // mutable borrow of data.v
8                                   starts
9     r = 7                        // last use of r: the borrow
10                                   ends HERE (NLL)
11     var y = data.v              // OK: the borrow has expired
12     return y
13 }

```

## 11.7 Reference bindings

- 1 A declaration `var r = var P`, where `P` is a place (a local, a field path, an element, or a `var` parameter), or `var r = f(...)` where `f` returns `var T`, binds `r` to a **mutable borrow** of that place. Reading `r` yields the value at the place; assigning to `r` (or to a field of it) writes the place; `var r` passes the borrow on. `var r2 = var r` is a *reborrow*: uses of `r2` are uses of `r`'s loan.
- 2 A slice already carries its data pointer, so `var r = var s` for a slice `s` lends the slice's *elements*: `r[i] = v` writes them. Reassigning `r` as a whole is *ill-formed* [E009], exactly as for a `var` slice parameter — it would rebind the view and leave `s` unchanged.
- 3 The loan lasts until the binding's last use (§11.6). While it is live, any other access to an overlapping place — a read, a write, or a borrow — violates the read-write lock invariant (§11.5) and shall be diagnosed [E004]. Distinct fields, and elements at distinct constant indices, are distinct places; elements at indices not proven distinct are assumed to overlap.
- 4 A binding to storage owned by the function shall not be returned [E010]. A function whose return type is `var T` shall return `var P` or a call that itself returns `var T` [E017]. A struct field shall not hold a mutable borrow [E126].
- 5 A borrow — a reference binding, or a `var` or shared parameter — lends a place; it does not own it. A linear value shall not be moved out through one, by `mov` or by a whole-value copy [E020], and a linear place reached through one shall not be assigned [E021]: the resource there is the owner's, and it would be lost. The same assignment to a place the function owns is refused while the place may still hold a resource [E021].

## Example

```

1 var p = P(1, 2)
2 var x = var p.x
3 var y = var p.y           // OK: p.x and p.y are disjoint
4 x = 7
5 y = 8
6 var t = p.x               // OK: x and y are dead here (NLL)
7
8 var q = var p
9 p.y = 5                   // error [E004]: q is used below
10 q.x = 3

```

## 11.8 Two-phase borrows

- 1 During argument evaluation for a function call, borrows of the receiver object are placed in the *RESERVED* phase. A *RESERVED* borrow allows concurrent shared reads of the same owner but blocks mutable writes and moves. After all arguments have been evaluated, *RESERVED* borrows are promoted to the *ACTIVE* phase.
- 2 This mechanism enables UFCS calls of the form `x.f(x.field)` where the first argument (the receiver) requires a mutable borrow of `x` while the second argument reads a field of `x`.

| Phase    | Allows on owner | Blocks on owner       |
|----------|-----------------|-----------------------|
| RESERVED | shared reads    | mutable writes, moves |
| ACTIVE   | (nothing)       | all other access      |

### Example

```

1 type Vec {
2     data int
3     cap int
4 }
5 func push_n(var v Vec, n int) { v.data = v.data +% n }
6
7 func main() int {
8     var v = Vec(0, 10)
9     v.push_n(v.cap)    // desugars to push_n(var v, v.cap)
10                        // var v enters RESERVED phase; v.cap
11                        // read is allowed
12     return v.data
13 }
```

## 11.9 Move semantics

- 1 The `mov` operator transfers ownership. The source variable transitions to Consumed state; any subsequent read or move of the source is *ill-formed* ([E001]).
- 2 Ownership transfer is the only way to pass a linear value to a consuming parameter. The `mov` keyword shall be present at the call site: `f(mov x)`.

### 11.10 Borrowed match

- 1 The `case &expr { ... }` form (see § 15) performs pattern matching without consuming the scrutinee. A temporary shared borrow of the scrutinee is registered for the duration of the match body and released when the match completes.

#### Constraint

Inside a `case &expr` body, the scrutinee shall not be mutated or moved. Attempting to do so violates the active shared borrow ([E004]).

### 11.11 Loop and ownership

- 1 A linear variable declared outside a loop may be consumed inside the loop body if and only if it is also reassigned (overwritten) on every path through the loop body before the next iteration begins.

**Example**

```

1 import std.fs.{open_file, close_file}
2
3 func main() int effects io, raises, alloc {
4     var f = open_file("data.txt", "r")
5     close_file(mov f)           // consumed before loop
6     var i = 0
7     while i < 3 decreasing 3 - i {
8         f = open_file("data.txt", "r") // reassign (Consumed
9         -> Unconsumed)
10        close_file(mov f)           // consume
11        i = i + 1
12    }
13    return 0
14 }

```

**11.12 Branch consistency**

- After an **if/else** chain or **case** match, the linear state of each variable from the enclosing scope is the *intersection* of its states across all branches:
  - If consumed in all branches: the variable is Consumed after the chain.
  - If consumed in no branches: the variable is Unconsumed after the chain.
  - If consumed in some but not all branches: *ill-formed* ([E016]).
- The same rule applies at field level: if a linear field of a struct is consumed on some branches but not on others, [E016] is emitted for that field.

**Example**

```

1 // ERROR [E016]: h is consumed in the then-branch but not in
  // the (implicit) else
2 type Handle { mov ptr *i32 }
3
4 func drop(mov h Handle) { }
5
6 func maybe_consume(mov h Handle, flag bool) {
7     if flag {
8         drop(mov h) // consumed in then-branch
9     }
10 }

```

**Note**

[E007] (generic ownership mismatch) was the diagnostic emitted before [E016] was introduced. [E016] is now the dedicated code for branch-inconsistency on linear values.

**11.13 Field-level tracking**

- The borrow checker tracks borrows at the field level. Borrowing non-overlapping fields of the same struct simultaneously is permitted:

```
1 type Pair {  
2     a i32  
3     b i32  
4 }  
5  
6 func main() i32 {  
7     var p = Pair(1, 2)  
8     var ra = var p.a           // mutable borrow of p.a  
9     var rb = var p.b           // mutable borrow of p.b -- OK:  
                                non-overlapping  
10    ra = 10  
11    rb = 20  
12    return p.a +% p.b  
13 }
```

### Note

The reference implementation tracks field borrows at one level of depth. Nested field borrows (e.g., `p.a.x` vs `p.a.y`) are conservatively treated as overlapping.



## Chapter 12

# Functions and effects

### 12.1 The effect row

- 1 A function has one introducer, `func`, and an **effect row**: a set drawn from `io`, `diverge`, `raises` and `alloc`. The row is inferred from the body across the call graph, its default is  $\emptyset$  — pure and total — and an `effects` clause widens that default for a function that deviates. The row is a *complete* upper bound: an effect the body has and the row does not name is *ill-formed*, so a caller may rely on what it reads (§ 12.2).

| Property                              | default row ( $\emptyset$ )    | with the effect named               |
|---------------------------------------|--------------------------------|-------------------------------------|
| Observable side effects               | forbidden                      | <code>effects io</code>             |
| Allocation                            | forbidden                      | <code>effects alloc</code>          |
| <code>panic</code>                    | forbidden                      | <code>effects raises</code>         |
| Unbounded <code>while</code>          | forbidden                      | <code>effects diverge</code>        |
| Bounded <code>while decreasing</code> | allowed                        | allowed                             |
| <code>for</code> loops                | allowed                        | allowed                             |
| Direct recursion (with measure)       | allowed                        | allowed                             |
| Mutual recursion                      | forbidden                      | <code>effects diverge</code>        |
| Global mutable state                  | does not exist in the language | —                                   |
| Termination guarantee                 | yes                            | waived by <code>diverge</code> only |

- 2 Naming an effect is not a local suppression. It enters this function’s row, which is what every caller reads, so the acknowledgement propagates outward until some function is genuinely entitled to it.

#### Note

`proc` was the form in which the introducer granted the effects instead of the row. It has been removed (§ 10.5); the row is the only mechanism.

- 3 A `func` is referentially transparent: calling it with the same arguments always produces the same result. Its evaluation has no observable effect on any state outside its own stack frame.
- 4 Local mutation through a `var T` parameter (exclusive mutable borrow) is permitted in a `func`. The borrow checker guarantees no aliasing, so the mutation is observable only through the parameter binding itself; the function remains deterministic and referentially transparent (same input borrow  $\Rightarrow$  same mutation pattern).

**Example**

```

1 type Counter { val i32 }
2
3 func increment(var c Counter) { // legal: mutation is local
4     c.val = c.val +% 1
5 }

```

- 5 [W130] advised downgrading an effect-free **proc** to a **func**. It was removed with the form it advised against: a warning whose subject cannot be written is dead weight that still has to be kept correct — and it had just been caught stating the opposite of the truth, because its eligibility test read the callee’s *keyword* rather than the callee’s row.

**12.2 Purity enforcement****Constraint**

A function that calls one whose effect row names an effect its own row does not is *ill-formed* ([E011]). The callee’s introducer plays no part: the effect is what is refused. Effects are transitive, so this reaches a callee three layers down that a rule about direct calls would miss.

**Constraint**

A function that contains a **while** loop whose termination the implementation can neither verify from an explicit **decreasing** measure nor establish by inferring one (§9.7.2) is *ill-formed* ([E011]) *unless its row names diverge*. Naming it is not a suppression: **diverge** enters this function’s row and every caller must acknowledge it in turn.

**Constraint**

A **func** that calls itself directly is permitted only if it carries a well-founded **decreasing** measure — supplied explicitly or inferred by the implementation (§12.3). A directly-recursive **func** with no measure and none inferable is *ill-formed* ([E011]); one whose measure is malformed is reported [E091], and one whose measure is not provably decreasing and non-negative at every self-call is reported [E082].

**Constraint**

A function that participates in a *mutual*-recursion cycle is *ill-formed* ([E011]) unless its row names **diverge**: no measure covers such a cycle, so termination cannot be established. The implementation shall detect mutual recursion by performing a depth-first search on the call graph.

**Note**

Generic functions (**func** with **type** parameters) are excluded from mutual-recursion detection until they are monomorphized.

**12.3 Termination analysis**

- 1 A **func** is guaranteed to terminate because:

1. every **while** loop carries a verified termination measure, supplied explicitly or inferred (§9.7.2);
  2. **for** loops iterate over finite ranges;
  3. every direct self-call strictly decreases a well-founded measure, and mutual recursion is forbidden.
- 2 **Recursion measure.** The measure for a directly-recursive **func** is a parameter (**decreasing** *n*) or a difference of two parameters (**decreasing** *hi* - *lo*). When omitted, the implementation attempts to infer a single parameter that strictly decreases in *every* self-call (each self-call passes it a syntactically smaller argument, such as *n* - 1 or *n* / 2). Each self-call is then verified to make the measure strictly smaller and to keep it non-negative (a base case); a call that does neither is reported [E082]. Inference never accepts a recursion an explicit measure could not.
- 3 No termination analysis is performed for a function whose row names **diverge**.

## 12.4 Universal Function Call Syntax (UFCS)

- 1 An expression of the form *x.f(args)* is desugared to *f(x, args)* before name resolution, provided *f* is not a declared field of *x*'s type. This is called Universal Function Call Syntax (UFCS).
- 2 **UFCS ownership:** the ownership mode of the first argument in the desugared call is determined by the declared mode of the first parameter of *f*:
  - If the first parameter is **var**, the receiver is automatically mutably borrowed: *x.f()* desugars to *f(var x)*.
  - If the first parameter is **mov**, the receiver is automatically moved: *x.f()* desugars to *f(mov x)*.
  - If the first parameter is shared (default), the receiver is shared: *x.f()* desugars to *f(x)*.
- 3 **UFCS resolution algorithm:**
  1. Check if *f* is a declared field of *x*'s type. If so, this is a field access, not UFCS.
  2. Search for a function named *f* whose first parameter type is compatible with the type of *x*.
  3. If found, desugar the call.
  4. If no matching function is found, the call is unresolved and the program is *ill-formed* (an unresolved receiver or callee is reported; a bare undeclared identifier is [E106], §6.3).

### Example

```

1 func is_empty(s u8[]) bool { return s.len == 0 }
2
3 func main() i32 {
4     var buf = "hello"
5     var empty = buf.is_empty()    // desugars to: is_empty(buf)
6     if empty { return 1 }
7     return 0
8 }

```

## 12.5 Calling conventions

- 1 The calling convention for Lain functions corresponds to the C99 calling convention of the target platform, as determined by the C compiler used for the translation output. The parameter passing rules are:

| Lain parameter      | C type   | Semantics                 |
|---------------------|----------|---------------------------|
| shared struct T     | const T* | pointer to read-only copy |
| <b>var</b> struct T | T*       | pointer, caller's copy    |
| <b>mov</b> struct T | T        | by value (bitwise copy)   |
| primitive any mode  | T        | by value                  |

## 12.6 Return values

- 1 A return statement has the form **return** <expr>. The mode of the returned value is determined by the function's declared return type and follows the same rules as call-site argument passing.
- 2 A **return** may be prefixed with **mov** or **var**: **return mov** x makes the ownership transfer explicit, and **return var** x returns a mutable borrow (the pattern for a **func** whose declared return type is **var** T). When the return type is owned, a bare **return** expr already moves expr, so **mov** is optional there.

### Note

A **return var** x / **return mov** x that would escape a borrow of a *local* variable is a dangling reference ([E010], §9.8). This supersedes an earlier design (Q-009) in which explicit **mov/var** in return position was rejected.

## 12.7 The panic builtin

- 1 **panic**(msg) is a builtin expression that terminates program execution immediately. Its type is *Never* (the bottom type: a subtype of every type). Because it does not return, **panic** may appear in any expression position—including as the body of a match arm whose declared type is otherwise unrelated.
- 2 **panic** may be called from any function whose row names **raises** contexts. The call is deterministic (same input yields the same termination), so it does not violate the purity of an enclosing **func**.

### Constraint

**Q-014, S16.** When **panic** is executed, the program terminates immediately via the platform abort primitive. No **defer** block is executed. No stack unwinding occurs.

### Note

The reference implementation emits, for **panic**(msg), the equivalent of `fprintf(stderr, "panic: %.*s\n", ...) ; abort()`. The message is written to the standard error stream before abort.

**Example**

```

1 func head(s u8[]) u8 effects raises {
2     if s.len == 0 {
3         panic("empty slice")    // Never; satisfies the return
                                type
4     }
5     return s[0]
6 }

```

**12.8 Function attributes**

1 A function declaration may be prefixed with one or more attributes (§10.1). The attributes applicable to functions in version 1.0 are:

- **[fast\_math]** — enable fast floating-point optimizations (FMA fusion, contraction) for the body of this function. This is an opt-out from the determinism guarantee (P4) and applies only locally; functions without **[fast\_math]** remain bit-deterministic within their target platform.
- **[private]** — module-private visibility (§16.5).

**Note**

The reference implementation translates **[fast\_math]** on a function  $f$  to a pair of pragmas surrounding the body of  $f$  in the emitted C source, enabling **fp-contract=fast** for  $f$  only. The default (i.e., absence of **[fast\_math]**) emits **-ffp-contract=off** flags, providing deterministic floating point within a single target platform and C compiler.



## Chapter 13

# Value range constraints

### 13.1 Overview

- <sup>1</sup> Value Range Analysis (VRA) is the static analysis that establishes, at every program point, sound facts about the integer values — and the relationships between values — that may arise at run time. It is the mechanism by which a conforming implementation discharges the language’s compile-time proof obligations, and thereby its safety guarantees and check-free code generation. It is used to:
  - prove array and slice accesses in bounds (§ 7.3);
  - prove the absence of division/modulo by zero (§ 8.3);
  - prove the absence of arithmetic overflow at a checked operation (§ 13.10);
  - verify parameter and return refinements, and propagate a return refinement to call sites (§ 13.6);
  - establish the loop and recursion facts that make termination measures provable (§ 12.3).
- <sup>2</sup> This clause specifies the *contract* of the VRA — what a conforming implementation shall guarantee — not the abstract domain or algorithm used to achieve it. The domain and algorithm are an implementation concern; the reference design (a relational abstract interpreter over the octagon domain, run as a fixpoint over a control-flow graph) is described, informatively, in § 13.13. Any analysis satisfying the contract below conforms, whether it reasons with intervals, difference bounds, octagons, or a stronger relational domain.

#### Constraint

**Soundness.** The VRA shall compute, at each program point, an *over-approximation* of the set of concrete states reachable there: every state that can occur at run time shall be admitted by the analysis result at that point. Consequently a proof obligation the VRA reports as *discharged* holds on every execution — an accepted access is in bounds, an accepted operation does not overflow or divide by zero, and a verified refinement holds for every result. A proof obligation the implementation cannot discharge shall be reported as an error, never silently assumed (*prove-or-reject*).

#### Constraint

**Determinism and decidability.** The VRA shall be deterministic and shall terminate on every input, in time polynomial in program size. No general SMT, Presburger, or unbounded search procedure is required or permitted in the analysis core; a conforming implementation reaches its verdict by a decidable, terminating computation.

## 13.2 What the analysis tracks

- 1 The analysis maintains, at each program point, a sound over-approximation of each live integer value's possible values and — for a relational implementation — of the relationships between values (for example  $i < len$ ,  $lo \leq hi$ , or  $o = i + j$ ). The per-value component is an interval  $[\ell, u]$  (with  $-\infty, +\infty$  for an absent bound, and  $unknown = (-\infty, +\infty)$ ); the relational component, when present, records linear relations among values. The concrete representation is implementation-defined; only the soundness contract above is normative.

## 13.3 Range propagation rules

- 1 The following table defines the result range for binary arithmetic operations on intervals  $[a, b]$  and  $[c, d]$ . All arithmetic is saturating: overflow wraps to  $\pm\infty$ .

| Operation                            | Result range                                   | Notes              |
|--------------------------------------|------------------------------------------------|--------------------|
| $[a, b] + [c, d]$                    | $[a + c, b + d]$                               | saturating         |
| $[a, b] - [c, d]$                    | $[a - d, b - c]$                               | saturating         |
| $[a, b] \times [c, d]$               | $[\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)]$ | saturating         |
| $[a, b] \div [c, d], c > 0$          | $[a \div d, b \div c]$                         | integer truncation |
| $[a, b] \div [c, d], 0 \in [c, d]$   | <b>[E015]</b>                                  | division by zero   |
| $[a, b] \% [c, d], a \geq 0, c > 0$  | $[0, d - 1]$                                   |                    |
| $[a, b] \% [c, d], \text{otherwise}$ | unknown                                        | conservative       |

## 13.4 Conditional refinement

- 1 After an **if**  $x < N$  test, the VRA refines the range of  $x$  in the then-branch to  $[\ell, N - 1]$  and in the else-branch to  $[N, u]$  (where  $[\ell, u]$  is the range before the test).
- 2 At the join point after an **if/else**, the range is the union of the ranges from both branches.
- 3 When conditions are chained with **and**, refinements from the left operand are active when evaluating the right operand (§8.7).
- 4 **Equality and disequality.** A test  $x == v$  refines  $x$  to the point range  $[v, v]$  in the then-branch. A disequality (the else-branch, or an explicit  $x != v$ ) refines  $x$  only when  $v$  lies *at a boundary* of the range  $[\ell, u]$  — to  $[v + 1, u]$  when  $v = \ell$ , or to  $[\ell, v - 1]$  when  $v = u$  — since removing an interior point cannot be expressed as a single interval. In particular an unsigned  $n$  guarded by **if**  $n == 0$  { **return** } is refined to  $[1, u]$  on the fall-through, so a subsequent  $a[0]$  on a length- $n$  slice is in bounds.
- 5 **Unsigned subtraction.** For  $r = a - b$  at an *unsigned* result type, the relation  $r = a - b$  holds over  $\mathbb{Z}$  but not over the type unless  $a \geq b$ : the difference wraps to a large positive value otherwise. An implementation shall record no relation between  $r, a$  and  $b$  unless it can show  $a \geq b$ ; where it can — **hi** - **lo** under a live **lo** < **hi**, or **n** - 1 behind **if**  $n == 0$  { **return** } — the relation is exact and may be recorded.

### Note

Recording the  $\mathbb{Z}$  relation unconditionally is unsound and not merely imprecise. Under it, **hi** = **mid** - 1 with **mid** zero appears to give **hi** the value -1 rather than the type's maximum, and every bound derived from that relation inherits the error.

- 6 **Early-return refinement.** When an **if** whose then-branch exits the enclosing scope (via **return**, and provided no **break** escapes) has condition  $C$ , the code that follows the **if** is reached only when  $C$  is false; the VRA therefore refines with  $\neg C$  for the remainder of the

scope. Likewise, after a `while cond { ... }` loop whose body contains no `break` at that loop's level, the VRA assumes  $\neg \text{cond}$ , since the loop exits only when the condition fails (e.g. after `while j > 0 { ... }`, `j` is refined to 0).

- 7 **Length guards.** A guard on a slice or array length — `i < a.len`, and also `a.len > 0`, `a.len >= k`, `a.len != 0` — refines the underlying length. For a fixed array `a T[N]` the length is the constant `N`; for a sized slice `a T[n]` it is the size parameter `n`. Thus `if a.len > 0 { a[0] }` is accepted exactly as `if n > 0 { a[0] }` would be.

### 13.5 In-guard semantics

- 1 An in-guard `idx in arr` (§8.7) introduces the constraint  $0 \leq \text{idx} < \text{arr.len}$  into the VRA's range environment. Inside the guarded body, `arr[idx]` is accepted without further proof.
- 2 When `idx in arr` is the condition of a `while decreasing M` loop, the VRA infers that  $\text{arr.len} - \text{idx} \geq 0$  (a valid non-negativity proof for the measure  $\text{arr.len} - \text{idx}$ ).

### 13.6 Constraint annotations

- 1 A constraint annotation of the form `T op bound` on a parameter or return type declares an invariant that the VRA shall enforce:
  - **Parameter constraint:** at every call site, the VRA shall prove that the argument satisfies the constraint.
  - **Return constraint:** at every `return` statement, the VRA shall prove that the return expression satisfies the constraint.

#### Constraint

A function call where the argument range cannot be proven to satisfy a parameter constraint is *ill-formed* ([E012]).

#### Constraint

A `return` expression whose range cannot be proven to satisfy the declared return constraint is *ill-formed* ([E086]) — the same boundary mechanism (§13.10) that checks assignments and narrowing conversions. (A *parameter*-constraint violation at a call site is the distinct code [E012], above.)

### 13.7 Interprocedural range propagation

- 1 When a call expression appears in a context where the VRA needs a range, it uses the callee's declared return constraint (if any) as a conservative approximation.

#### Example

```

1 func nonzero() int >= 1 { return 42 }
2
3 func divide(a int) int {
4     return a / nonzero() // the divisor is known >= 1, so no
                          divisor diagnostic
5 }
```

- 2 When the callee has no declared return constraint, the call expression has an unknown range.

**Note**

Interprocedural VRA is conservative: only declared constraints are used, not body analysis. This preserves decidability and ensures that changing a function’s body cannot silently affect its callers’ safety proofs.

**13.8 Limits and known gaps**

- 1 The following limitations are known and accepted:
  1. A struct field is tracked like a variable from the point the function itself stores it. Its value on ENTRY — through a **var** parameter or a pointer — is unknown beyond what its declared invariant (§7.6) states.
  2. VRA does not track ranges across loop-carried dependences with non-linear measures.
  3. The analysis reasons over  $\mathbb{Z}$ . This is consistent with the language, because an operation that could overflow is *ill-formed* (§5.10) and the operators that *do* wrap or saturate are written explicitly and carry no obligation. Such an operator’s result is known exactly when the analysis can show it does not wrap, and not at all otherwise.
  4. Floating-point values are not tracked by VRA.

**13.9 Loop body affine recap**

- 1 For **for** **V** **in** *start*..*end* loops whose body contains assignments of the form  $x = x + c$  or  $x = x - c$  with constant step  $c$ , the implementation computes the post-loop range of  $x$  precisely:

$$x_{\text{post}} \in [x_{\text{init}} + c \cdot n_{\text{min}}, x_{\text{init}} + c \cdot n_{\text{max}}]$$

where  $n_{\text{min}}, n_{\text{max}}$  is the iteration count derived from the loop’s range. This avoids the conservative widening that would otherwise mark  $x$  as unknown after the loop.

**Example**

```

1 var x = 0
2 for i in 0..10 {
3     x = x + 1
4 }
5 // Post-loop: x is exactly 10 (range [10, 10]).

```

- 2 Patterns not matching the affine form (mixed updates in branches, non-constant step, multiple variables updated together) fall back to the conservative widening (unknown post-loop range).

**13.10 Overflow-at-boundary check**

- 1 The Phase 5 boundary check applies at every integer-type boundary where a value crosses into a narrower or different-domain target. The check sites are:
  - Variable declaration (**var**  $x$   $T = \text{expr}$ ).
  - Assignment to identifier, field, or indexed element ( $x = \text{expr}$ ,  $p.f = \text{expr}$ ,  $\text{arr}[i] = \text{expr}$ ).

- Return value vs the function’s declared return type.
  - Call argument vs the parameter’s declared type.
  - Struct constructor field initialization.
- At each site, if the source value’s VRA range exceeds the target type’s natural range, the program is *ill-formed* ([E086]). The diagnostic offers these resolutions: widen the target; choose an explicit overflow policy via a cast tier or operator — checked (`as?`, `+`?), wrapping (`as%`, `+`%), or saturating (`as|`, `+`|); or tighten input constraints so VRA can prove safety (§7.2.6).
  - The check is skipped:
    - Inside an `unsafe` block;
    - When the source range is unbounded (the analysis cannot rule out overflow without further information; the user must annotate);
    - For integer-literal sources (literals are polymorphic across `iN/uN` and trusted to fit when written explicitly).
  - The wrapping/saturating operators (`+`%, `*`%, `+`|, `*`|) produce a result whose range is automatically clamped to the LHS type’s natural range, so they pass the boundary check unconditionally.

### Example

```

1 var a u8 = 200
2 var b u8 = 100
3 // var bad u8 = a + b           // [E086]: range [300, 300] exceeds
   u8
4 var w u8 = a +% b              // OK: wrapping clamps to [0, 255]
5 var s u8 = a +| b              // OK: saturating clamps to [0, 255]
```

## 13.11 Sub-slice bounds

- A sub-slice expression `arr[a..b]` produces a slice covering the half-open range  $[a, b)$  of `arr`. The inferred type of the result is `T[b-a]`, a sized slice (§7.4.1) whose length is statically known. When both endpoints are compile-time integer literals, the compiler verifies  $a \leq b$ ; if the check fails the program is *ill-formed* ([E087]).
- When the endpoints are variables, each is range-checked against the container length ([E085], the standard bounds check). A runtime guarantee on the ordering of variable endpoints is not emitted at compile time; the programmer must establish it through refinement constraints or by limiting the range expression to literal bounds.

### Example

```

1 var arr i32[10] = [0,1,2,3,4,5,6,7,8,9]
2 var ok = arr[2..5]           // type i32[3], half-open, length 3
3 // var bad = arr[5..2]       // [E087]: start 5 > end 2
4 var empty = arr[3..3]        // OK: empty slice of length 0
```

### 13.12 Sized slice call-site verification

- 1 When a function parameter is declared with a sized slice type (§7.4.1), the compiler verifies at every call site that the supplied argument satisfies the parameter’s length constraint. A proved violation is *ill-formed* ([E087]). The check is conservative: it fires only when VRA can *prove* the constraint is not met; it is silent when the constraint cannot be determined statically.
- 2 The cases checked are:
  - $T[\geq k]$  or  $T[> k]$  with integer literal  $k$ : fires if the argument length’s VRA upper bound is less than the required minimum ( $\geq k$  requires  $len \geq k$ ;  $> k$  requires  $len \geq k + 1$ ).
  - $T[k]$  (exact length, literal  $k$ ): fires if the argument length’s VRA range does not contain  $k$ .
  - $T[ref.len]$  (cross-parameter equality): fires if both sides have a concrete singleton VRA range and they differ.

#### Example

```

1 func pair_sum(arr i32[>= 2]) i32 { return arr[0] +% arr[1] }
2
3 func main() i32 {
4     var one i32[1] = [42]
5     // pair_sum(one)           // [E087]: len 1 < required 2
6     var two i32[2] = [1, 2]
7     return pair_sum(two)      // OK: len 2 satisfies >= 2
8 }

```

- 3 **Sized slice VRA entry.** At function entry, for each parameter name  $T[expr]$  the compiler seeds the range table with the constraint derived from the size expression:
  - $T[k]$  (literal):  $__len\_name \in [k, k]$ .
  - $T[\geq k]$  (literal):  $__len\_name \in [k, +\infty)$ .
  - $T[> k]$  (literal):  $__len\_name \in [k+1, +\infty)$ .
  - $T[a.len]$  (member reference): difference constraint  $__len\_name = __len\_a$ .
  - $T[a.len + b.len]$  (arithmetic): constraints  $__len\_a \leq __len\_name$  and  $__len\_b \leq __len\_name$ .

These constraints are available to the VRA throughout the function body (§13.13 shows two patterns they prove).

### 13.13 The reference implementation (informative)

- 1 This section is informative. It says how the reference compiler meets the contract above, so that the precision a program may rely on is concrete. Every example in it is compiled by the specification’s gate.
- 2 **Domain.** An abstract interpreter over the *octagon* domain. Every tracked integer quantity — an SSA value, a local variable’s cell, a field of a local struct, the length of a slice — is a dimension, and a state is a conjunction of constraints  $\pm x \pm y \leq c$ . That includes intervals ( $x \leq c$ ,  $-x \leq c$ ) and differences ( $x - y \leq c$ ). Unary bounds are kept exactly within  $\pm 2^{60}$  and

only weakened beyond it, never strengthened. A table of compile-time constants gives exact values alongside.

- 3 **Closure.** The constraints are kept under integer *tight closure* (shortest paths, then unary bounds tightened to integers, then strengthening), so a fact implied by others is as available as one stated directly. After a step adds a few constraints to a closed state, the closure is updated through the new constraints only.
- 4 **Control flow.** The analysis runs to a fixpoint over the control-flow graph of the IR, refining the state along each branch by its condition, and widening at loop headers with thresholds taken from the loop's own constants. A branch the state makes impossible is dead: its obligations hold vacuously and it takes no part in a termination argument.
- 5 **Arithmetic.** A checked operation is modelled over  $\mathbb{Z}$ , which is exact because its overflow is an obligation. A wrapping or saturating operation is modelled exactly when the state shows it cannot wrap: `x +% 1` with `x` known to be 6 is 7. When it may wrap, its relation to the operands is dropped.
- 6 **Beyond the domain.** Facts the octagon cannot hold are computed alongside and handed back as bounds: a running total's range as the loop's trip count times its step; a divisor's non-zero-ness from a `!= 0` refinement or guard; a remainder's and a logical shift's bound by their operand; the range of `@movemask`.
- 7 **Examples** the reference implementation proves. `reverse` proves as written. `concat` needs its inputs' lengths bounded: without a bound, the parameter type's own `a.len + b.len` may overflow a `usize`, and that is refused ([E086]).

### Example

```

1 func reverse(arr i32[], out i32[arr.len]) {
2     var n = arr.len
3     for i in 0..n {
4         out[n - i - 1] = arr[i]    // n - i - 1 is in [0, n - 1]
5                                     since i < n
6     }
7
8 func concat(a i32[<= 1000000], b i32[<= 1000000], out i32[a.len
9     + b.len]) {
10     for j in 0..b.len {
11         out[j + a.len] = b[j]    // j + a.len < b.len + a.len
12                                     == out.len
13     }
14 }
```



## Chapter 14

# Generics and compile-time evaluation

### 14.1 Overview

- 1 Lain implements genericity through compile-time function evaluation (CTFE). Generic types and functions accept *type parameters* (declared as `name type`) or *compile-time value parameters* (declared as `comptime name non-type`). The compiler evaluates them at compile time and monomorphizes the result. There is no type erasure and no runtime polymorphism.

#### Note

**Implementation status.** The reference compiler implements the *core* of this chapter and monomorphizes it: generic `funcs` with `type` parameters (bound by inference from a value argument or by an explicit type argument), and generic types written `type Name(T type)= type { ... }`. The remaining constructs below are *planned* and not yet accepted — compile-time *value* parameters / `const` generics (`comptime N int`), `comptime { ... }` expression blocks, and the `func`-returning-`type` spelling; each is flagged at its section, and is presently reported as [E100]. (`comptime if`, §14.7, *is* available.)

### 14.2 Type parameters

- 1 A parameter declared `name type` is a type parameter. Types are implicitly compile-time — the `comptime` keyword shall *not* precede a type parameter. At every call site, the argument shall be a type expression known at compile time.

```
1 func identity(T type, x T) T {
2     return x
3 }
4
5 func main() i32 {
6     var n = identity(int, 42)           // T = int, given;
6         monomorphized
7     var s = identity("hi")              // T = u8[:0], inferred; a
7         separate monomorphization
8     return 0
9 }
```

- 2 Type parameters are evaluated entirely at compile time; they produce no runtime code.

#### Constraint

The `comptime` keyword shall not appear before a parameter whose type is `type`. `func f(comptime T type)` is ill-formed; write `func f(T type)` instead.

### 14.3 Compile-time value parameters

- 1 A parameter declared `comptime` name non-`type` is a compile-time value parameter. The argument shall be a constant expression of the declared type known at compile time.

#### Note

**Planned (not yet implemented).** Compile-time value parameters (const generics) are not accepted by the reference compiler: a declaration such as the one below is presently [E100].

```

1 // ERROR: E100 -- planned: a compile-time value parameter is not
  yet accepted at a use
2 type RingBuffer(T type, N usize) {
3     items T[N]
4     head  usize
5     tail  usize
6 }
7 func main() i32 {
8     var r RingBuffer(i32, 8) = RingBuffer(i32, 8)([0, 0, 0, 0, 0,
9         0, 0, 0], 0, 0)
10    return r.items[0]
11 }
```

#### Constraint

A compile-time evaluation context (CTFE) shall not call a function whose effect row is non-empty; only calls with an empty row are admissible. A violation is *ill-formed* ([E101]).

### 14.4 Type as value

- 1 A type declared with type parameters is a type constructor: `Option(int)` names the instance for `int`, which is generated once (monomorphized) and may be used in type annotations and variable declarations. A variant of an instance is constructed as `Option(int).Some(42)` where the declaration states the instance.

```

1 type Option(T type) {
2     Some { value T }
3     None
4 }
5
6 func main() i32 {
7     var x Option(int) = Option(int).Some(42)    // the variable
8         states the instance
9     return 0
10 }
```

- 2 Type constructors may also be written as `func` declarations returning `type`:

#### Note

**Planned (not yet implemented).** The `func`-returning-`type` spelling below is presently [E100]; use the `type` `Name(T type){ ... }` form above, which is accepted. (The spelling

`type` `Name(T type)= type { ... }` declares an alias, not a generic type: its instances are refused, [E124].)

```

1 // ERROR: E100 -- the planned spelling, not yet accepted
2 func Option(T type) type {
3     return type {
4         Some { value T }
5         None
6     }
7 }
```

### Constraint

A function returning `type` shall have an empty effect row: a type-level value cannot depend on an effect.

## 14.5 Monomorphization

- 1 For each unique combination of comptime arguments at call sites, the implementation produces a separate instantiation of the function. The name of each instantiation in the translation output is *implementation-defined*<sup>1</sup>.
- 2 Monomorphization is a compile-time-only operation; at runtime, each call site calls a specific, non-generic function.

## 14.6 Comptime expression blocks

- 1 A `comptime { expr }` expression evaluates *expr* at compile time. The result is a compile-time constant that may be used as a type annotation, array size, or parameter constraint value.

### Note

**Planned (not yet implemented).** `comptime { ... }` expression blocks are not accepted by the reference compiler (presently [E100]). The statement form `comptime if` (§14.7) is available.

### Constraint

A `comptime` expression shall be pure and terminating (i.e., it may only use `func` calls and the constructs permitted in `func` bodies).

## 14.7 Comptime if

- 1 `comptime if` *cond* { ... } `else` { ... } is a compile-time conditional. Only the taken branch is type-checked and emitted; the other branch is entirely discarded. *cond* shall evaluate to a compile-time boolean constant.

### Note

`comptime if` enables platform-specific code selection without dead-code warnings. It is the preferred way to write code that varies based on compile-time parameters.

<sup>1</sup>Implementation-defined behavior: determined by the implementation

## 14.8 Comptime recursion depth

- 1 The planned compile-time evaluation engine (§ 14.3, § 14.6) will bound the recursion depth of comptime function evaluation *implementation-defined*<sup>2</sup> and reject an evaluation that exceeds it. Since that engine is not yet implemented, no depth diagnostic is currently emitted.

## 14.9 Mode preservation in generics

- 1 When a type parameter `T` is substituted with a concrete type, the ownership mode of the parameter is preserved. If a parameter was declared `var T`, and `T` is substituted with a type that has mode `shared` in its declaration, the substituted parameter retains the `var` mode.

### Note

This rule ensures that mode annotations in generic functions behave consistently regardless of the concrete type argument.

---

<sup>2</sup>Implementation-defined behavior: by an implementation-defined limit

## Chapter 15

# Pattern matching

### 15.1 Overview

- 1 The **case** expression matches a value against a sequence of patterns and evaluates the body of the first matching arm. Every **case** expression shall be exhaustive: the patterns shall collectively cover all possible values of the scrutinee type. The grammar production is *match-expression* (Annex A).

### 15.2 Syntax

```
1 // SYNOPSIS: the form of a case expression
2 case expr {
3     Pattern1: body1
4     Pattern2: body2
5     else:      default_body
6 }
```

- 1 Each arm consists of a pattern (or comma-separated list of patterns) and a body. The body may be a single expression or a block. The arms are tested in declaration order; the first matching arm is taken.

### 15.3 Patterns

- 1 The following pattern forms are supported:

Matches any value. Must be the last arm. Required for integer and character scrutinees.

**Wildcard (Literal pattern)** Matches a specific integer or character value: `42:`, `'a':`.

**Enum/ADT variant pattern** Matches a specific variant. If the variant has fields, they are bound: `Circle(r):` ... binds the radius to `r`.

**Range pattern** Matches a range of integer or character values. Consistent with expression ranges and sub-slices (§5.8, §8.13), `a..b` is *exclusive* of the upper bound and `a..=b` is *inclusive*: `1..10:` matches 1 through 9, and `1..=10:` matches 1 through 10.

**Multi-pattern (OR)** A comma-separated list of patterns on a single arm: `1, 2, 3: ....`

### 15.4 Exhaustiveness

- 1 A **case** expression is exhaustive if the set of patterns covers all values of the scrutinee type. The rules by type are:

| Scrutinee type   | Exhaustiveness requirement                           |
|------------------|------------------------------------------------------|
| Enum             | All variants covered, or <b>else</b> arm present     |
| ADT              | All variants covered, or <b>else</b> arm present     |
| Integer          | <b>else</b> arm always required                      |
| <b>u8</b> (char) | <b>else</b> arm always required                      |
| <b>bool</b>      | <b>true</b> and <b>false</b> covered, or <b>else</b> |

**Constraint**

A non-exhaustive **case** expression is *ill-formed* ([E014]).

**Example**

```

1  type Color {
2      Red
3      Green
4      Blue
5  }
6
7  func code(c Color) i32 {
8      case c {
9          Red:    return 1
10         Green: return 2
11         Blue:  return 3
12     } // exhaustive: all three variants covered
13 }

```

## 15.5 Ownership and match

- 1 By default, **case** *expr* may consume the scrutinee if it is a linear type. Bindings introduced by variant patterns in consuming arms hold ownership of the bound values.
- 2 **case** *&expr* performs a non-consuming match: the scrutinee is not consumed but instead temporarily borrowed for the duration of the match body (see §11.10).

**Constraint**

Inside a **case** *&expr* body, the scrutinee shall not be mutated or moved ([E004]).

## 15.6 Case expression result type

- 1 When **case** is used as an expression, all arms shall produce values of compatible types (§7.11). The type of the **case** expression is the common type.

**Constraint**

Arms of a **case** expression that produce incompatible types are *ill-formed*.

## 15.7 String pattern matching

- 1 String literals may appear as patterns. A string pattern matches if the scrutinee has the same byte sequence and length as the pattern literal. The comparison is performed byte-by-byte (analogous to `memcmp`).

**Note**

String pattern matching in the translation output is implemented as a sequence of byte comparisons or `memcmp` calls, depending on the implementation. Its asymptotic cost is  $O(n)$  in the pattern length.



## Chapter 16

# Module system

### 16.1 Overview

- 1 Each Lain source file defines one module. Modules are the unit of code organization and compilation. All names declared at the top level of a source file are in that module's scope and are accessible from other modules that import the module.

### 16.2 Module identity

- 1 A module's identity is *implementation-defined*<sup>1</sup>; there is *no* module declaration keyword (a leading `module ...` line is [E100]). The reference implementation names a module by its path relative to the working directory, with directory separators mapped to dots — so `std/io.ln` is the module `std.io`.

### 16.3 Import declarations

- 1 `import module.path` makes the declarations of the named module available in the current source file. The grammar production is *import-declaration* (Annex A).
- 2 **Selective import.** `import module.path.{name1, name2, ...}` brings only the listed names into scope rather than the whole module. This is the form used throughout the standard library, e.g. `import std.mem.{mem_alloc, mem_free}` or `import std.fs.{File, open_file, close_file}`.
- 3 The `use` statement (§ 5.5) provides a *local-scope* import/alias inside a function body. *implementation-defined*<sup>2</sup>; the standard library uses top-level `import` rather than `use`.
- 4 **Path resolution:** a module path is converted to a file system path by replacing dots with directory separators and appending `.ln`. The search is relative to the compiler's working directory. *implementation-defined*<sup>3</sup>.

| Module path              | File path                   |
|--------------------------|-----------------------------|
| <code>std.c</code>       | <code>std/c.ln</code>       |
| <code>std.io</code>      | <code>std/io.ln</code>      |
| <code>std.fs</code>      | <code>std/fs.ln</code>      |
| <code>std.math</code>    | <code>std/math.ln</code>    |
| <code>mylib.utils</code> | <code>mylib/utils.ln</code> |

<sup>1</sup>Implementation-defined behavior: derived from its file path by the implementation

<sup>2</sup>Implementation-defined behavior: Its precise semantics are implementation-defined in this revision

<sup>3</sup>Implementation-defined behavior: The exact search algorithm, including support for search paths, is implementation-defined

## 16.4 Module loading

- 1 When the compiler encounters an **import** declaration:
  1. The module path is converted to a file path.
  2. If the module has already been loaded, the import is a no-op (deduplication).
  3. Otherwise, the file is read, lexed, parsed, and recursively resolved.
  4. The loaded module's declarations are merged into the importing module's declaration list.
- 2 Each module is loaded at most once. Circular imports are therefore implicitly handled: if module A imports module B which imports module A, the second import of A is a no-op because A is already loaded.

### Note

The reference implementation maintains a linked list of loaded modules keyed by their dot-separated name.

## 16.5 Name visibility

- 1 By default, every name declared at module scope is *public*: visible to any module that imports the declaring module. This default matches the typical usage where module declarations form an API surface.
- 2 A declaration prefixed with the attribute **[private]** (§ 10) is *module-private*: it is visible only within the defining module and may not be referenced from any importing module.

```
1 // In module foo.bar:
2 func public_api() int { return internal_helper() } // public by
   default
3
4 [private]
5 func internal_helper() int { return 0 } // only foo.bar may
   reference this
```

### Constraint

**Q-018.** A reference to a **[private]** declaration from a module other than the one where it is defined is *ill-formed* ([E084]). The reference may appear in any expression position—identifier lookup, member access, type instantiation, etc.

### Note

Compilation under the reference implementation translates each **[private]** declaration to a C declaration with internal linkage (**static** storage class), and each public declaration to one with external linkage.

## 16.6 Standard library module paths

- 1 The following module paths are defined by this specification for the standard library:

| Module path           | Description                 |
|-----------------------|-----------------------------|
| <code>std.c</code>    | C standard library bindings |
| <code>std.io</code>   | Console I/O                 |
| <code>std.fs</code>   | File system operations      |
| <code>std.math</code> | Mathematical functions      |

## 16.7 Name mangling

- <sup>1</sup> The name of a Lain declaration in the translation output is *implementation-defined*<sup>4</sup>. The reference implementation uses the function name prefixed with the module path, with dots replaced by underscores. Name collisions between modules are *implementation-defined*<sup>5</sup>.

---

<sup>4</sup>Implementation-defined behavior: determined by the implementation

<sup>5</sup>Implementation-defined behavior: detected by the implementation; the reference implementation appends a numeric suffix to resolve conflicts



## Chapter 17

# C interoperability

### 17.1 Overview

- 1 Lain compiles to C99 and provides direct, zero-overhead interoperability with C libraries through three mechanisms:
  1. `c_include` — injects C `#include` directives;
  2. `extern` declarations — declares C functions for use in Lain;
  3. type mapping — Lain types map to their C99 counterparts.

### 17.2 The `c_include` declaration

- 1 `c_include "header.h"` causes the compiler to emit a `#include` directive in the generated C code. The string content is emitted verbatim.

```
1 c_include "<stdio.h>"
2 c_include "<stdlib.h>"
```

#### Constraint

`c_include` shall appear only at module top level.

### 17.3 Extern function declarations

- 1 `extern func name(params) return` declares a C function. Because an extern has no body, its effect row cannot be inferred; the row is therefore *believed*, and its default is the set of *all* effects (`io`, `diverge`, `raises`, `alloc`). An `effects` clause on an extern *narrows* that default.
- 2 This is the inverse of a declaration with a body, whose default row is  $\emptyset$  and whose clause *widens* it (§ 10.4). The direction differs for one stated reason: inference is available in the one case and impossible in the other. A caller of an extern acknowledges whatever the extern's row names, by the ordinary rule.

#### Constraint

An `extern` with no `effects` clause has the row `{io, diverge, raises, alloc}`. A caller whose own bound does not cover it is *ill-formed* ([E011]).

#### Note

**Trust boundary.** An extern's `effects` clause is a *programmer assertion* that the referenced C function does no more than the row names. The compiler cannot verify it and believes it, so a wrong narrowing—declaring `libc_rand` with an empty row, say—may invalidate

the guarantees that depend on purity (notably P4 determinism for any function that transitively calls it). What has changed is the cost of *silence*: omitting the clause once meant "trusted pure" (the rule once labelled I-016), so forgetting it on a C function that performs I/O was a silent fail-open. Omitting it now claims everything, so forgetting costs precision instead. The empty row, **effects** with no names, is how purity is asserted deliberately.

- 3 Ownership annotations on extern parameters work identically to those on native Lain functions; the borrow checker tracks ownership of resources obtained via extern calls.

## 17.4 Type mapping

- 1 The following table defines the normative mapping between Lain types and their C99 equivalents. The mapping is the ABI used by the translation output.

| Lain type                    | C99 type                                     | Notes                                                         |
|------------------------------|----------------------------------------------|---------------------------------------------------------------|
| <code>u8</code>              | <code>uint8_t</code>                         |                                                               |
| <code>u16</code>             | <code>uint16_t</code>                        |                                                               |
| <code>u32</code>             | <code>uint32_t</code>                        |                                                               |
| <code>u64</code>             | <code>uint64_t</code>                        |                                                               |
| <code>usize</code>           | <code>size_t</code>                          | pointer-sized                                                 |
| <code>i8</code>              | <code>int8_t</code>                          |                                                               |
| <code>i16</code>             | <code>int16_t</code>                         |                                                               |
| <code>i32</code>             | <code>int32_t</code>                         |                                                               |
| <code>i64</code>             | <code>int64_t</code>                         |                                                               |
| <code>isize</code>           | <code>ptrdiff_t</code>                       | pointer-sized                                                 |
| <code>int</code>             | <code>int</code>                             | at least 32-bit                                               |
| <code>f32</code>             | <code>float</code>                           | IEEE 754                                                      |
| <code>f64</code>             | <code>double</code>                          | IEEE 754                                                      |
| <code>bool</code>            | <code>_Bool</code>                           | C99 bool (1 byte); <i>implementation-defined</i> <sup>1</sup> |
| <code>shared struct T</code> | <code>const T*</code>                        |                                                               |
| <code>var struct T</code>    | <code>T*</code>                              |                                                               |
| <code>mov struct T</code>    | <code>T</code>                               | by value                                                      |
| <code>T[]</code>             | <code>struct { size_t len; T* data; }</code> | fat slice                                                     |
| <code>*T</code>              | <code>const T*</code>                        | unsafe only                                                   |
| <code>var *T</code>          | <code>T*</code>                              | unsafe only                                                   |

## 17.5 ABI for function parameters

- 1 This table is normative and applies to all Lain function calls, including those to **extern** functions.

<sup>1</sup>Implementation-defined behavior: 0 or 1

| Lain param        | Type      | C param  | Direction     |
|-------------------|-----------|----------|---------------|
| name T            | struct    | const T* | in            |
| <b>var</b> name T | struct    | T*       | in/out        |
| <b>mov</b> name T | struct    | T        | in (by value) |
| any mode          | primitive | T        | in            |

## 17.6 Pointer operations

- 1 Raw pointer operations (`*ptr` dereference, `&x` address-of) require an **unsafe** block (see §18).

### Constraint

Pointer-to-pointer or integer-to-pointer casts shall occur only inside an **unsafe** block ([E012]).

## 17.7 String interop

- 1 Lain strings (`u8[]`) are byte slices, not null-terminated C strings. To pass a Lain string to a C function expecting `char*`, the `.data` pointer must be extracted explicitly:

```

1 c_include "<stdio.h>"
2 extern func puts(s *u8) int effects io
3
4 func main() int effects io {
5     var s = "hello"
6     puts(s.data)    // OK: .data is *u8 = const char*
7     return 0
8 }
```

- 2 String literals are null-terminated (§5.7.5), so `literal.data` is a valid `const char*` for C functions that expect a C string.

## 17.8 Opaque C types

- 1 An opaque C type is declared with **extern type** Name. The type's layout is unknown to Lain; it may only be used behind a pointer.

```

1 extern type FILE
2 extern func fopen(path *u8, mode *u8) mov *FILE effects io, alloc
```

### Constraint

An opaque type shall not be used as a value type (only as a pointer target). `var f FILE` is *ill-formed*.



## Chapter 18

# Unsafe code

### 18.1 Purpose

- 1 The **unsafe** block provides a controlled escape hatch for operations that a conforming implementation cannot statically verify. Unsafe blocks are explicitly delimited in the source text and are therefore auditable. Unsafe code cannot accidentally spread outside the enclosing block.

### 18.2 Syntax

- 1 An unsafe block is written as **unsafe** { body }. It may appear wherever a statement is valid (§9.10).

### 18.3 What unsafe enables

- 1 Inside an **unsafe** block, the following operations are permitted:
  1. **Raw pointer dereference:** `*ptr` reads the value pointed to by a raw pointer; `*ptr = v` writes through a mutable raw pointer.
  2. **Address-of:** `&x` creates a raw pointer to the storage of `x`.
  3. **Pointer and integer casts:** `p as *T`, `n as *T`, `p as usize`, etc.
  4. **Direct ADT field access:** reading a variant's field with the qualified form `value.Variant.field` without going through a **case** arm. Outside **unsafe** this is *ill-formed* ([E125]); the safe idiom is to destructure with **case**.
  5. **Bounds-check suppression:** array indexing inside an **unsafe** block does not require a VRA proof (the programmer asserts the access is in bounds).

### 18.4 What unsafe does NOT disable

- 1 The following safety features remain active inside **unsafe** blocks:

| Feature                           | Disabled by <b>unsafe</b> ? |
|-----------------------------------|-----------------------------|
| Pointer dereference ([E060])      | Yes                         |
| Pointer/integer casts ([E012])    | Yes                         |
| Direct ADT field access ([E125])  | Yes                         |
| Bounds check requirement ([E085]) | Yes                         |

| Feature                            | Disabled by <code>unsafe</code> ? |
|------------------------------------|-----------------------------------|
| Ownership tracking ([E001]–[E007]) | No                                |
| Borrow checking ([E004])           | No                                |
| Move semantics                     | No                                |
| Type checking                      | No                                |
| Division-by-zero check ([E015])    | No                                |
| Exhaustiveness checking ([E014])   | No                                |
| Purity enforcement ([E011])        | No                                |

### Constraint

Ownership and linearity rules (§ 11) are not suspended inside `unsafe` blocks. Use-after-move, double-move, and borrow conflicts are still constraint violations inside `unsafe` ([E001]–[E007]).

## 18.5 Undefined behavior in `unsafe`

- 1 The following operations inside an `unsafe` block may produce **undefined behavior**<sup>1</sup>:
  - Dereferencing a null pointer.
  - Dereferencing a pointer to freed or uninitialized memory.
  - Writing to memory through a pointer derived from a read-only (shared) source.
  - Accessing an array element with an out-of-range index.
  - Accessing a field of an ADT through a tag that does not match the active variant.
- 2 A conforming implementation that generates C99 translation output inherits the undefined behavior rules of ISO/IEC 9899 (C17) for operations within `unsafe` blocks.

## 18.6 Borrow scope for `unsafe` blocks

- 1 An `unsafe` block is a regular block scope for the purposes of borrow checking. Borrows registered within the block are released on block exit, and the borrow checker’s invariants hold at the block boundaries.

## 18.7 Nesting

- 1 `unsafe` blocks may be nested. The inner block does not extend the permissions of the outer block beyond what is already enabled. *implementation-defined*<sup>2</sup>.

---

<sup>1</sup>Undefined behavior: undefined behavior if the programmer’s safety assertion is incorrect

<sup>2</sup>Implementation-defined behavior: Whether the implementation issues a diagnostic for redundant nested `unsafe` blocks is implementation-defined

## Chapter 19

# Memory model

### 19.1 Overview

- 1 Lain’s memory model is founded on three invariants:
  1. **Deterministic allocation** — every allocation is explicit and predictable; the implementation never implicitly allocates heap memory.
  2. **Zero runtime overhead** — there is no garbage collector and no reference-counting mechanism.
  3. **C99 layout compatibility** — the physical layout of all objects follows ISO/IEC 9899 (C17) rules exactly.
- 2 This clause specifies the abstract notions of *object*, *storage duration*, *object representation*, and *address*. The abstract machine described here is realised by any conforming translation output (§4).

### 19.2 Objects

- 1 An *object* is a contiguous region of storage that holds a value of a declared type. Every variable declaration introduces exactly one object. Fields of a struct introduce sub-objects. Elements of a fixed-size array introduce element sub-objects.
- 2 Every object has:
  - a *declared type* (§7);
  - an *ownership mode* — shared, **var**, or **mov** (§11.1);
  - a *linear state* — UNINIT, UNCONSUMED, or CONSUMED (§11.2);
  - a *storage duration* (§19.3);
  - an *address* — the byte address of its first storage unit, aligned as required by the target platform.

### 19.3 Storage duration

- 1 Three storage durations are defined:

**Automatic duration** Objects declared inside a function or block body (including function parameters) have automatic storage duration. Storage is allocated when the enclosing scope is entered and released when that scope exits.

**Static duration** Objects declared at module scope have static storage duration. Their storage persists for the entire program execution. *implementation-defined*<sup>1</sup>.

**Heap (explicit) duration** Objects allocated through C interop (`malloc` or equivalent) have programmer-controlled duration. The ownership system (§11) enforces that `mov` pointers to heap memory are consumed exactly once ([E003]).

### Constraint

A `return` statement shall not return a reference whose referent has automatic storage duration ([E010]).

## 19.4 Stack allocation

- 1 All local variables, struct values, and fixed-size arrays are allocated on the call stack. No implicit heap allocation is ever performed by the compiler.

```
1 var x int = 42           // stack: sizeof(int32_t) bytes
2 var arr int[100]         // stack: 100 * sizeof(int32_t) bytes
3 var p Point              // stack: sizeof(Point) bytes
```

- 2 Each function call produces a *stack frame* containing:
  - all local variables declared in the function;
  - all function parameters (by value or by pointer, per the calling convention of §12.1);
  - compiler-generated temporaries required for expression evaluation.
- 3 The stack frame is automatically reclaimed when the function returns. *implementation-defined*<sup>2</sup>.

## 19.5 Heap allocation

- 1 Heap allocation is performed exclusively through C interop (§17). The idiomatic pattern declares the C `malloc/free` pair as an `extern func` with its row:

```
1 c_include "<stdlib.h>"
2 extern func malloc(size usize) mov *void effects io, alloc
3 extern func free(ptr mov *void) effects io
```

- 2 Because `malloc` returns `mov *void`, the caller receives sole ownership of the allocated block. Because `free` consumes `mov *void`, calling `free` transfers that ownership to the allocator. The ownership system therefore enforces statically:
  - **No use-after-free:** using the pointer after `free` is a use-after-move ([E001]).
  - **No double-free:** a second call to `free` on the same pointer is a double-consume ([E002]).
  - **No leak:** failing to call `free` on a `mov` pointer is an unconsumed linear value ([E003]).

<sup>1</sup>Implementation-defined behavior: The value of a static-duration object whose declaration lacks an initializer is implementation-defined; the reference implementation zero-initializes such objects via C99 static initialization rules

<sup>2</sup>Implementation-defined behavior: The implementation's behaviour when the call stack is exhausted is implementation-defined; the reference implementation inherits the platform's stack overflow signal

## 19.6 Value semantics

- 1 The behaviour of assignment and function argument passing is determined by the type's linearity and the ownership mode:

| Type                                                     | Assignment $b = a$                 | Notes                     |
|----------------------------------------------------------|------------------------------------|---------------------------|
| Primitives ( <code>int</code> , <code>bool</code> , ...) | Bitwise copy                       | Always allowed            |
| Non-linear struct                                        | Field-by-field copy                | Allowed                   |
| Linear struct (has <code>mov</code> field)               | Forbidden                          | Requires <code>mov</code> |
| Non-linear array                                         | Element-wise copy                  | Allowed                   |
| Linear array                                             | Forbidden                          | Requires <code>mov</code> |
| Slice ( <code>T[]</code> )                               | Copy <code>{len, data}</code> pair | Always allowed            |
| Raw pointer ( <code>*T</code> )                          | Copy address                       | Always allowed            |

### Constraint

A linear value cannot be implicitly copied: transferring it (by assignment, argument, or return) without an explicit `mov` is an implicit move and is *ill-formed* ([E007], § 11.3).

## 19.7 Object representation

- 1 The *object representation* of a Lain type is its C99 equivalent as defined in § 17.4. Every object is a sequence of `u8` bytes whose count equals the C99 `sizeof` of the corresponding C type.
- 2 *Value equality* is defined per type:
  - Primitives: bitwise equality of the value representation.
  - Structs: field-by-field value equality.
  - Slices: equality of both the data pointer and the length.
  - Raw pointers: address equality.
  - ADTs: tag equality and, when tags match, field-by-field equality of the active variant.

## 19.8 Alignment and layout

- 1 Lain inherits ISO/IEC 9899 (C17) alignment and padding rules:
  - Struct fields are laid out in declaration order.
  - Each field is aligned to its natural alignment (the alignment of its C99 equivalent type).
  - The compiler inserts padding between fields as necessary.
  - The total size of a struct is rounded up to a multiple of its largest field alignment.
- 2 Lain provides one layout control annotation: `[packed]`, which lays a struct of integer fields out bit-exactly inside one integer (§ 7.2.4). There is no `#pragma pack` equivalent and no `alignas`; every other struct has the platform-default C99 layout.

*implementation-defined*<sup>3</sup>.

- 3 `@sizeof(T)` and `@alignof(T)` are expressions of type `usize` whose values are the size and alignment of `T` as laid out for the target: the C99 `sizeof` and `_Alignof` of the type the implementation emits for `T`. A layout claim written as a comment is not checked; one written as a module-scope `assert @sizeof(Token)== 8` (§10.7) is, every time the program is built.

## 19.9 ADT and enum layout

- 1 A *plain enumeration* — an enum type none of whose variants carries a payload — is emitted as the smallest unsigned integer type that holds its variants: `uint8_t` for up to 256 variants, `uint16_t` for up to 65536, `uint32_t` otherwise. The variants are the values  $0, 1, \dots, n-1$  in declaration order. Each enum type is still nominally distinct in the language; the C translation names it with its own `typedef`:

```
typedef uint8_t Color;           // Red = 0, Green = 1, Blue = 2
```

- 2 Because struct fields keep their declaration order (§19.8), the programmer controls the padding a narrow enumeration leaves: `{pos u32, len u16, kind Kind}` is 8 bytes where `{kind Kind, pos u32, len u16}` is 12.
- 3 An ADT type uses a niche-based encoding whenever a zero-cost representation is derivable. The compiler performs *aggressive multi-slot niche optimization* (D-Niche): it explores the full sentinel space of the payload and packs as many empty variants as fit, without any tag byte. When the pool is insufficient to cover every empty variant, the layout falls back to a tagged union and the compiler emits `[W120]` (best-effort warning; not an error).

```
// Tagged-union fallback (only when no niche is derivable):
typedef struct {
    enum { Shape_Circle_tag, Shape_Rect_tag } tag;
    union {
        struct { float r; } Circle;
        struct { float w; float h; } Rect;
    };
} Shape;
```

- 4 *Niche encoding (zero-cost)*. When the payload of one variant contains a bit pattern that cannot occur in valid values—for instance, `0x0` for a pointer field, or a value outside a VRA-proven range for an integer field—that pattern is reserved to encode an alternative variant. No explicit tag is emitted.

```
// Niche encoding for Option(*File): size = sizeof(*File)
typedef File *Option_Ptr_File;
// Some(p)  -> non-null pointer p
// None     -> 0x0
```

- 5 The sentinel pool size depends on the target. For pointer payloads, the pool is the zero-page-unmappable region of the target’s address space, stepped by the pointer alignment (e.g. 16384 slots for `x86_64-linux` with alignment 8). For integer payloads, the pool is the type’s full range minus any VRA-narrowed sub-range: `type Pct = i32 >= 0 and <= 100` as a payload yields a pool of  $\approx 4.3 \times 10^9$  slots (every integer outside `[0, 100]`). For `bool`, the pool is the 254 byte patterns outside `{0, 1}`.

---

<sup>3</sup>Implementation-defined behavior: The exact size and alignment of each type is implementation-defined, following the C99 ABI of the target platform

- 6 Niche allocation cascades through nested enum payloads. If `Option(*T)` reserves the value 0 for `None`, a surrounding `Result(Option(*T), Err4)` sees the remaining slots (8, 16, 24, ...) and packs its own empty variants into them. Layout of the entire result still occupies one pointer-sized word.
- 7 *Multi-payload niche*. An enum with exactly two payload variants where one payload (the *primary*) has a non-empty sentinel pool and the other payload (the *secondary*) has a single field whose type is an *enum of pure-empty variants* is encoded by mapping each sub-variant into a sentinel of the primary's pool. For `Result(*T, FileErr)` with `FileErr = enum { NotFound, Permission, OOM }`, the encoding on a host with pointer alignment 8 is `Ok(f) = any valid pointer`, `Err(NotFound) = sentinel 0`, `Err(Permission) = sentinel 8`, `Err(OOM) = sentinel 16`. The secondary constructor emits a stride-multiplication; the case-match discriminator emits a corresponding stride-division for binding-style patterns.
- 8 The compiler never refuses compilation to satisfy a layout optimization: when the pool is insufficient, [W120] is emitted with suggestions (constrain the payload, change to a type with larger sentinel space, or split the enum), and a tag byte is added. The programmer decides whether to accept the overhead.
- 9 The CLI flag `-target=<triple>` selects the target's sentinel pool size (host auto-detection is the default). Supported triples include `x86_64-linux-gnu`, `aarch64-linux-gnu`, `x86_64-windows-msvc`, and `cortex-m4-bare` (the last has zero pool: no MMU means address 0 may be a valid vector table entry).

## 19.10 No garbage collector

- 1 Lain has no garbage collector. Consequently:
  - execution is never interrupted for memory management;
  - deallocation occurs at statically deterministic program points;
  - no reference counting or heap tracing is performed at runtime.
- 2 This makes Lain suitable for real-time, embedded, and safety-critical applications where garbage-collection pauses are unacceptable.

## 19.11 Memory safety in the safe fragment

- 1 In the *safe fragment* (code outside `unsafe` blocks), the following memory safety properties hold by construction:

| Property                 | Enforcement mechanism                     |
|--------------------------|-------------------------------------------|
| No use-after-free        | [E001] (use-after-move)                   |
| No double-free           | [E002] (double-consume)                   |
| No leak of linear values | [E003] (unconsumed at scope exit)         |
| No dangling references   | [E010] (ref to automatic-duration object) |
| No out-of-bounds access  | [E085] (VRA proof failure)                |
| No null-pointer deref    | No raw pointers in safe fragment          |
| No data races            | No concurrency in current version         |

### Constraint

Inside **unsafe** blocks, pointer operations may violate the above properties if the programmer's safety assertion is incorrect (§ 18.5).

## 19.12 Compiler-internal arena (informative)

### Note

The reference implementation uses a bump-pointer arena internally for all AST nodes, symbol table entries, and diagnostic objects. This arena is an implementation detail; it is never exposed to user code. User programs may implement arena allocators of their own through C interop.

# Chapter 20

# Standard library

## 20.1 Overview

- 1 The Lain standard library is a collection of modules under the `std` namespace (§ 16.6). All standard library modules are written in Lain, using C interop (§ 17) for system-level operations. They are subject to the same ownership, borrow-checking, and purity rules as user code.
- 2 This clause is normative for the modules listed in Table 20.1. All other modules described in informative documentation are not part of this specification.

Table 20.1: Standard library module index

| Module path           | Description                 | Status    |
|-----------------------|-----------------------------|-----------|
| <code>std.c</code>    | C standard library bindings | Normative |
| <code>std.io</code>   | Console I/O                 | Normative |
| <code>std.fs</code>   | File system operations      | Normative |
| <code>std.math</code> | Pure mathematical functions | Normative |

### Constraint

A conforming implementation shall provide all modules listed in Table 20.1 at the specified module paths.

## 20.2 `std.c` — C standard library bindings

### 20.2.1 Purpose

- 1 `std.c` provides low-level bindings to selected C standard library functions. It is the foundation on which all other standard library modules that require system calls are built.

### 20.2.2 Required C headers

- 2 `std.c` emits the following C include directives into the translation output:

```
1 c_include "<stdio.h>"
2 c_include "<stdlib.h>"
```

### 20.2.3 Opaque types

- 3 `std.c` declares the following opaque C type:

```
1 extern type FILE
```

FILE is an opaque type representing a C standard I/O stream. It may only be used through pointers and passed to the `extern` functions declared below.

### Constraint

FILE shall not be used as a value type. A declaration such as `var f FILE` is *ill-formed* (§17.8).

## 20.2.4 Extern declarations

4 The following C functions are declared in `std.c`:

| Declaration                                                                       | Ownership notes                    |
|-----------------------------------------------------------------------------------|------------------------------------|
| <code>extern func libc_printf(fmt *u8, ...)int effects io</code>                  | Shared <code>fmt</code> ; variadic |
| <code>extern func libc_puts(s *u8)int effects io</code>                           | Shared <code>s</code>              |
| <code>extern func fopen(filename *u8, mode *u8)mov *FILE effects io, alloc</code> | Returns owned handle               |
| <code>extern func fclose(stream mov *FILE)int effects io</code>                   | Consumes handle                    |
| <code>extern func fputs(s *u8, stream *FILE)int effects io</code>                 | Shared args                        |
| <code>extern func fgets(s var *u8, n int, stream *FILE)var *u8 effects io</code>  | Mutable buffer                     |

### Note

`libc_printf` and `libc_puts` are renamed to avoid conflicts with user-defined functions named `printf` and `puts`.

## 20.3 std.io — Console I/O

### 20.3.1 Dependencies

1 `std.io` imports `std.c`.

### 20.3.2 print

---

**print** (*module std.io*)

*Synopsis:* `proc print(s *u8[:0])`

*Effect:* Writes the null-terminated byte string `s` to standard output. No trailing newline is appended.

*Parameter s:* Shared, null-terminated C string (`*u8[:0]`).

---

### 20.3.3 println

---

**println** (*module std.io*)

*Synopsis:* `proc println(s *u8[:0])`

*Effect:* Writes the null-terminated byte string `s` to standard output, followed by a newline character (`0x0A`).

*Parameter s:* Shared, null-terminated C string (`*u8[:0]`).

## 20.4 std.fs — File system

### 20.4.1 Dependencies

- 1 std.fs imports std.c.

### 20.4.2 Type: File

- 2 File is a linear struct that wraps a C FILE\* handle:

```
1 type File {
2     mov handle *FILE
3 }
```

- 3 Because handle is annotated **mov**, File is a linear type. It must be consumed exactly once, typically by calling close\_file.

### 20.4.3 open\_file

**open\_file** (module std.fs)

*Synopsis:* **proc** open\_file(path \*u8[:0], mode \*u8[:0])**mov** File

*Effect:* Opens the file at path with the access mode mode (C fopen semantics). Returns an owned File value. The caller is responsible for closing the file by consuming the File.

*Parameter path:* Shared borrow of a null-terminated path string.

*Parameter mode:* Shared borrow of a mode string ("r", "w", "a", etc.).

#### Note

If fopen returns null, the handle field is a null pointer. Subsequent operations on such a file produce C-level undefined behavior. A future version may return Result(File, int) instead.

### 20.4.4 close\_file

**close\_file** (module std.fs)

*Synopsis:* **proc** close\_file(**mov** {handle} File)

*Effect:* Closes the file by calling C fclose on the extracted handle pointer. Consumes the File value.

*Parameter:* The File value is consumed by the call; it shall not be used after this point.

### 20.4.5 write\_file

**write\_file** (module std.fs)

*Synopsis:* `proc write_file(f File, s *u8[:0])`

*Effect:* Writes the byte sequence `s` to the file `f` using C `fputs`.

*Parameter `f`:* Shared borrow of the file handle.

*Parameter `s`:* Shared borrow of the string to write.

---

### Example

```
1 import std.fs.{open_file, close_file, write_file}
2
3 func main() int effects io, raises, alloc {
4     var f = open_file("log.txt", "w")
5     defer close_file(mov f)
6     write_file(f, "Hello, world!\n")
7     return 0
8 }
```

## 20.5 std.math — Mathematical functions

### 20.5.1 Overview

- 1 All functions in `std.math` are declared `func` (pure and terminating). They may be called from any context, including other `func` bodies. `std.math` has no dependencies.

### 20.5.2 min

---

**min** (module *std.math*)

*Synopsis:* `func min(a int, b int)int`

*Returns:* The smaller of `a` and `b`. If `a == b`, either may be returned (they are equal).

---

### 20.5.3 max

---

**max** (module *std.math*)

*Synopsis:* `func max(a int, b int)int`

*Returns:* The larger of `a` and `b`.

---

### 20.5.4 abs

---

**abs** (module *std.math*)

*Synopsis:* `func abs(x int)int`

*Returns:* The absolute value of `x`, i.e., `x` if `x >= 0`, `-x` otherwise.

---

**Note**

`abs(INT_MIN)` produces implementation-defined behaviour because `|INT_MIN|` is not representable as `int` on two's complement platforms.

**20.5.5 clamp**

**clamp** (module *std.math*)

*Synopsis:* `func clamp(x int, lo int, hi int) int`

*Returns:* `x` clamped to the inclusive range `[lo, hi]`, i.e., `max(lo, min(x, hi))`.

**Constraint**

Behaviour is **undefined behavior**<sup>a</sup> if `lo > hi`.

<sup>a</sup>Undefined behavior: undefined

**20.6 Generic types (informative)**

- 1 This section describes generic type constructors that are part of the standard library but are not normative in the current version of this specification.

**20.6.1 Option(T)**

- 2 `Option(T)` is a type constructor for optional values:

```
1 type Option(T type) {           // std/option.ln
2     Some { val T }
3     None
4 }
```

- 3 Usage:

```
1 import std.c.{libc_printf}
2 import std.option.{Option}
3
4 type OptionInt = Option(int)
5
6 func main() int effects io {
7     var x = OptionInt.Some(42)
8     case x {
9         Some(v): libc_printf("value: %d\n", v)
10        None:    libc_printf("no value\n")
11    }
12    return 0
13 }
```

**20.6.2 Result(T, E)**

- 4 `Result(T, E)` is a type constructor for fallible computations:

```
1 type Result(T type, E type) { // std/result.ln
2     Ok { val T }
3     Err { err E }
4 }
```

5 Usage:

```
1 import std.result.{Result}
2
3 type Digit = Result(i32, int)
4
5 func digit(c u8) Digit {
6     if c >= 48 and c <= 57 { return Digit.Ok((c - 48) as i32) }
7     return Digit.Err(1)
8 }
```

- 6 **Result**'s fields are not linear, so a linear value (a **File**) cannot be its payload: **Result(File, int)** is refused (**[E083]**). A fallible open declares its own sum type whose payload field is **mov**.

# Grammar summary

- 7 This annex collects all grammar productions defined in this specification. It is normative. Productions are listed in the order they are introduced in the body of the specification. Each section heading identifies the originating clause.
- 8 The grammar uses an extended BNF notation. `::=` introduces a production. Quoted strings are terminals. Unquoted names are non-terminals. `{ X }` denotes zero or more repetitions. `[ X ]` denotes an optional item. `A | B` denotes alternatives.

## §5 Lexical grammar

### §5.5 Tokens

```
token ::=
    keyword
    | identifier
    | integer-literal
    | float-literal
    | string-literal
    | char-literal
    | bool-literal
    | operator
    | punctuator
```

### §5.6 Keywords

```
keyword ::=
    "func" | "effects" | "var" | "mov" | "type"
    | "return" | "if" | "else" | "while"
    | "for" | "in" | "case" | "import" | "use"
    | "extern" | "c_include" | "unsafe" | "comptime" | "defer"
    | "true" | "false" | "and" | "or" | "as"
    | "decreasing" | "continue" | "break"
```

### §5.7 Identifiers

```
identifier      ::= identifier-start { identifier-continue }
identifier-start ::= letter | "_"
identifier-continue ::= letter | decimal-digit | "_"
letter          ::= "A" .. "Z" | "a" .. "z"
```

### §5.8 Literals

```
integer-literal ::=
    decimal-literal
    | hex-literal
    | binary-literal
    | octal-literal
```

```
decimal-literal ::= nonzero-digit { decimal-digit | "_" }
                  | "0"
hex-literal     ::= "0x" hex-digit { hex-digit | "_" }
binary-literal  ::= "0b" binary-digit { binary-digit | "_" }
octal-literal   ::= "0o" octal-digit { octal-digit | "_" }

decimal-digit   ::= "0" .. "9"
nonzero-digit   ::= "1" .. "9"
hex-digit       ::= decimal-digit | "a" .. "f" | "A" .. "F"
binary-digit    ::= "0" | "1"
octal-digit     ::= "0" .. "7"

float-literal    ::= decimal-literal "." decimal-digit { decimal-digit | "_" } [ float-exponent ]
float-exponent  ::= ( "e" | "E" ) [ "+" | "-" ] decimal-digit { decimal-digit | "_" }

string-literal  ::= ''' { string-char } '''
string-char     ::= any-source-char-except-'''-or-\' | escape-sequence
escape-sequence ::= '\ ( '\'' | '\ ' | "n" | "t" | "r" | "0" )

bool-literal    ::= "true" | "false"
```

## §5.9 Operators and punctuators

```
operator        ::=
    "+" | "-" | "*" | "/" | "%"
    | "==" | "!=" | "<" | ">" | "<=" | ">="
    | "and" | "or" | "!" | "in"
    | "&" | "|" | "^" | "<<" | ">>"
    | "=" | "+=" | "-=" | "*=" | "/=" | "%="
    | "&=" | "|=" | "^=" | "<<=" | ">>="
    | "->" | "." | ".."

punctuator      ::=
    "(" | ")" | "{" | "}" | "[" | "]"
    | ":" | "," | "@"
```

## §6 Program structure grammar

```
program          ::= { top-level-declaration }
top-level-declaration ::=
    import-declaration
    | type-declaration
    | function-declaration
    | procedure-declaration
    | extern-declaration
    | c-include-declaration

import-declaration ::= "import" qualified-identifier [ "." "{" identifier { "," identifier } "}" ]
type-declaration  ::= "type" identifier [ "(" type-param-list ")" ]
                  ( "=" type | struct-body | enum-body )
type-param-list   ::= type-param { "," type-param }
type-param        ::= identifier "type"
qualified-identifier ::= identifier { "." identifier }
```

## §7 Type grammar

```

type ::=
    primitive-type
    | array-type
    | slice-type
    | pointer-type
    | qualified-type
    | constrained-type
    | user-defined-type

primitive-type ::=
    "int"   | "u8"   | "u16"  | "u32"  | "u64"  | "usize"
    | "i8"   | "i16"  | "i32"  | "i64"  | "isize"
    | "f32"  | "f64"
    | "bool" | "str"  | "void"

array-type      ::= type "[" integer-literal "]"
slice-type      ::= [ "*" ] type "[" "]"           % fat slice; "*" optional
                  | [ "*" ] type "[" size-constraint "]" % sized fat slice
pointer-type    ::= "*" type                       % raw pointer
                  | "*" type "[" integer-literal "]" % thin ptr + static length
                  | "*" type "[" ":" sentinel-literal "]" % sentinel-terminated
qualified-type  ::= ownership-qualifier type
ownership-qualifier ::= "var" | "mov"
user-defined-type ::= qualified-identifier
constrained-type ::= type relational-op integer-literal
                  { ( "and" | "or" ) type relational-op integer-literal }
relational-op   ::= ">=" | "<=" | ">" | "<" | "=="

```

## §8 Expression grammar

```

expression      ::= assignment-expression

assignment-expression ::=
    logical-or-expression
    | lvalue-expression assignment-operator assignment-expression

assignment-operator ::=
    "=" | "+=" | "-=" | "*=" | "/=" | "%="
    | "&=" | "|=" | "^=" | "<<=" | ">>="

logical-or-expression ::= logical-and-expression { "or" logical-and-expression }

logical-and-expression ::= in-expression { "and" in-expression }

in-expression      ::= equality-expression { "in" equality-expression }

equality-expression ::= relational-expression
                    { ( "==" | "!=" ) relational-expression }

relational-expression ::= shift-expression
                    { ( "<" | ">" | "<=" | ">=" ) shift-expression }

shift-expression    ::= additive-expression
                    { ( "<<" | ">>" ) additive-expression }

additive-expression ::= multiplicative-expression
                    { ( "+" | "-" ) multiplicative-expression }

```

```
multiplicative-expression ::= cast-expression
                             { ( "*" | "/" | "%" ) cast-expression }

cast-expression             ::= unary-expression { "as" type }

unary-expression           ::=
    postfix-expression
  | unary-operator unary-expression

unary-operator              ::= "-" | "!" | "mov" | "var"

postfix-expression         ::=
    primary-expression
  | postfix-expression "." identifier
  | postfix-expression "[" expression "]"
  | postfix-expression "(" [ argument-list ] ")"

argument-list              ::= expression { "," expression }

primary-expression         ::=
    identifier
  | literal
  | "(" expression ")"
  | block-expression
  | if-expression
  | match-expression
  | constructor-expression
  | comptime-expression

literal                    ::=
    integer-literal
  | float-literal
  | string-literal
  | bool-literal

block-expression           ::= "{" { statement } expression "}"

if-expression              ::=
    "if" expression block-expression
  [ "else" ( if-expression | block-expression ) ]

match-expression           ::=
    "case" [ "&" ] expression "{"
    { match-arm }
    "}"

match-arm                  ::= ( pattern-list | "else" ) ":" arm-body

arm-body                   ::= expression newline | block-expression

pattern-list               ::= pattern { "," pattern }

pattern                   ::=
    literal
  | literal ".." literal % inclusive range pattern
  | string-literal
  | qualified-identifier % unit variant
  | qualified-identifier "(" [ binding-list ] ")" % variant with bound fields
```

```

binding-list          ::= identifier { "," identifier }

constructor-expression ::=
    qualified-identifier "{" [ field-init-list ] "}"
  | qualified-identifier "(" [ argument-list ] ")"

field-init-list       ::= identifier ":" expression { "," identifier ":" expression }

comptime-expression  ::= "comptime" block-expression

lvalue-expression    ::=
    identifier
  | postfix-expression "." identifier
  | postfix-expression "[" expression "]"

```

## §9 Statement grammar

```

statement             ::=
    block-statement
  | immutable-declaration
  | var-declaration
  | assignment-statement
  | expression-statement
  | if-statement
  | while-statement
  | bounded-while-statement
  | return-statement
  | defer-statement
  | unsafe-block
  | comptime-if-statement

block-statement        ::= "{" { statement } "}"

immutable-declaration  ::= identifier [ type ] "=" expression newline

var-declaration        ::= "var" identifier [ type ] "=" expression newline

assignment-statement  ::= lvalue-expression assignment-operator expression newline

expression-statement  ::= expression newline

if-statement           ::=
    "if" expression block-statement
  { "else" "if" expression block-statement }
  [ "else" block-statement ]

while-statement        ::= "while" expression block-statement

bounded-while-statement ::=
    "while" expression "decreasing" expression block-statement

return-statement       ::= "return" [ expression ] newline

defer-statement        ::= "defer" block-statement

unsafe-block           ::= "unsafe" block-statement

```

```
comptime-if-statement ::=
  "comptime" "if" expression block-statement
  [ "else" block-statement ]
```

## §10 Declaration grammar

```
function-declaration ::=
  { attribute } "func" identifier
  "(" [ param-list ] ")" return-annotation
  [ effects-clause ] [ "decreasing" expression ]
  block-statement

effects-clause        ::= "effects" [ effect-name { "," effect-name } ]

effect-name           ::= "io" | "diverge" | "raises" | "alloc"

(* REMOVED: 'proc' granted the effects that the row now states. The keyword remains
   reserved and diagnoses the replacement. *)

attribute             ::= "[" identifier [ "(" argument-list ")" ] "]" newline

return-annotation     ::= type [ return-constraint { "and" return-constraint } ]
                        | ( empty )
return-constraint     ::= relational-op constraint-term { ( "+" | "-" ) constraint-term }
constraint-term       ::= integer-literal | identifier | identifier "." "len"
                        % identifier: a parameter

param-list            ::= param { "," param }
param                 ::=
  [ ownership-qualifier ] identifier type           % value parameter
  | identifier "type"                               % type parameter (generics)
  | [ ownership-qualifier ] "{" identifier { "," identifier } "}" type
                        % field-destructuring parameter

struct-body           ::= "{" { field-declaration } "}"
field-declaration     ::= [ "mov" ] identifier type newline

enum-body             ::= "{" { variant-declaration } "}"
variant-declaration   ::=
  identifier newline
  | identifier "{" field-declaration-list "}" newline

field-declaration-list ::= field-declaration { field-declaration }
```

# Diagnostic codes

- 9 This annex is normative. It lists every diagnostic code defined by this specification, the clause that requires the diagnostic, the category (constraint violation, warning, or informational), and a representative message text. A conforming implementation shall issue a diagnostic for every constraint violation in the table below.
- 10 Diagnostic codes are divided into the following ranges:

| Range                        | Domain                                                                    |
|------------------------------|---------------------------------------------------------------------------|
| [E001]–[E010], [E016]–[E021] | Ownership and lifetime                                                    |
| [E011]–[E015], [E060]        | Purity, safety, and verification                                          |
| [E082]                       | Bounded while verification                                                |
| [E083]–[E087]                | Array/slice safety and integer boundaries                                 |
| [E084], [E101]–[E134]        | Modules, attributes, generics, packing, ADT access, effects, verification |
| [E100]                       | Miscellaneous                                                             |
| [VRA]                        | Value range analysis                                                      |

## Ownership and lifetime diagnostics ([E001]–[E010], [E016], [E020]–[E021])

| Code   | Clause | Message and trigger                                                                                                                                                                                                                                                                                                                                                                        |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E001] | § 11.3 | <i>use of moved value 'x'</i> — Issued when an expression reads or passes a variable whose linear state is CONSUMED.                                                                                                                                                                                                                                                                       |
| [E002] | § 11.3 | <i>value 'x' consumed twice</i> — Issued when a second <b>mov</b> expression is applied to a variable already in state CONSUMED. This subsumes the retired E006 ( <i>consumed inside a loop</i> ): consuming a value declared outside a loop from inside its body <i>is</i> consuming it twice, and [E002] names the violation where E006 named only the syntactic setting it occurred in. |
| [E003] | § 11.3 | <i>linear value 'x' not consumed before end of scope</i> — Issued when a <b>mov</b> -mode variable exits scope in state UNCONSUMED.                                                                                                                                                                                                                                                        |

| Code   | Clause | Message and trigger                                                                                                                                                                                                                                                                                                                                  |
|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E004] | § 11.3 | <i>cannot borrow 'x' as mutable; it is already borrowed</i> — Issued when a mutable borrow conflicts with an existing borrow (reads or write).                                                                                                                                                                                                       |
| [E005] | § 11.3 | <i>use of uninitialized variable 'x'</i> — Issued when a variable, struct field, or array element is read before it has been initialized (definite-assignment analysis).                                                                                                                                                                             |
| [E007] | § 11.3 | <i>moving linear value 'x' requires an explicit <b>mov</b></i> — Issued when a linear value is passed to an owned ( <b>mov</b> ) parameter (or otherwise transferred) without the explicit <b>mov</b> keyword: an implicit move.                                                                                                                     |
| [E010] | § 19.3 | <i>cannot return reference to local variable 'x'</i> — Issued when a <b>return</b> statement returns a reference whose referent has automatic storage duration.                                                                                                                                                                                      |
| [E016] | § 11.3 | <i>linear variable 'x' is used inconsistently in the branches of stmt</i> — Issued when a linear value is consumed in some control-flow paths of an <b>if/case</b> but not in others, leaving an ambiguous state at the join point. Field-level inconsistency on linear struct fields is reported with the same code.                                |
| [E020] | § 11.7 | <i>cannot move out of a borrow</i> — Issued when a <b>mov</b> , or a whole-value copy of a linear value, takes its operand from a place the function does not own: a <b>var</b> or shared parameter, a reference binding, a returned borrow, or a field or element of one. The owner still holds the resource; moving it out would leave two owners. |
| [E021] | § 11.3 | <i>assignment overwrites a linear value that still holds a resource</i> — Issued when a linear place is assigned while it may still hold a resource, which would be lost. Consume it first. Through a borrow the old value can never have been consumed ([E020]), so any such assignment is refused.                                                 |

## Purity, safety, and verification ([E011]–[E015])

| Code   | Clause | Message and trigger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E011] | § 12.2 | <i>unacknowledged effect in 'f'</i> — Issued when a function's body has an effect its row does not name, while its bound is $\emptyset$ — that is, while no <code>effects</code> clause was written, or an empty one was. A bound of $\emptyset$ is a function's default <i>guarantee</i> , so violating it is this code; understating a <i>non-empty</i> row is [E130]. The effects are <code>io</code> , <code>diverge</code> , <code>raises</code> and <code>alloc</code> . Also issued for a <code>while</code> loop whose measure is neither given nor inferable, and for a direct self-recursion for which no measure can be inferred — both sources of <code>diverge</code> . (A direct recursion <i>with</i> a well-founded measure is permitted; a present-but-failing measure is [E082] / [E091].) |
| [E012] | § 17.6 | <i>type or constraint violation</i> — Issued when a value does not have the type its context requires (an argument, a constructor field, a return, an operand), when a refinement at a call site cannot be proven, and when a pointer-to-pointer or integer-to-pointer cast occurs outside an <code>unsafe</code> block. The message names which; a test that expects [E012] should say which it means, since the code alone does not.                                                                                                                                                                                                                                                                                                                                                                       |
| [E013] | § 6.2  | <i>redeclaration or shadowing of 'x' is forbidden</i> — Issued for a duplicate parameter name, a local that repeats or shadows an outer binding, a <code>main</code> declared as <code>func</code> , or a binding (variable, parameter, or loop variable) whose name is a builtin type ( <code>i32</code> , <code>u8</code> , <code>bool</code> , ...): a type name denotes a first-class type value and cannot also name a binding.                                                                                                                                                                                                                                                                                                                                                                         |
| [E014] | § 15.4 | <i>non-exhaustive case</i> — Issued when the arms of a <code>case</code> do not cover all values of the scrutinee type. (An out-of-bounds <i>index</i> is a different code, [E085].)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| [E015] | § 8.3  | <i>division by zero</i> — Issued when VRA can prove statically that a division or modulo expression has a zero divisor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## Bounded while verification ([E082])

| Code   | Clause          | Message and trigger                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E082] | § 9.7.1, § 12.3 | <i>termination measure not proven well-founded</i> — Issued when the implementation cannot verify that every looping-back path through the loop body strictly decreases the measure by at least 1 (paths that <b>break</b> / <b>return</b> out are exempt), or that every self-call of a directly-recursive <b>func</b> makes its measure strictly smaller, or that the measure is bounded below. A measure the implementation cannot analyse at all is this code as well. <div><b>Note</b><br/>E080 (<i>measure not proven non-negative</i>) and E081 (<i>cannot extract variables from measure</i>) were separate codes for two sub-cases of this one, issued by a second termination checker in the front end. That checker is removed and the obligation has a single source, so the two codes are withdrawn: an absent or uninferable measure is [E011], a present one that cannot be shown well-founded is [E082], and nothing else is reported. A conforming implementation shall not issue E080 or E081.</div> |
| [E091] | § 12.3          | <i>malformed recursion measure</i> — Issued when a directly-recursive <b>func</b> ’s <b>decreasing</b> measure is not a parameter or a difference of two parameters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Miscellaneous ([E100])

| Code   | Clause   | Message and trigger                                                                                                                                                                                                                                                                                                            |
|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E100] | § 5, § 7 | <i>ill-formed token / semantic constraint violation</i> — A catch-all code used for lexical and syntactic errors without a dedicated code, including: an unexpected token, using <b>==</b> on enum types (§ 7.7), an empty or multi-character character literal, a keyword used as an identifier, and a zero-field array size. |

### Array/slice safety and integer boundaries ([E083]–[E087])

| Code   | Clause          | Message and trigger                                                                                                                                                                                                                                                                                                                  |
|--------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [E083] | § 7.6           | <i>linear field 'f' missing mov annotation</i> — Issued when a struct or enum variant field has a linear type but lacks the <b>mov</b> keyword. (Q-008 transitivity rule.)                                                                                                                                                           |
| [E085] | § 8.12, § 8.13  | <i>index or slice endpoint not proven in bounds</i> — Issued when VRA cannot prove a non-negativity or upper-bound requirement for an array/slice index or sub-slice endpoint.                                                                                                                                                       |
| [E086] | § 13.10         | <i>integer overflow at boundary: range exceeds target type</i> — Issued when the VRA range of a source value provably exceeds the natural range of the target type at a type boundary (declaration, assignment, call argument, return, struct constructor).                                                                          |
| [E087] | § 8.13, § 13.12 | <i>sized slice constraint violated</i> — Issued (1) when both endpoints of a sub-slice <code>arr[a..b]</code> are integer literals and $a > b$ ; (2) at a call site where VRA can prove that the argument does not satisfy the formal parameter's sized slice constraint ( $T[k]$ , $T[>= k]$ , $T[> k]$ , or $T[\text{ref.len}]$ ). |

### Value range analysis ([VRA])

| Code  | Clause | Message and trigger                                                                                                                                                                                                                                                                                         |
|-------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [VRA] | § 13   | <i>value range proof required</i> — Issued when VRA cannot prove that an operation satisfies its required constraint and more specific code is not available. Appears as a diagnostic prefix in messages such as: “[VRA] cannot prove index in bounds” and “[VRA] return value may not satisfy constraint”. |

### Additional diagnostics ([E008], [E009], [E017]–[E125])

The following diagnostics are emitted by the reference implementation and are normative. Each was verified against the compiler's diagnostic strings during the § 8 reconciliation pass.

| Code   | Message and trigger                                                                                                |
|--------|--------------------------------------------------------------------------------------------------------------------|
| [E008] | cannot move a value that is currently borrowed                                                                     |
| [E009] | cannot mutate a field through a raw (shared) pointer — use a <b>var</b> borrow                                     |
| [E017] | a mutable borrow is spelled <b>var</b> : passing to a <b>var</b> parameter requires it at the call site, and a fun |
| [E018] | a function can reach its end without returning a value                                                             |

| Code   | Message and trigger                                                                                                                             |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| [E019] | use of a partially-initialized value                                                                                                            |
| [E060] | pointer dereference outside an <code>unsafe</code> block                                                                                        |
| [E063] | a possibly-marker union value used without testing it first                                                                                     |
| [E064] | a <code> </code> -union cannot be niche-optimized (no spare bit pattern)                                                                        |
| [E084] | access to a <code>[private]</code> declaration from another module                                                                              |
| [E101] | calling a <code>proc</code> from a <code>comptime</code> context (only <code>func</code> is allowed)                                            |
| [E102] | expected an attribute name after the opening bracket                                                                                            |
| [E103] | unknown attribute                                                                                                                               |
| [E104] | a type contains itself by value (infinite size)                                                                                                 |
| [E105] | identifier is defined in another module — add an <code>import</code>                                                                            |
| [E106] | use of an undeclared identifier (diagnosed at translation-output generation)                                                                    |
| [E121] | a struct field invariant — <code>pos usize in text</code> (a valid index), <code>pos usize &lt;= src.len</code> or <code>len usize &lt;=</code> |
| [E122] | a function pointer must be initialized with a matching function                                                                                 |
| [E123] | wrong number of arguments in a call through a function pointer                                                                                  |
| [E124] | generic instantiation: non-generic type, missing or uninferable type argument, wrong arity                                                      |
| [E125] | direct ADT field access outside <code>unsafe</code> — destructure with <code>case</code>                                                        |
| [E065] | a <code>try</code> may propagate a marker the enclosing return union does not list                                                              |
| [E066] | <code>try/else</code> requires a <code>T   markers</code> union value                                                                           |
| [E067] | a checked operator ( <code>+, -?, *, as?</code> ) must be handled by <code>else</code>                                                          |
| [E088] | a function returns a fixed-size array by value — return a slice or write through a <code>var</code> parameter                                   |
| [E089] | a non-terminated value used where a sentinel-terminated type is required                                                                        |
| [E090] | assignment between fixed-length strings of different lengths                                                                                    |
| [E126] | a mutable borrow ( <code>var</code> ) stored in a struct field — a field would outlive any loan the borrow checker                              |
| [E127] | call to an undeclared function                                                                                                                  |
| [E128] | the receiver has no such member, and no function of that name is in scope for UFCS                                                              |
| [E129] | <code>assume</code> outside an <code>unsafe</code> block — use <code>assert</code> to have the fact proven                                      |
| [E130] | a declared, non-empty <code>effects</code> row does not cover what the body does (an empty or absent row is [                                   |
| [E131] | an <code>extern</code> declares a slice parameter — a slice has no C representation; pass a pointer and a length                                |
| [E132] | a struct field's refinement is not one that is enforced — a relation other than <code>f CMP g</code> ( <code>g</code> an integer f              |
| [E133] | a module-scope <code>assert</code> whose operand is not a <code>bool</code> constant expression (§ 10.7) — a run-time fact i                    |
| [E134] | a module-scope assertion is false — reported by the implementation, or, for one that measures a typ                                             |

# Language design rationale

- 11 This annex is informative. It explains the reasoning behind key design decisions in the Lain language. Understanding the rationale helps readers interpret edge cases in the normative text and guides implementors when the specification is ambiguous.

## .1 The `func`/`proc` distinction

- 1 **Decision.** Separate pure functions (`func`) from side-effecting procedures (`proc`) at the declaration level.
- 2 **Rationale.**
- *Termination guarantees:* a `func` is guaranteed to terminate (no unbounded `while`, no unchecked recursion). This enables the compiler to reason statically about function completion and to use `func` bodies inside `comptime` expressions.
  - *Purity propagation:* the type system propagates purity constraints — a `func` cannot call a `proc`, so pure code stays pure by construction.
  - *Optimisation:* the compiler may freely reorder, memoize, or eliminate calls to `func` because they have no observable side effects.
  - *Readability:* the declaration keyword tells the reader whether a callable has side effects without inspecting its body.
- 3 **Alternatives considered.** Single keyword with a purity annotation (`func(pure)`) was rejected as less visible. A Haskell-style IO monad was rejected as too complex for a systems language.

## .2 Explicit `mov` at call sites

- 1 **Decision.** Require explicit `mov` at call sites when transferring ownership.
- 2 **Rationale.**
- *Readability:* `close_file(mov f)` makes ownership transfer visible at the call site.
  - *Intentionality:* accidental ownership transfer is a common source of bugs. Requiring `mov` forces the programmer to acknowledge the transfer.
  - *Symmetry:* the annotation appears in both declarations (`mov x T`) and at call sites (`mov x`), creating a uniform language.
- 3 **Alternatives considered.** Rust-style implicit moves were rejected because they are invisible at call sites. A library function `std::move` (C++) was rejected because ownership transfer should be a language primitive, not a library function.

### .3 Implicit NLL lifetimes (no annotations)

- 1 **Decision.** Borrows have implicit lifetimes determined by non-lexical lifetime analysis. No explicit lifetime annotations are required.
- 2 **Rationale.**
  - *Simplicity:* Rust’s lifetime annotations are widely cited as its steepest learning curve. Lain targets programmers who want safety without that complexity.
  - *Sufficient expressiveness:* NLL handles the vast majority of real-world borrow patterns.
  - *Predictability:* borrows expire at last use — a single, intuitive rule.
- 3 **Trade-offs.** Some programs that Rust can verify with explicit lifetime annotations may be rejected by Lain’s more conservative analysis. Higher-order borrowing patterns are constrained.

### .4 C99 translation output

- 1 **Decision.** Compile to C99 source code rather than LLVM IR, machine code, or bytecode.
- 2 **Rationale.**
  - *Portability:* C99 compilers exist for every platform, from server-class x86 to microcontrollers. Lain inherits this portability for free.
  - *Toolchain reuse:* users get GCC/Clang optimisation passes, debuggers, profilers, and sanitisers without custom tooling.
  - *Interoperability:* C interop is zero-cost because the output *is* C.
  - *Simplicity:* a C emitter is orders of magnitude simpler than an LLVM backend or a native code generator.
  - *Auditability:* the generated C code is human-readable.
- 3 **Trade-offs.** Compilation is two-phase (Lain  $\rightarrow$  C  $\rightarrow$  binary). Some optimisations are impossible without direct IR control. The C type system constrains what Lain can express (e.g., no guaranteed tail calls).

### .5 No exception mechanism

- 1 **Decision.** Lain has no exception mechanism. All error handling uses return values.
- 2 **Rationale.**
  - *Zero overhead:* no unwinding tables, catch tables, or RTTI.
  - *Determinism:* error handling follows the same control flow as normal code; no invisible jumps.
  - *Embedded compatibility:* stack unwinding is unavailable or unreliable on many embedded platforms.
  - *Explicit error paths:* every error is visible in the function signature (via `Result` return types).
  - *Compatibility with `defer`:* the `defer` mechanism provides deterministic cleanup without exceptions.

## .6 Explicit unsafe blocks

- 1 **Decision.** Pointer dereference and direct ADT field access require an explicit `unsafe` block.
- 2 **Rationale.**

- *Auditability:* safety-critical codebases can grep for `unsafe` { to enumerate all potentially dangerous operations.
- *Minimal scope:* Lain’s `unsafe` is narrow — it enables only pointer dereference, address-of, casts, direct ADT access, and bounds suppression. Unlike C (where everything is unsafe by default) or some languages (where `unsafe` enables a much larger capability set), Lain’s `unsafe` is a surgical escape hatch.
- *Social contract:* the keyword communicates to code reviewers that extra scrutiny is warranted.

## .7 Value Range Analysis instead of SMT

- 1 **Decision.** Use interval-based VRA for static verification instead of an SMT solver.
- 2 **Rationale.**

- *Decidability:* VRA always terminates in polynomial time. SMT solving is NP-hard in the general case and may time out.
- *Predictability:* programmers can reason about what the compiler will accept by thinking about integer ranges. SMT solver reasoning is opaque.
- *Simplicity:* the VRA implementation is approximately 600 lines. An SMT integration would be orders of magnitude more complex and would introduce an external dependency.

- 3 **Trade-offs.** VRA is incomplete — it may reject valid programs that an SMT solver could prove correct. Non-linear constraints produce conservatively wide ranges.

## .8 Comptime generics instead of <T> syntax

- 1 **Decision.** Use compile-time function evaluation with `comptime` parameters instead of traditional generic syntax.
- 2 **Rationale.**

- *Unification:* generics, type aliases, and compile-time assertions all use the same mechanism.
- *Expressiveness:* comptime functions can contain arbitrary logic, enabling type-level programming that traditional generics cannot express.
- *Simplicity:* no angle brackets, no trait bounds, no where clauses. Just functions that take types and return types.
- *Precedent:* inspired by Zig’s comptime approach, which has proven ergonomic in practice.

## .9 Universal Function Call Syntax (UFCS)

- 1 **Decision.** Support UFCS: `x.f(args)` desugars to `f(x, args)`.
- 2 **Rationale.**

- *No classes needed:* UFCS provides method-like syntax without class hierarchies, virtual dispatch, or inheritance.

- *Chainability*: enables fluent APIs.
- *Discoverability*: IDE auto-completion works naturally — typing a dot after a value shows available operations.
- *Simplicity*: syntactic sugar only; no new concepts, no method tables, no implicit `this` pointer.

## .10 Bounded while in func

- 1 **Decision.** Allow `while` loops in `func` bodies when a `decreasing` termination measure is supplied.
- 2 **Rationale.** Without `decreasing`, any `while` loop in a `func` would require a proof of termination that goes beyond what the compiler can verify automatically. Requiring the programmer to supply a *measure* — a non-negative integer expression that strictly decreases on every iteration — is a well-established technique from program verification (*ranking functions*). The compiler's three checks (non-negativity, variable dependency, strict decrease) are sufficient to verify termination without an external solver.
- 3 This allows algorithms such as binary search and GCD computation to be expressed as `func` without resorting to `proc`.

# Comparison with related languages

<sup>4</sup> This annex is informative. It places Lain in the context of other systems-programming languages to help readers understand its design choices. Comparisons are accurate as of the date of this specification; language features evolve and readers should verify current status.

## .11 Feature matrix

| Feature          | Lain                      | Rust          | C      | Zig         | Go            |
|------------------|---------------------------|---------------|--------|-------------|---------------|
| Memory safety    | Compile-time              | Compile-time  | Manual | Runtime+OFC | OFC           |
| Null safety      | Option type               | Option type   | None   | Optional    | Nil           |
| Exceptions       | None (Result)             | None (Result) | errno  | Error union | panic/recover |
| Generics         | Comptime                  | Trait-based   | Macros | Comptime    | Type params   |
| Lifetime annots. | None (NLL)                | Required      | N/A    | None        | N/A           |
| Purity tracking  | <b>func</b> / <b>proc</b> | None          | None   | None        | None          |
| Pattern matching | Exhaustive                | Exhaustive    | switch | switch      | switch        |
| UFCS             | Yes                       | No            | No     | No          | No            |
| Backend          | C99                       | LLVM          | Native | LLVM        | Go compiler   |
| GC / runtime     | None                      | Minimal       | None   | Minimal     | GC            |

## .12 Lain versus Rust

### Similarities

- Ownership-based memory safety with no GC.
- Move semantics with explicit transfer.
- No null pointers (Option type for optional values).
- Pattern matching with exhaustiveness checking.
- No exceptions; Result type for error propagation.
- Zero-cost abstractions.

## Differences

| Aspect                | Lain                            | Rust                                    |
|-----------------------|---------------------------------|-----------------------------------------|
| Lifetime annotations  | Implicit (NLL only)             | Explicit ('a, 'static)                  |
| Mutable borrow syntax | <code>var x</code> at call site | <code>&amp;mut x</code>                 |
| Move syntax           | Explicit <code>mov x</code>     | Implicit (default)                      |
| Generics              | Comptime functions              | Trait-based + monomorphization          |
| Purity                | <code>func/proc</code> enforced | Convention only                         |
| Async                 | Not supported                   | <code>async/await</code>                |
| Traits / Interfaces   | Not supported                   | Full trait system                       |
| Macros                | Not supported                   | <code>macro_rules!</code> + proc macros |
| Backend               | C99                             | LLVM                                    |
| Learning curve        | Lower                           | Higher                                  |

- 1 **Key insight.** Lain trades Rust's expressiveness (explicit lifetime annotations, trait system, `async/await`) for simplicity. The result is a language that is easier to learn but less expressive for complex borrowing patterns.

## .13 Lain versus C

### Similarities

- Compiles to native code; no GC.
- Explicit memory management.
- No runtime overhead.
- Systems-level programming: raw pointers accessible in `unsafe`.

### Differences

| Aspect           | Lain                         | C                                  |
|------------------|------------------------------|------------------------------------|
| Memory safety    | Compile-time guarantees      | Manual (error-prone)               |
| Type safety      | Strong, narrow widening      | Weak, implicit conversion          |
| Ownership        | Tracked by compiler          | Programmer discipline              |
| Bounds checking  | Static (VRA)                 | None                               |
| Division by zero | Static detection (E015)      | Undefined behavior                 |
| Pattern matching | Exhaustive <code>case</code> | <code>switch</code> (fall-through) |
| Null safety      | Option type                  | Null everywhere                    |
| Strings          | Slices with length           | Null-terminated arrays             |
| Generics         | Comptime functions           | Macros / <code>void*</code>        |
| Header files     | Module system                | <code>#include</code>              |

- <sup>1</sup> **Key insight.** Lain provides C-level performance with compile-time safety guarantees. The C99 backend enables seamless interop with existing C libraries while eliminating the most common classes of C undefined behavior from the safe fragment.

## .14 Lain versus Zig

### Similarities

- Comptime-based generics.
- No hidden control flow; no hidden allocations.
- C interoperability as a first-class concern.
- Explicit error handling with no exceptions.

### Differences

| Aspect           | Lain                            | Zig                             |
|------------------|---------------------------------|---------------------------------|
| Memory safety    | Compile-time ownership          | Runtime (debug)                 |
| Ownership system | Borrow checker                  | Manual                          |
| Backend          | C99                             | LLVM + self-hosted              |
| Comptime scope   | Type-parameter functions        | Full expression evaluation      |
| Error handling   | Result type + <code>case</code> | Error unions + <code>try</code> |
| Allocator model  | C interop (malloc/free)         | Custom allocators everywhere    |

- <sup>1</sup> **Key insight.** Both Lain and Zig use comptime for generics. Zig’s comptime is more powerful (arbitrary expression evaluation); Lain’s ownership system provides stronger compile-time memory safety guarantees than Zig’s debug-mode runtime checks.

## .15 Lain versus Go

### Similarities

- Simple, readable syntax.
- Fast compilation.
- No class hierarchies.

### Differences

| Aspect            | Lain               | Go                       |
|-------------------|--------------------|--------------------------|
| Memory management | Ownership (no GC)  | Garbage collected        |
| Runtime           | None               | GC + goroutine scheduler |
| Generics          | Comptime functions | Type parameters          |
| Error handling    | Result type        | Multiple return values   |
| Null safety       | Option type        | Nil (null)               |

| Aspect      | Lain               | Go                    |
|-------------|--------------------|-----------------------|
| Concurrency | Not supported      | Goroutines + channels |
| Determinism | Yes (no GC pauses) | GC pauses possible    |

- 1 **Key insight.** Go achieves simplicity through garbage collection. Lain achieves similar simplicity goals through ownership-based safety, without a runtime — making Lain suitable for embedded and real-time applications where Go’s GC is problematic.

## .16 Features unique to Lain

- 1 The following features distinguish Lain from all languages in the comparison:

| Feature                                                          | Description                                                                                                              |
|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <code>func/proc</code> split                                     | Compiler-enforced purity with termination guarantees. No other listed language enforces purity at the declaration level. |
| Caller-site ownership annotations                                | <code>mov</code> and <code>var</code> are visible at every call site, making ownership transfer explicit and auditable.  |
| Interval-based VRA                                               | Polynomial-time static verification of array bounds and arithmetic constraints, without an SMT solver.                   |
| <code>in</code> operator for index constraints                   | <code>idx in arr</code> establishes the bounds guard $0 \leq \text{idx} < \text{arr.len}$ in the containing scope.       |
| Bounded <code>while</code> with <code>decreasing</code>          | Allows terminating loops in pure functions via programmer-supplied ranking functions, verified by the compiler.          |
| Non-consuming <code>match</code> ( <code>case &amp;expr</code> ) | Pattern matching that does not consume the scrutinee; the scrutinee is borrowed for the duration of the match body.      |

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